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(54) **HEALTH EVENT LOGGING AND COACHING USER INTERFACES**

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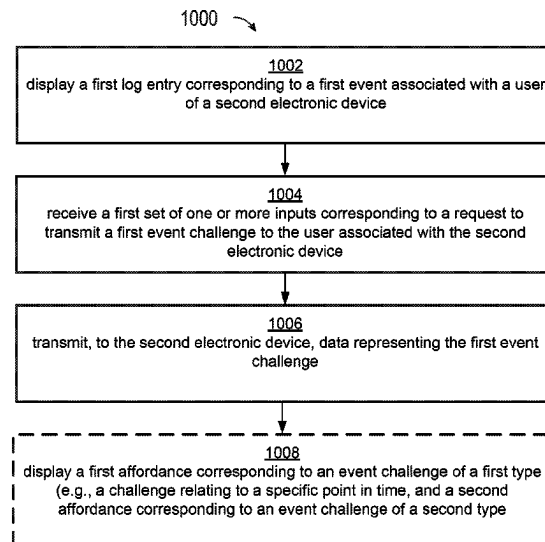
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ABSTRACT

The present disclosure generally relates to logging health-related events and for health-related coaching. In some embodiments, an event is logged by associating an action with a specific time and a user sentiment. In some embodiments, an event challenge, associated with a log entry, is transmitted to an external electronic device.

39 Claims, 36 Drawing Sheets



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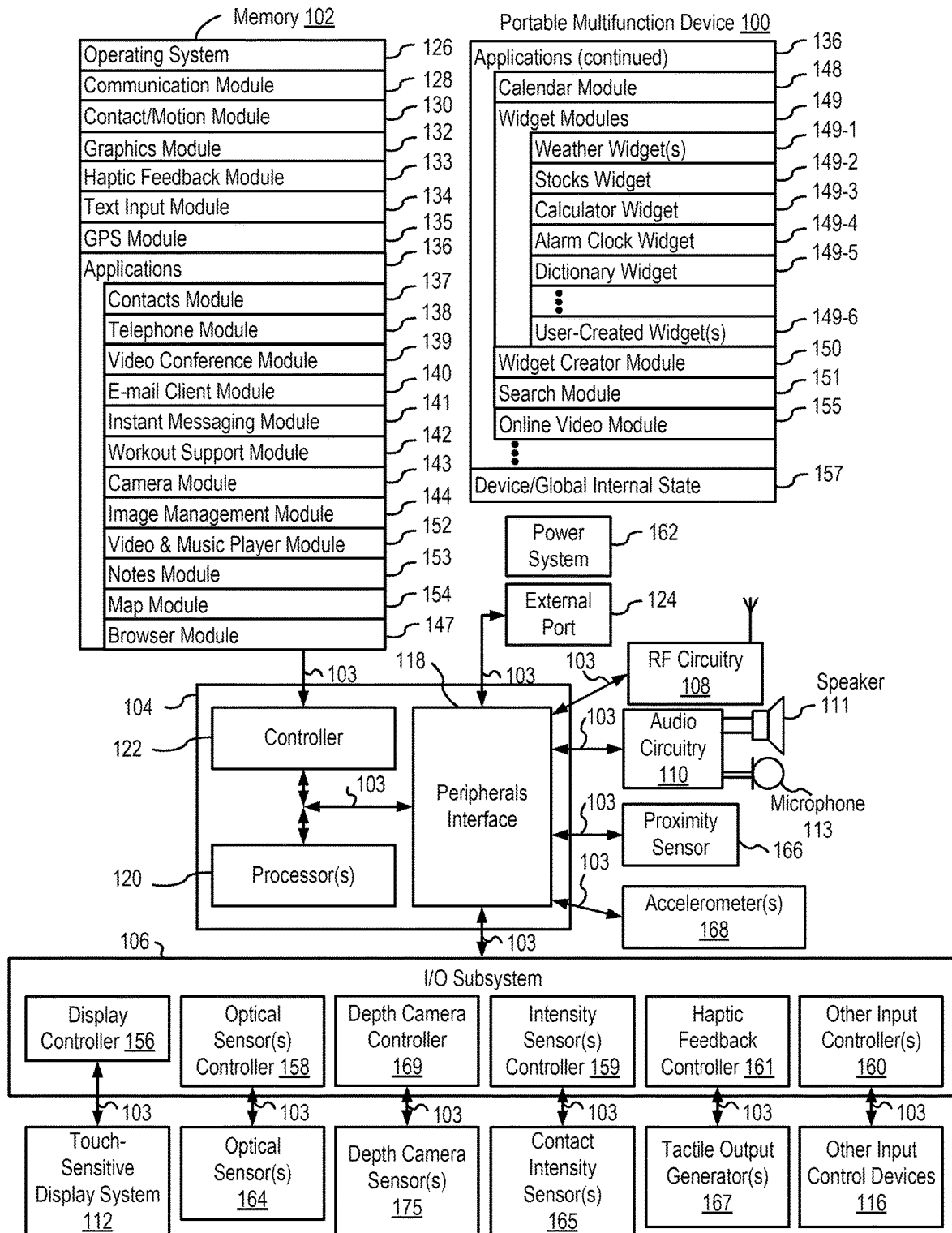


FIG. 1A

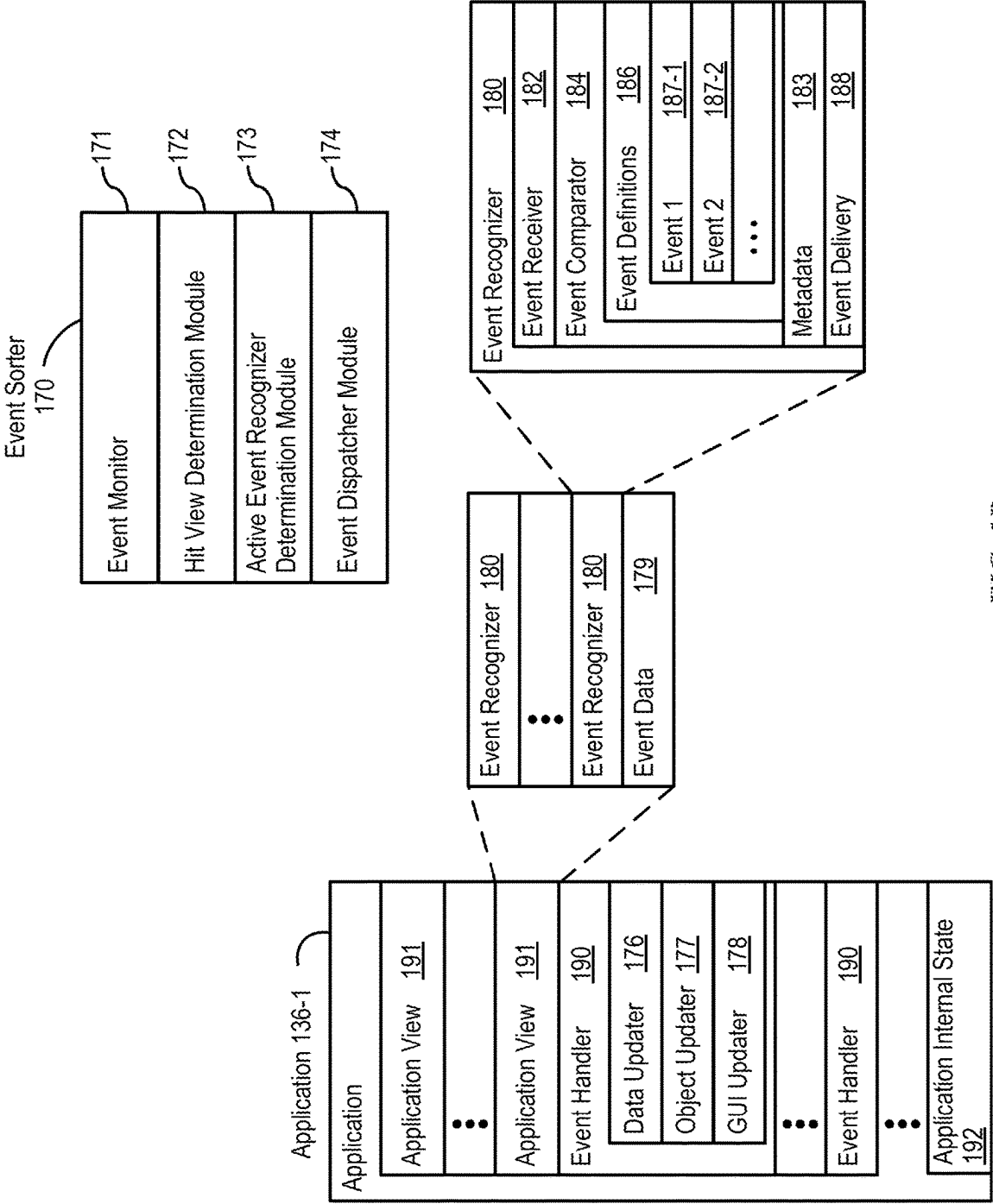


FIG. 1B

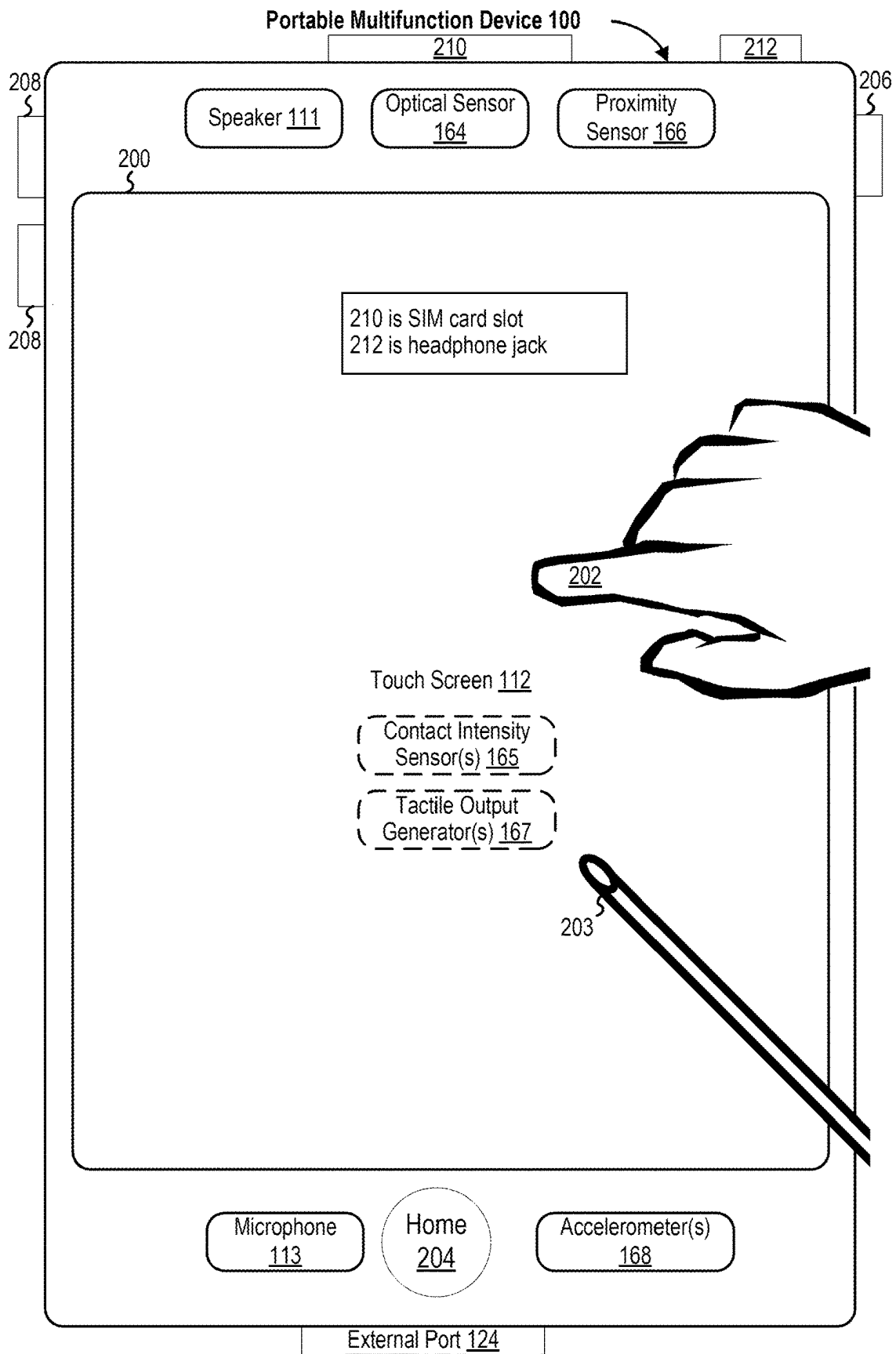


FIG. 2

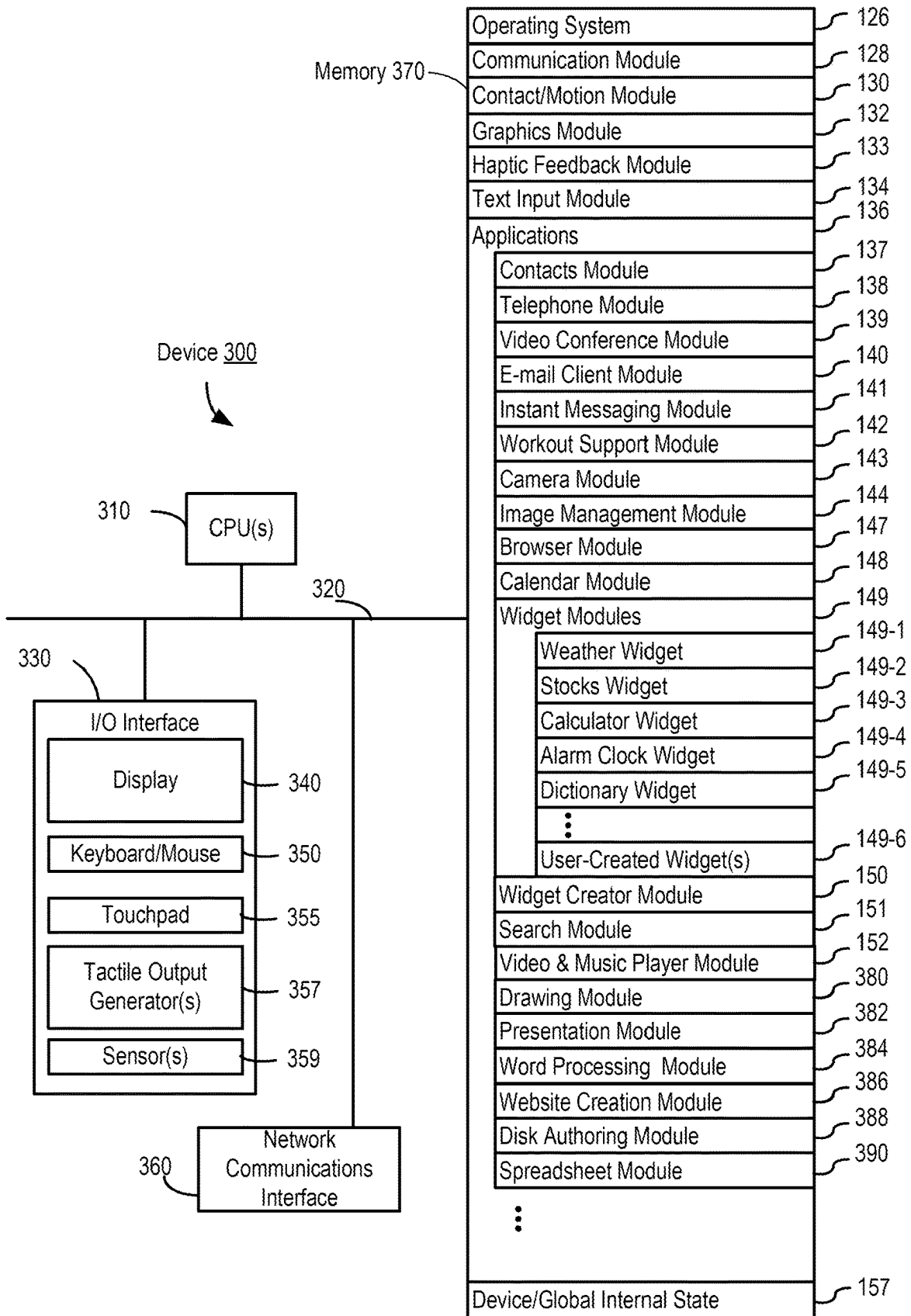


FIG. 3

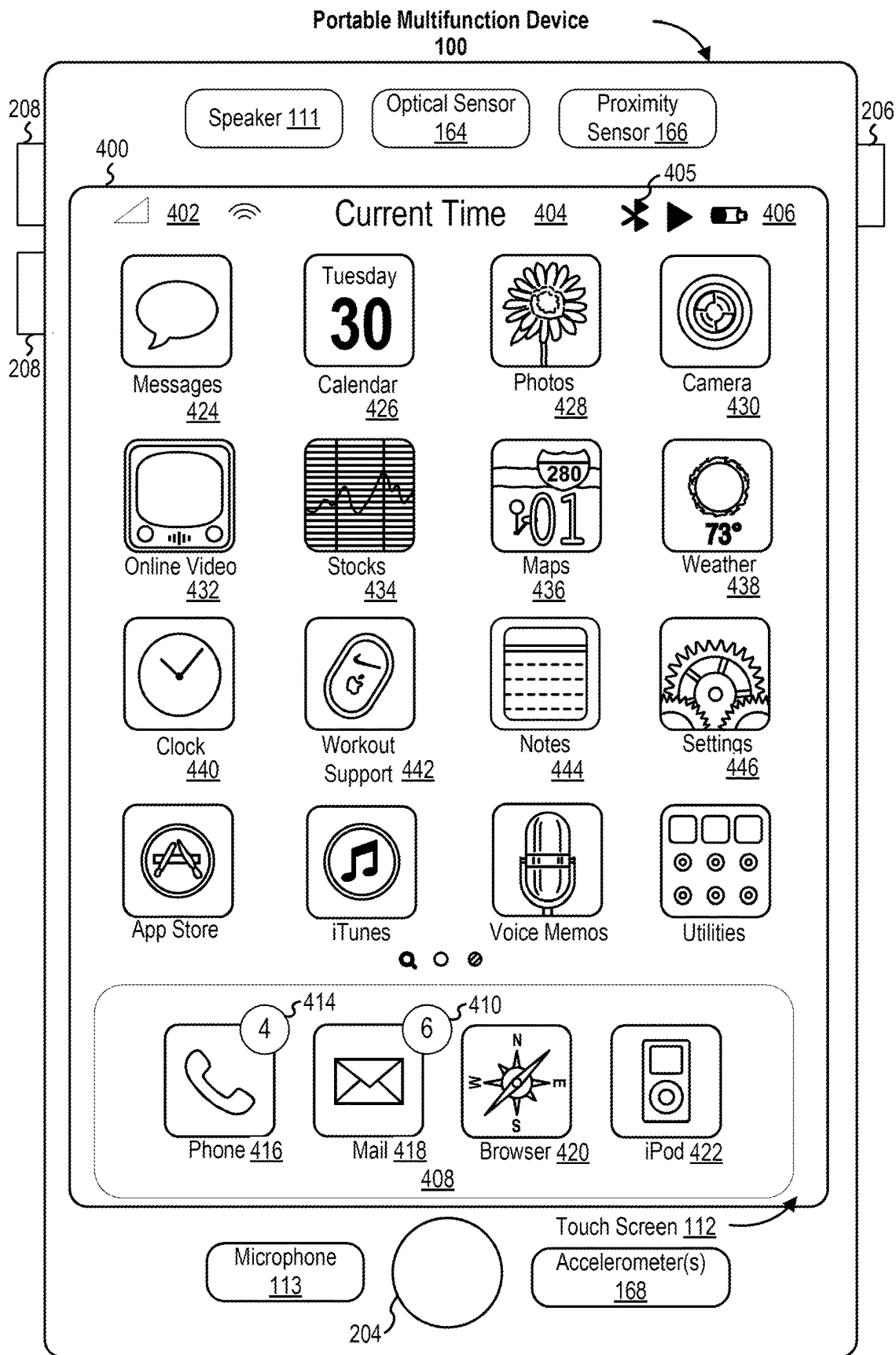


FIG. 4A

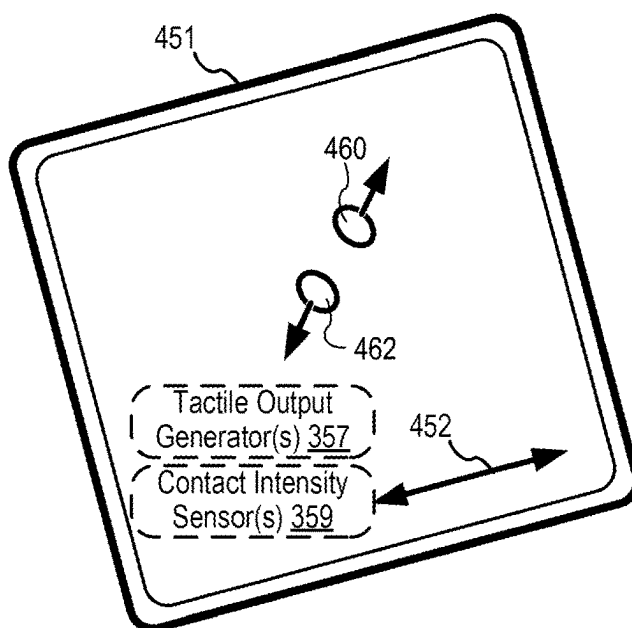
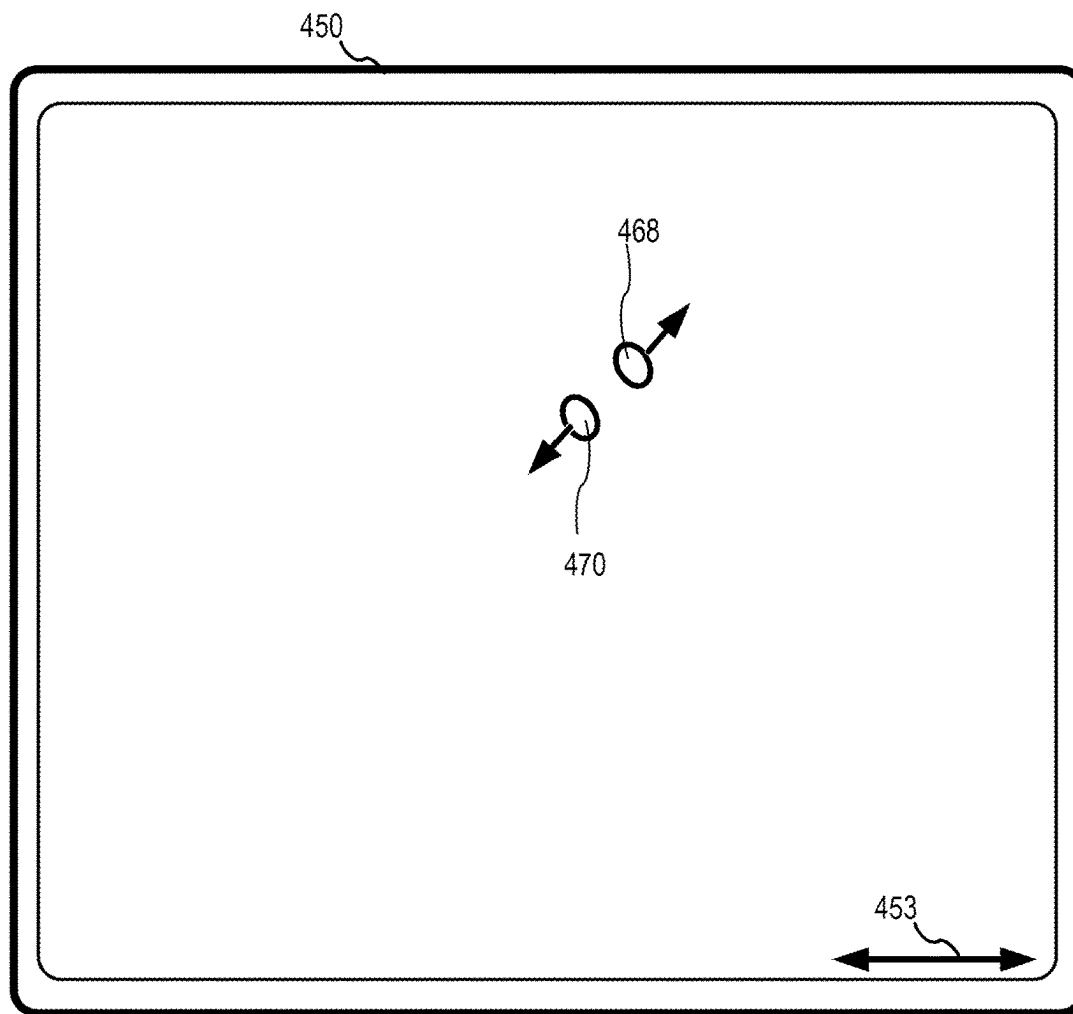


FIG. 4B

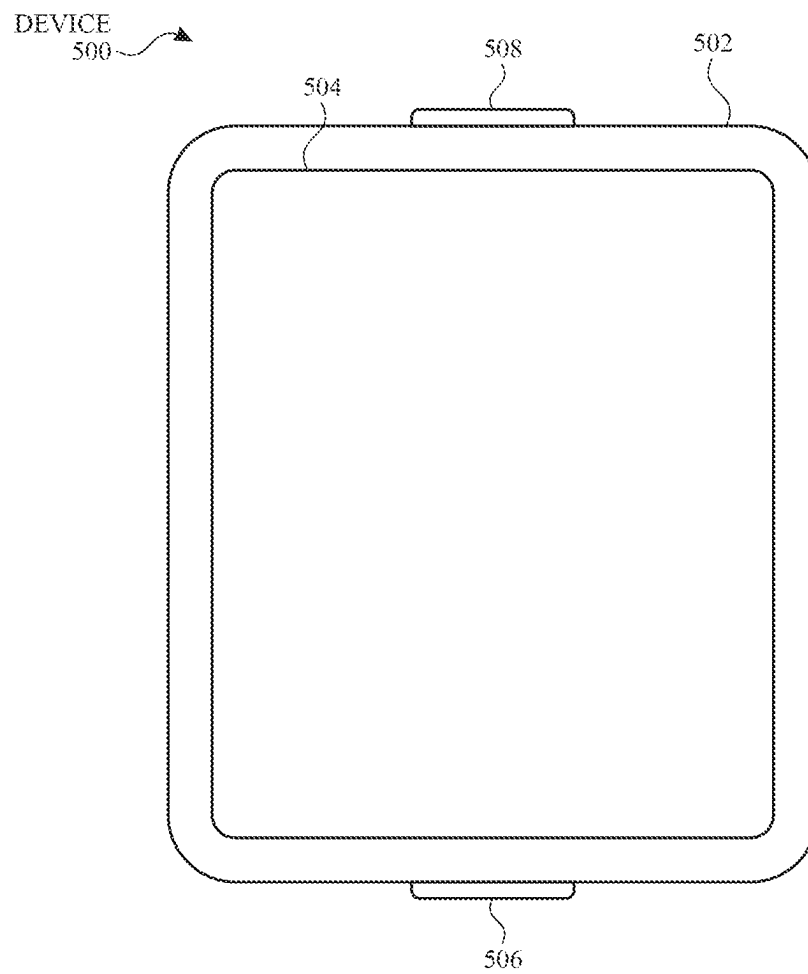


FIG. 5A

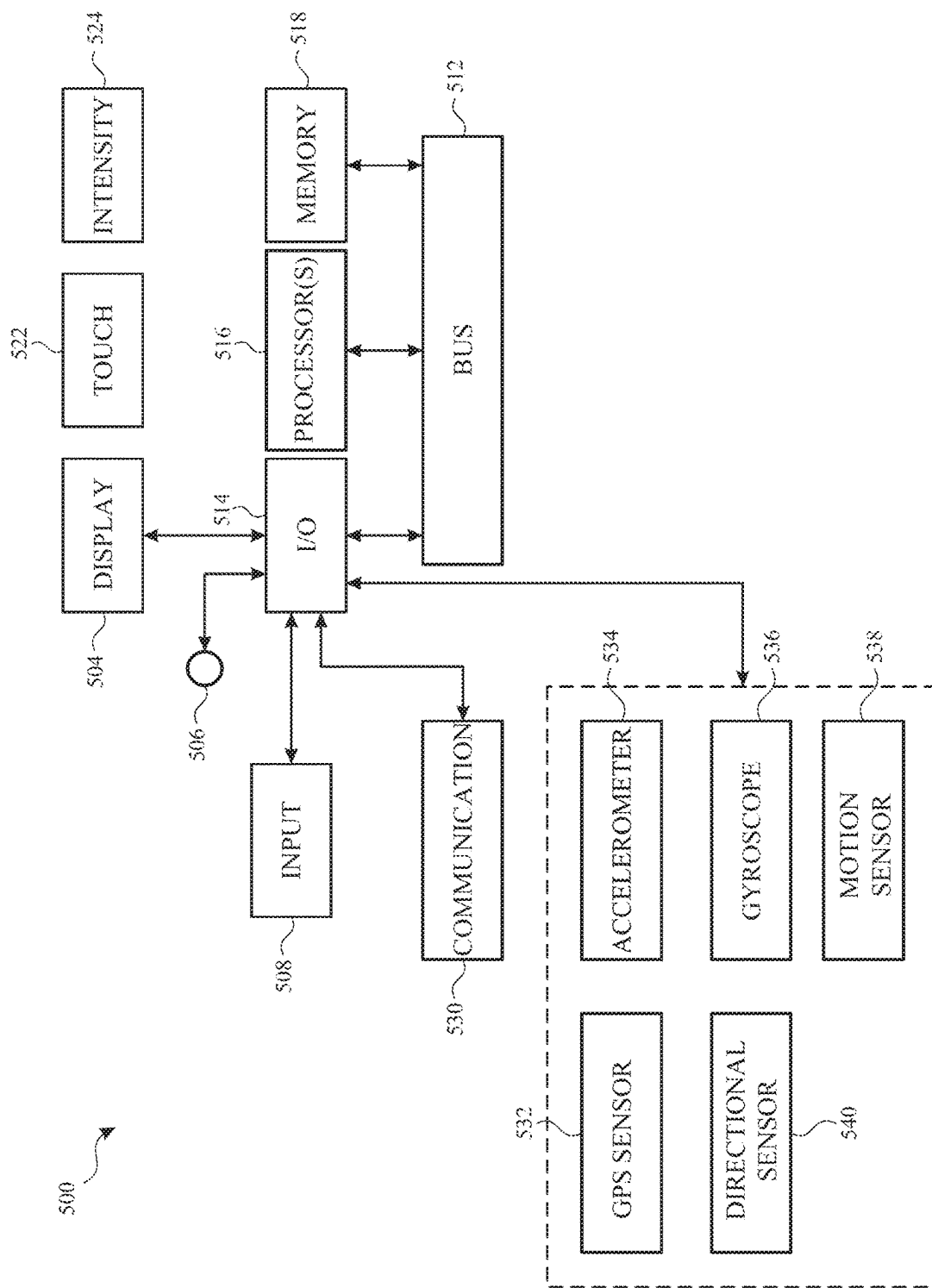


FIG. 5B

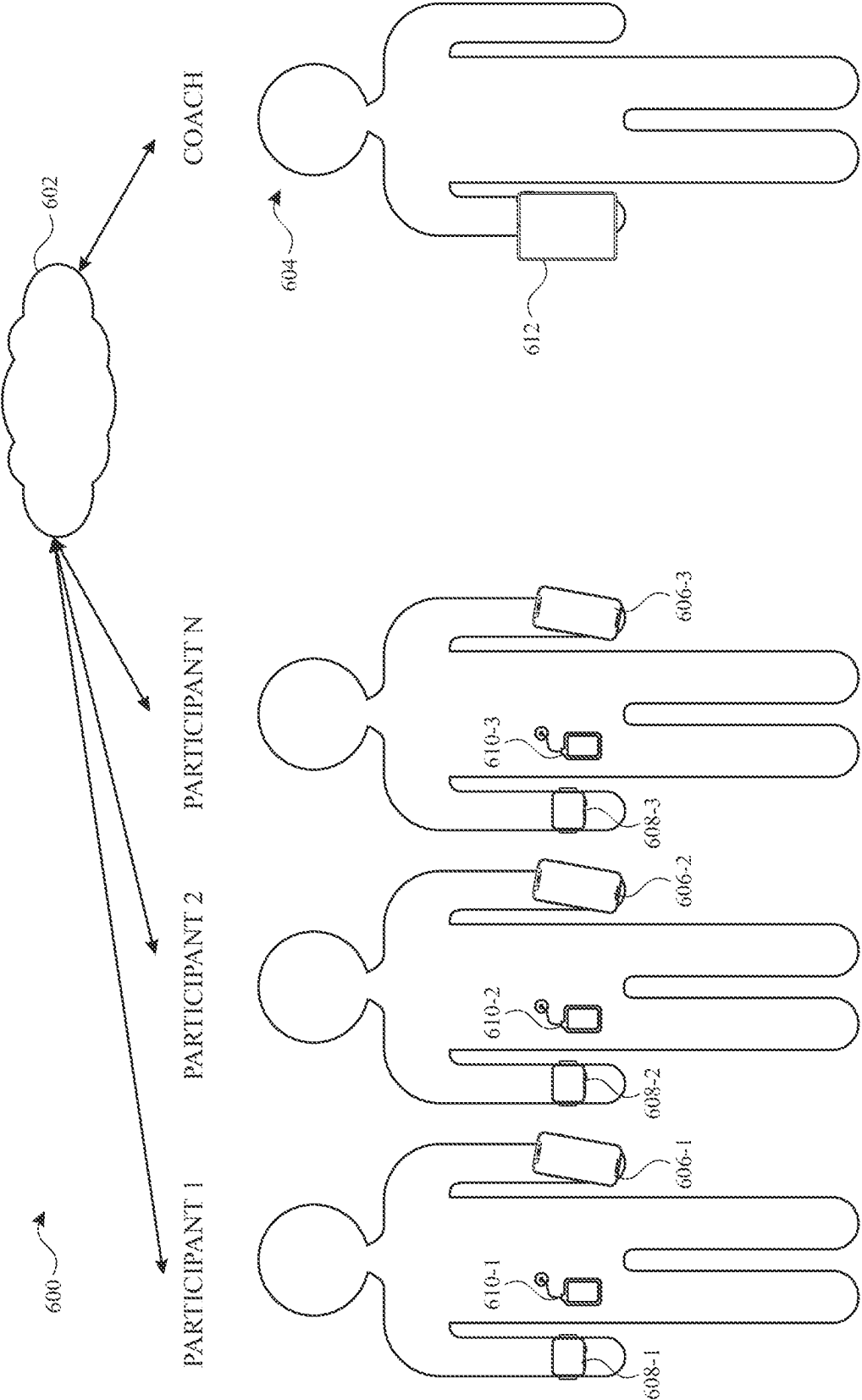


FIG. 6

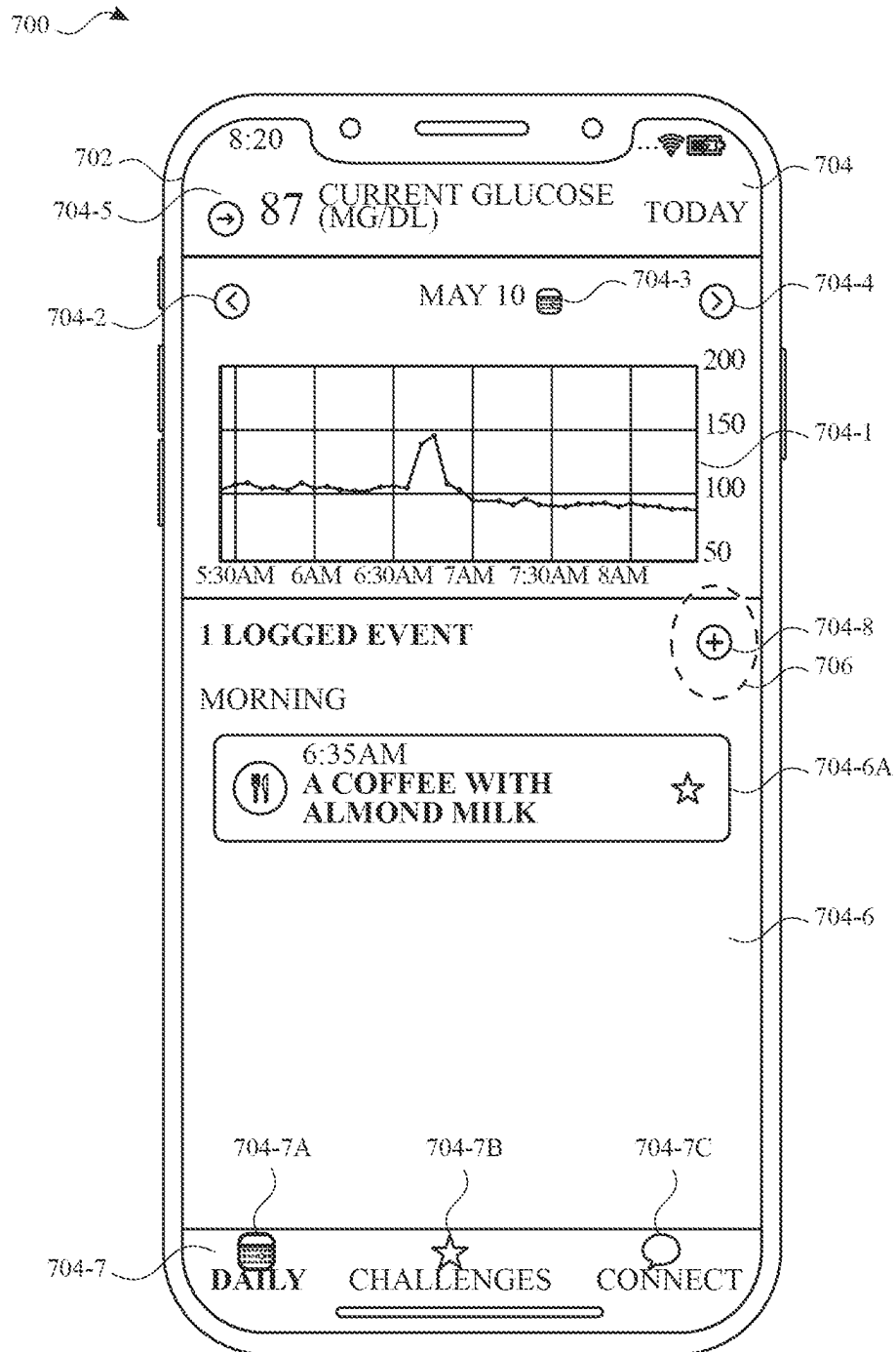


FIG. 7A

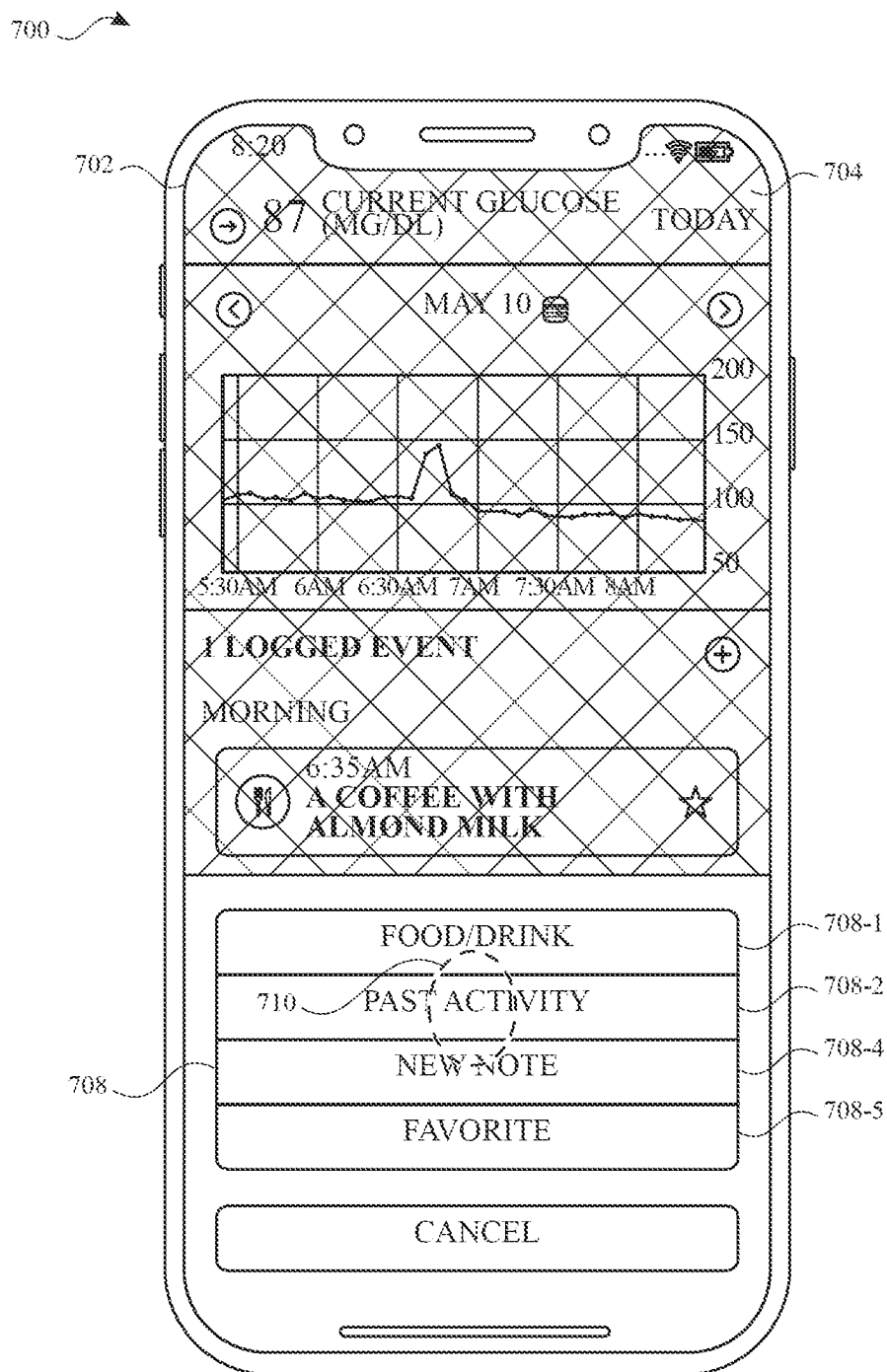


FIG. 7B

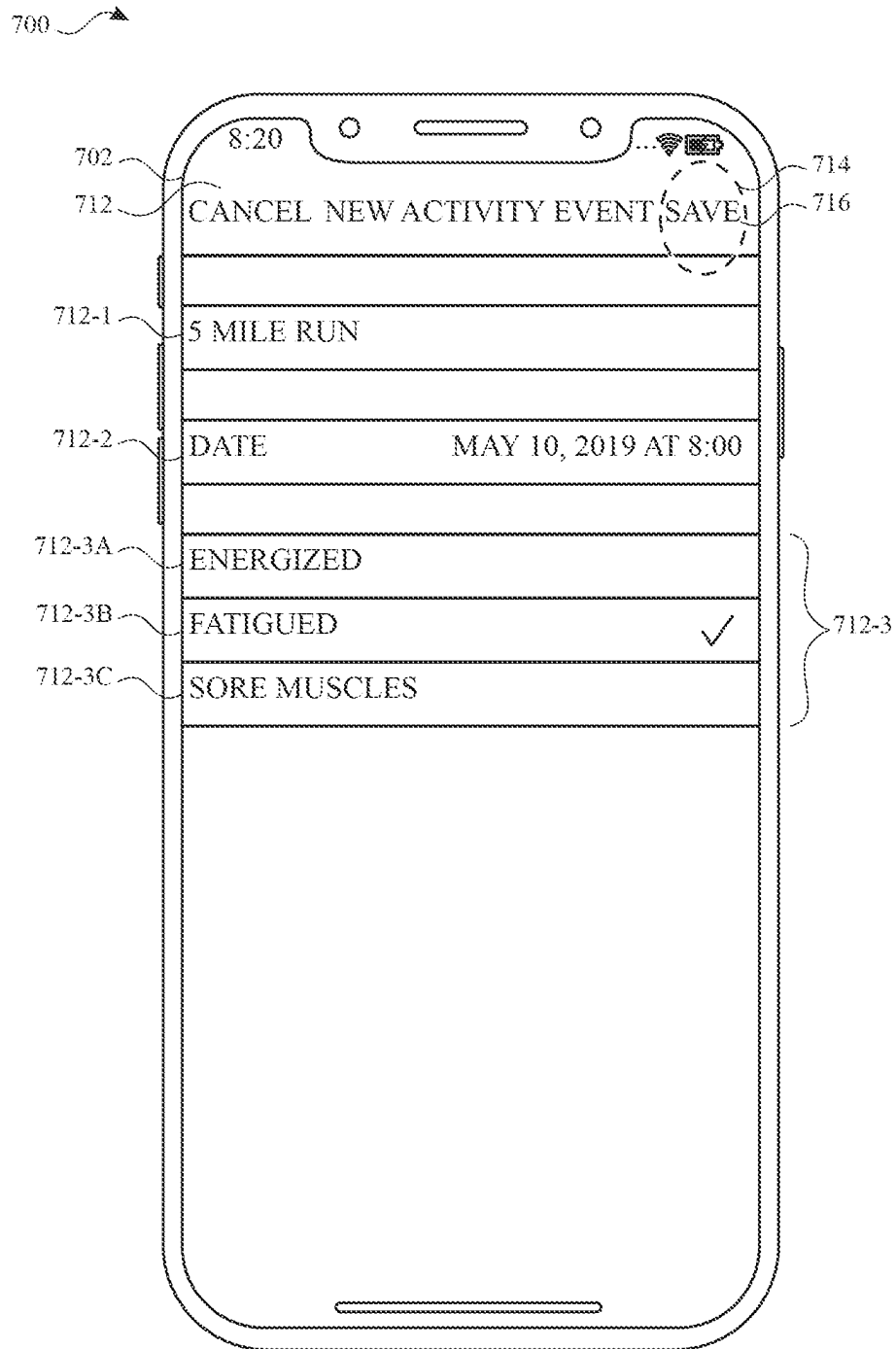


FIG. 7C

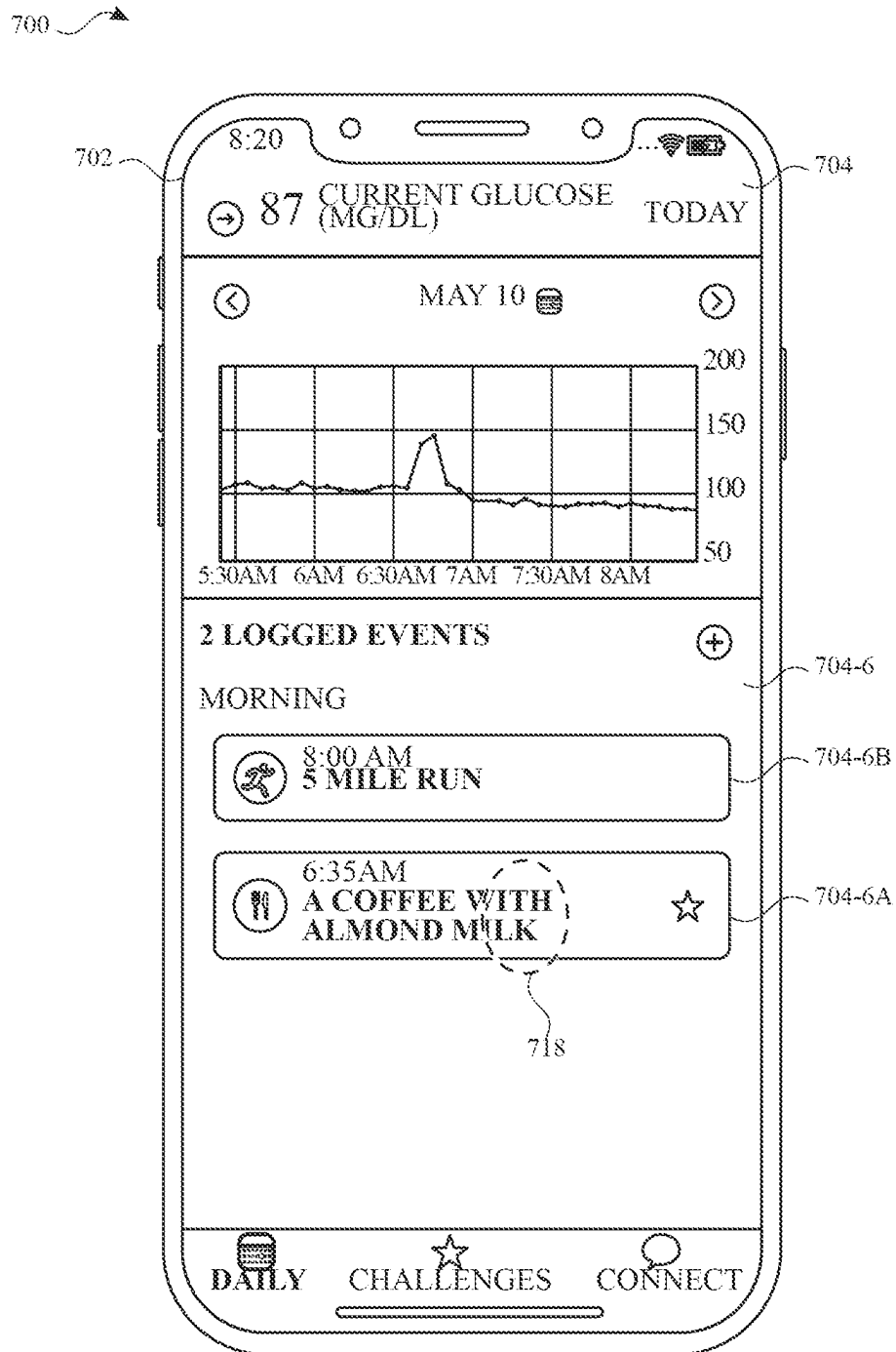


FIG. 7D

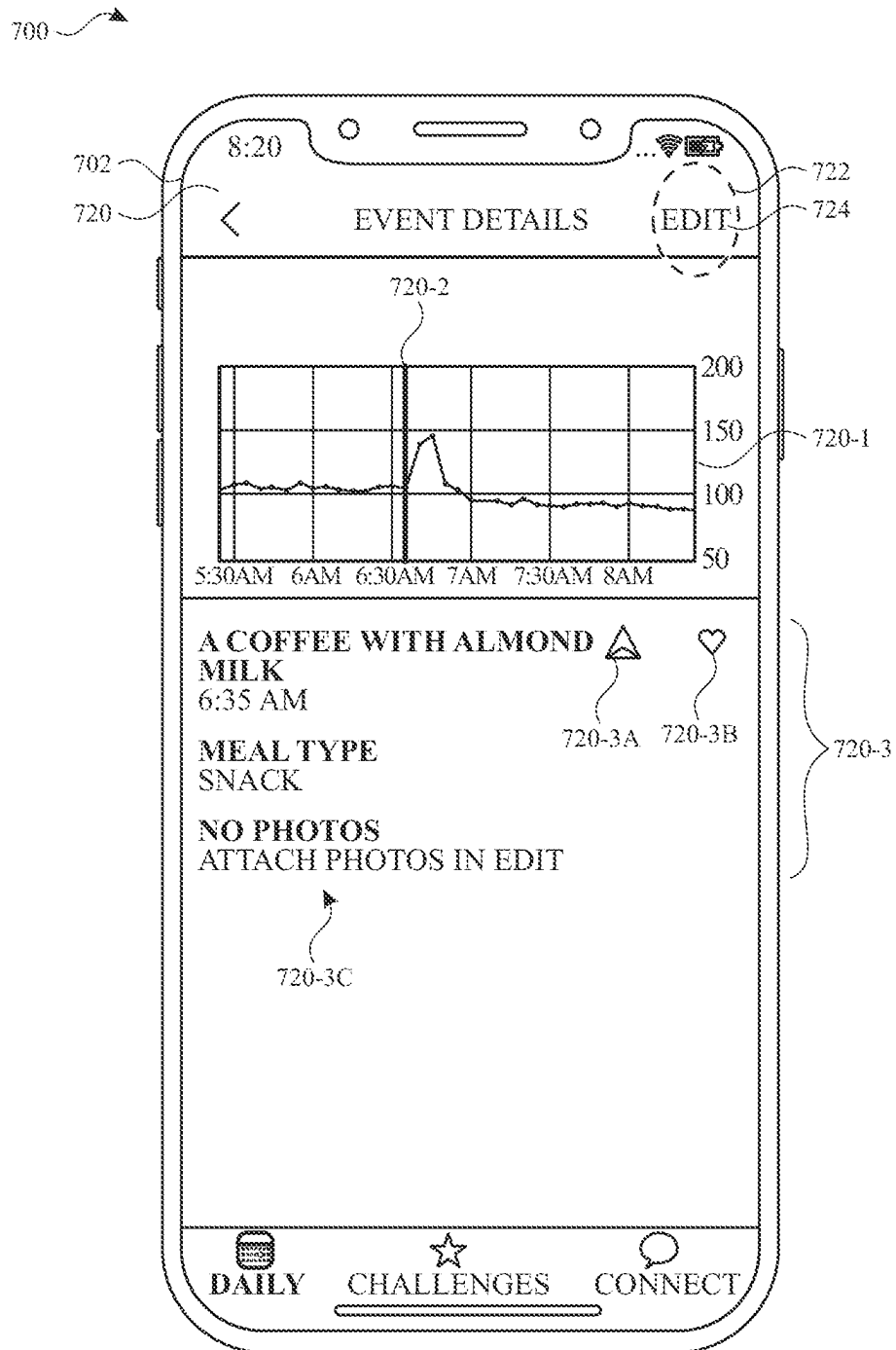


FIG. 7E

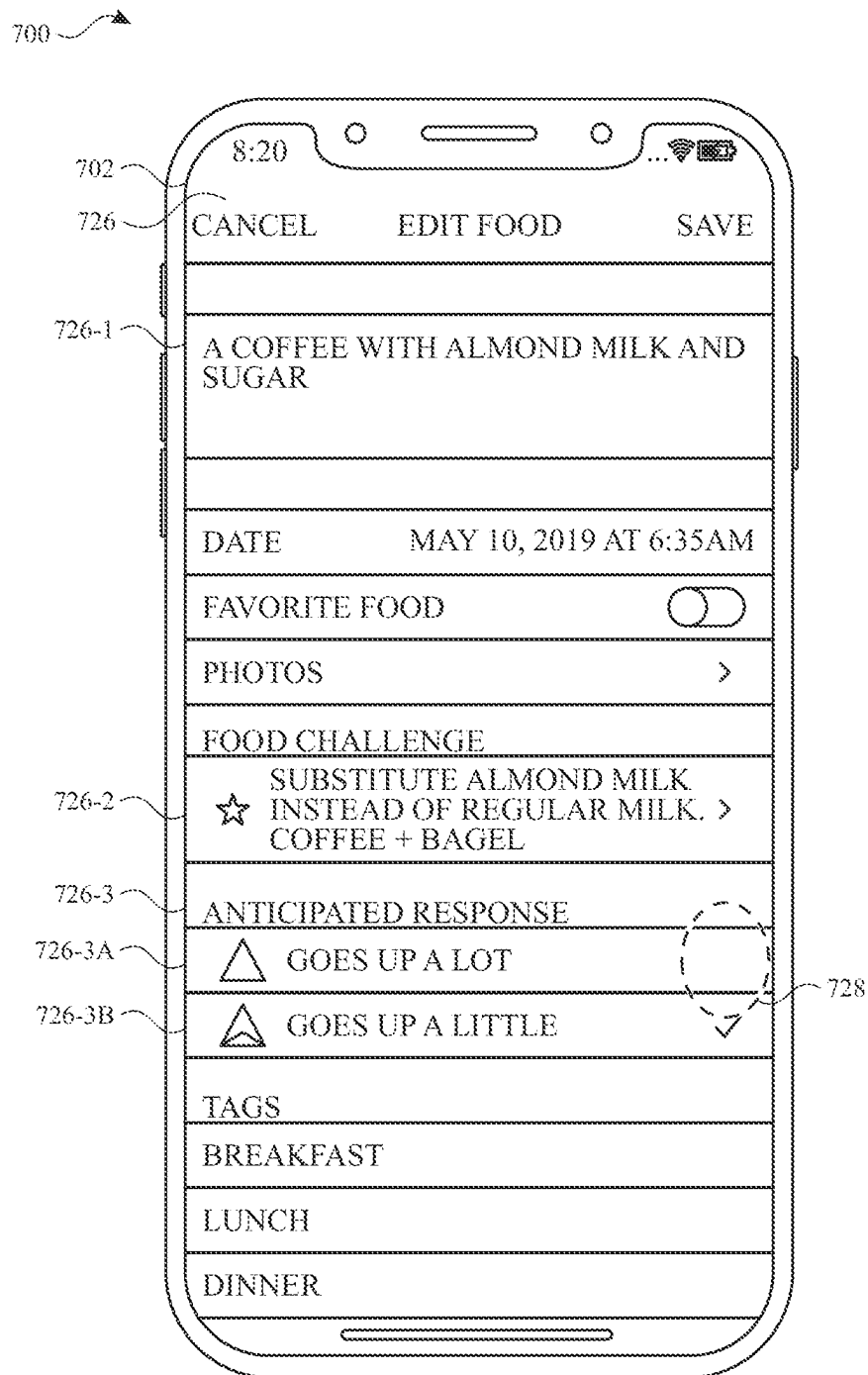


FIG. 7F

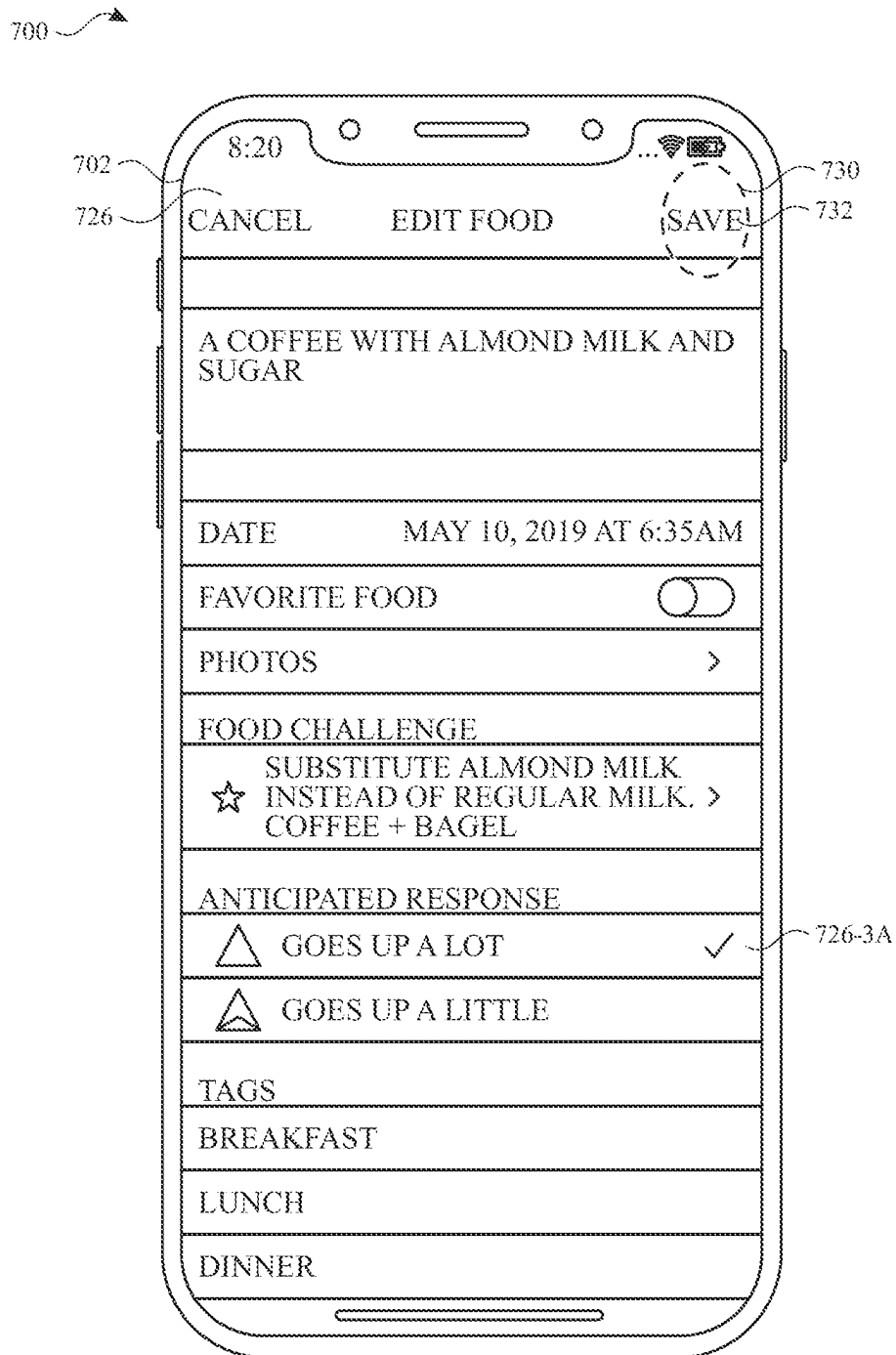


FIG. 7G

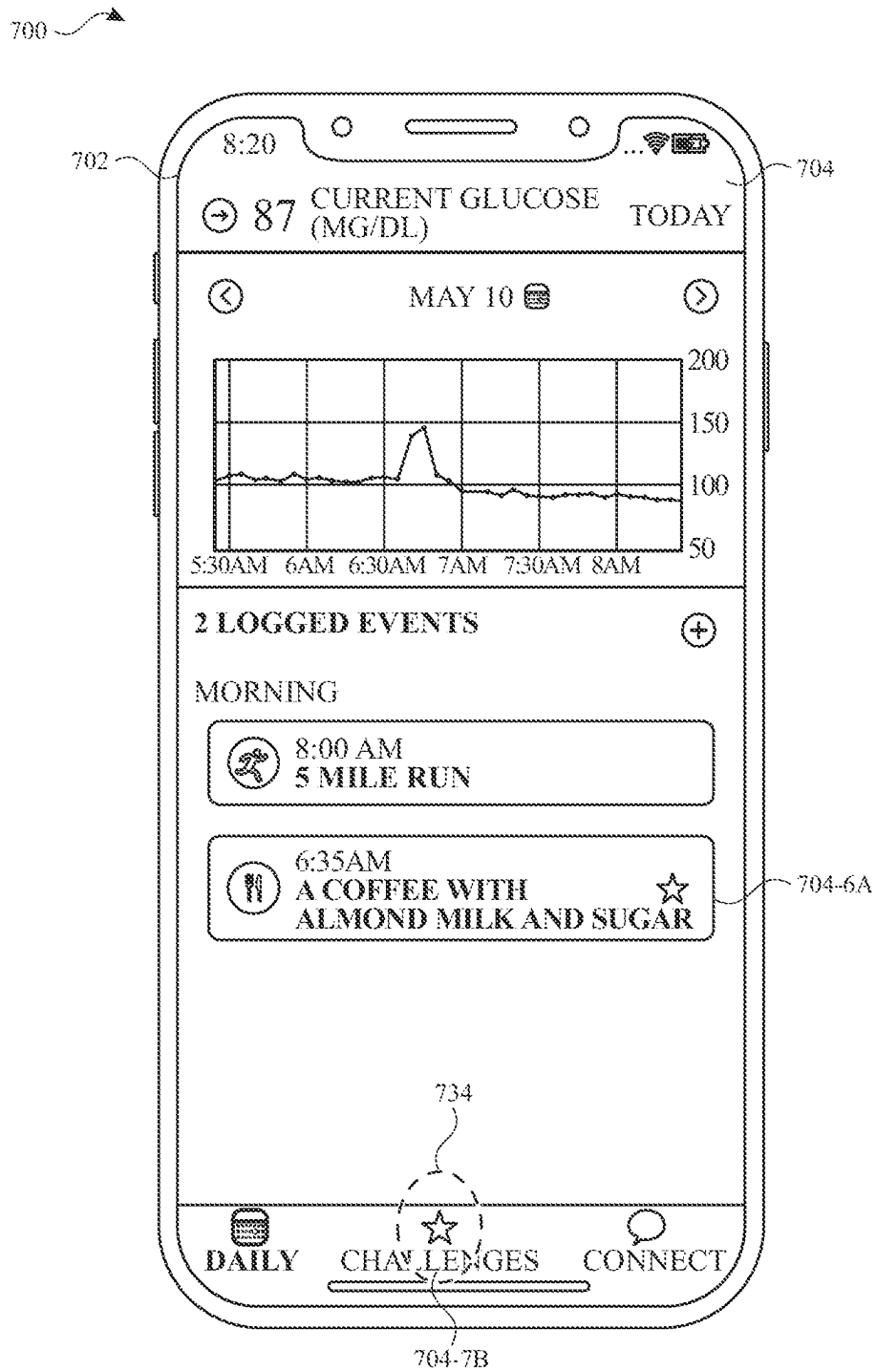


FIG. 7H

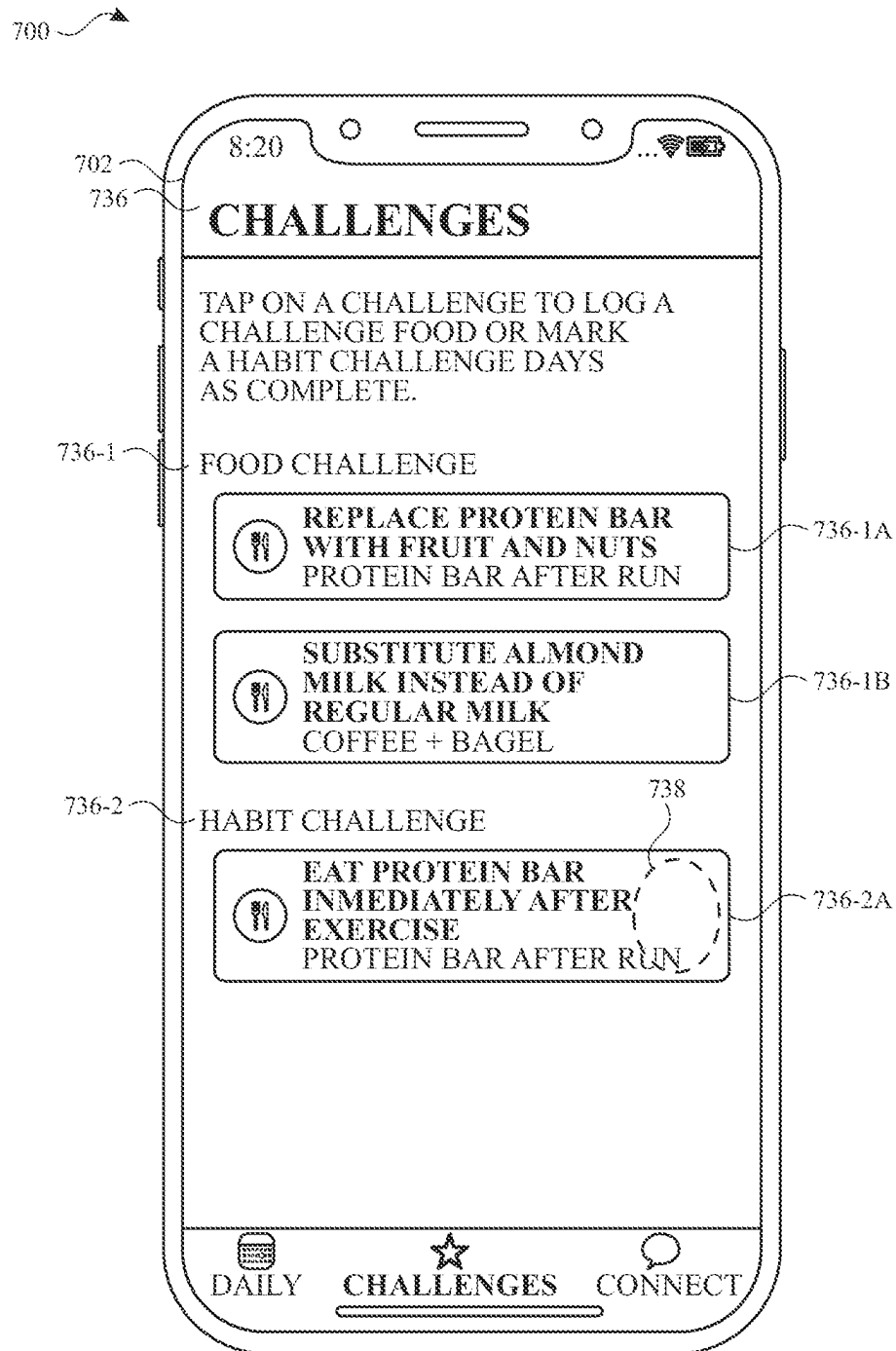


FIG. 7I

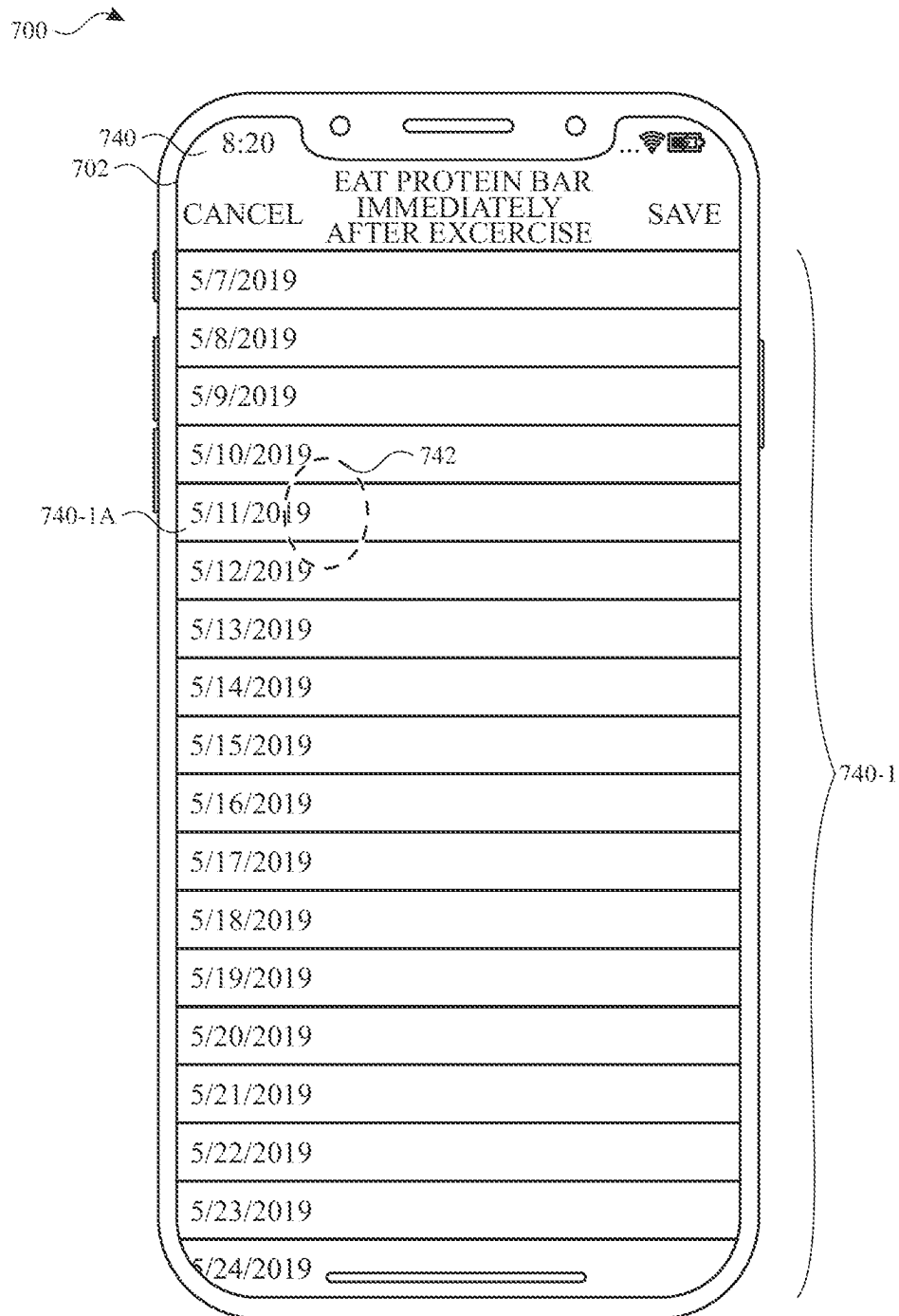


FIG. 7J

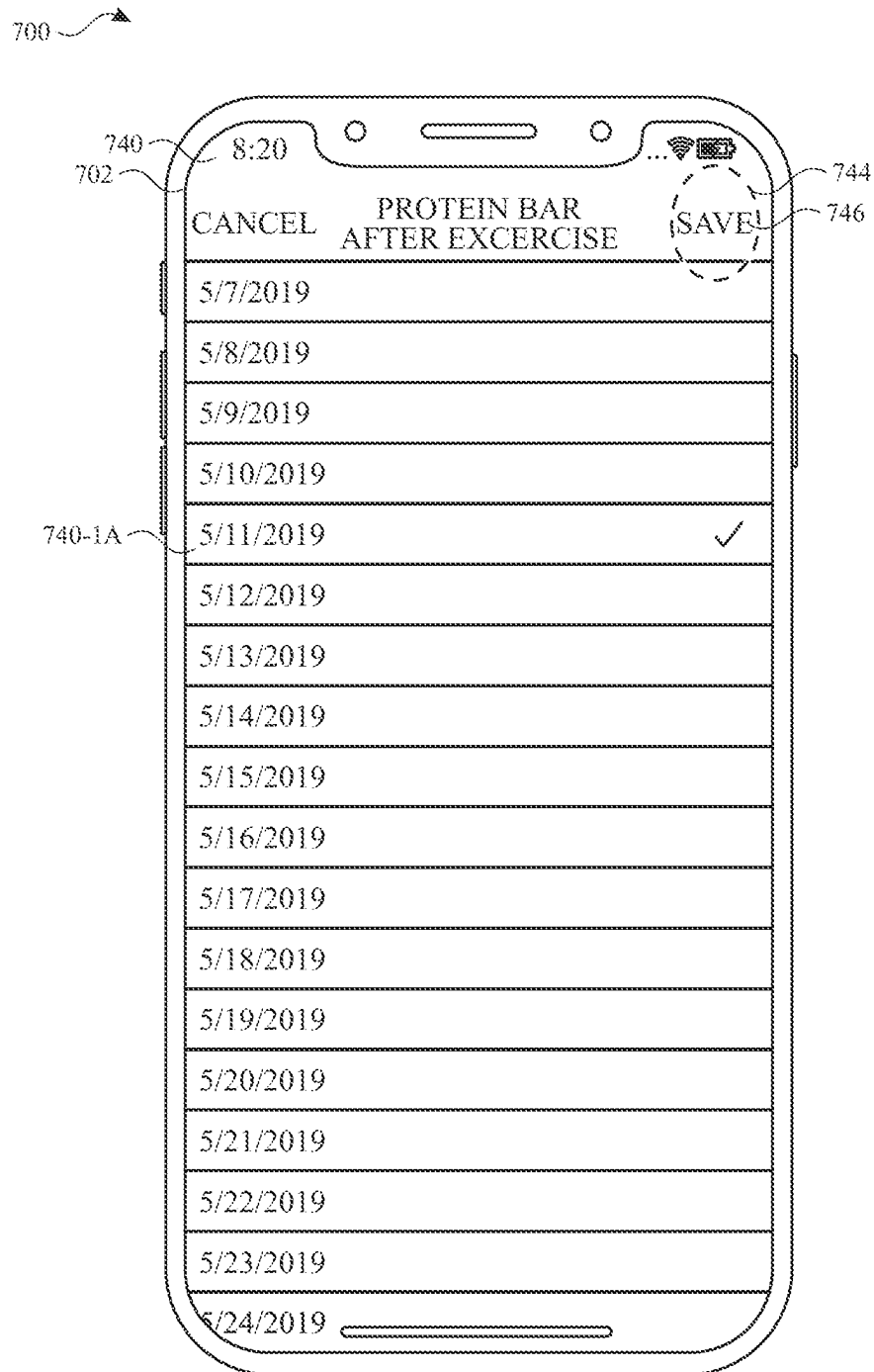


FIG. 7K

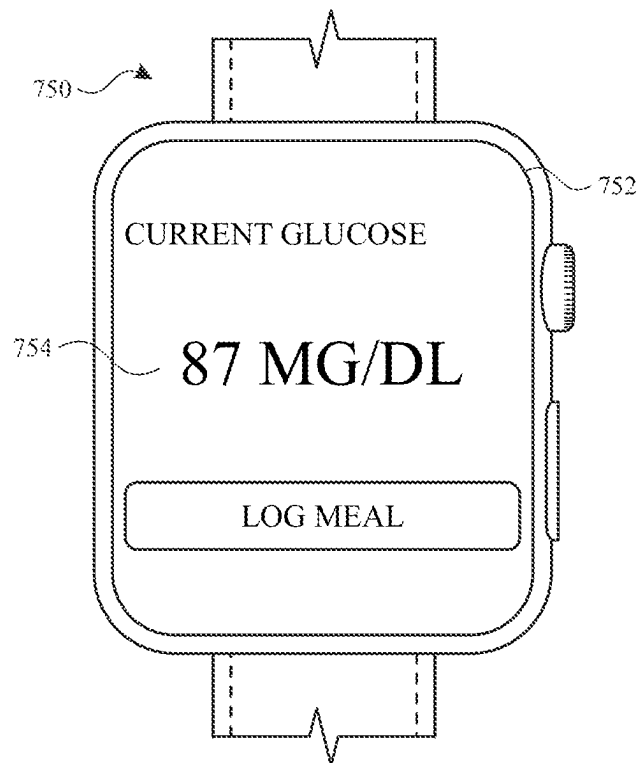


FIG. 7L

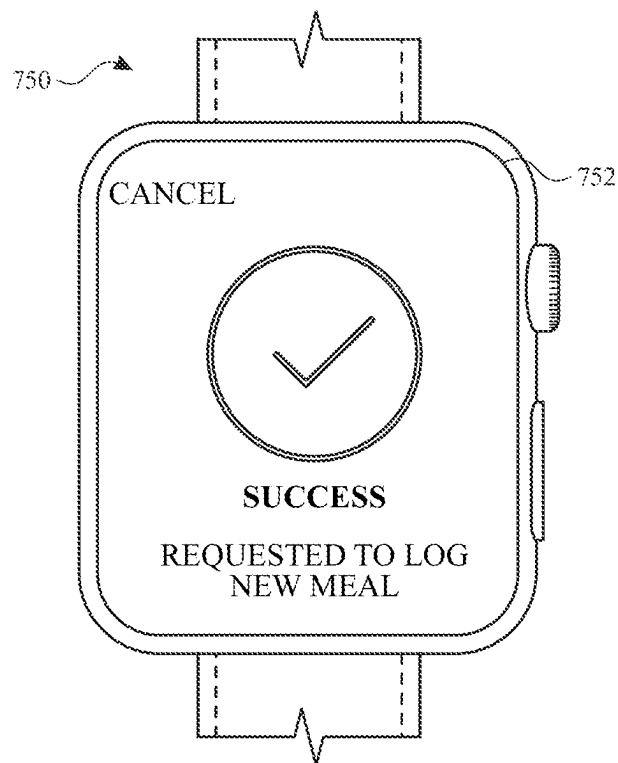
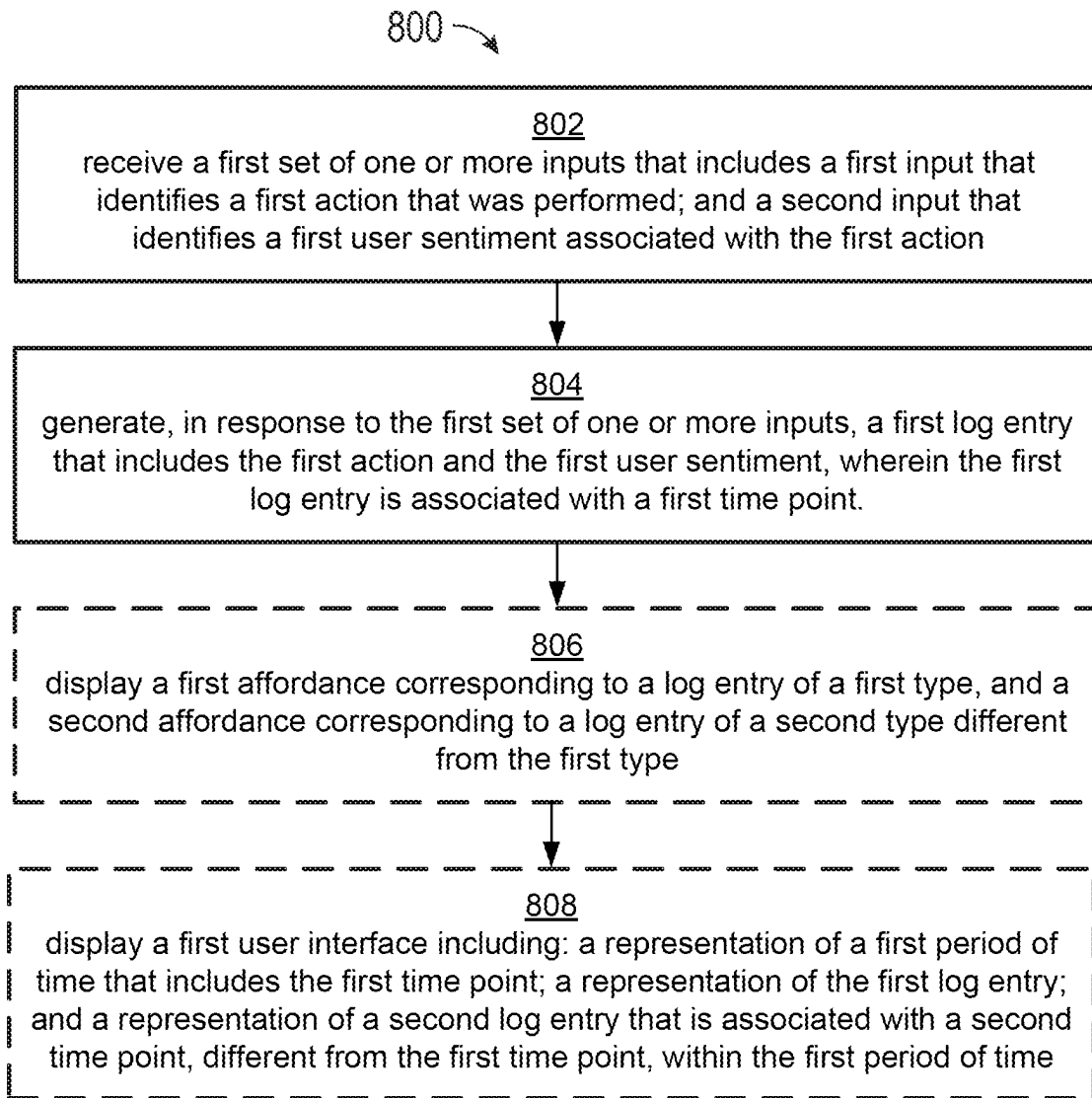


FIG. 7M

*FIG. 8*

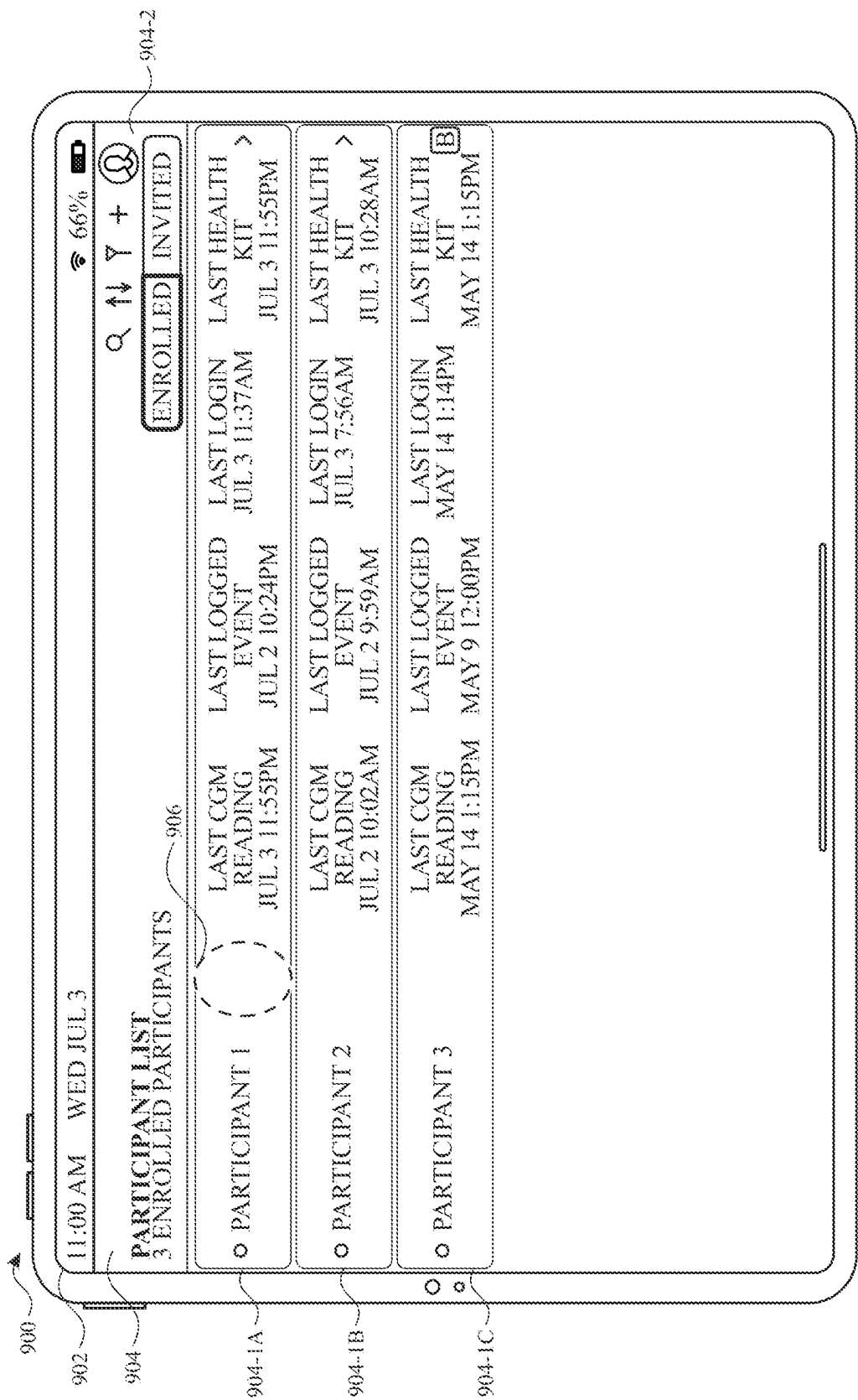


FIG. 9A

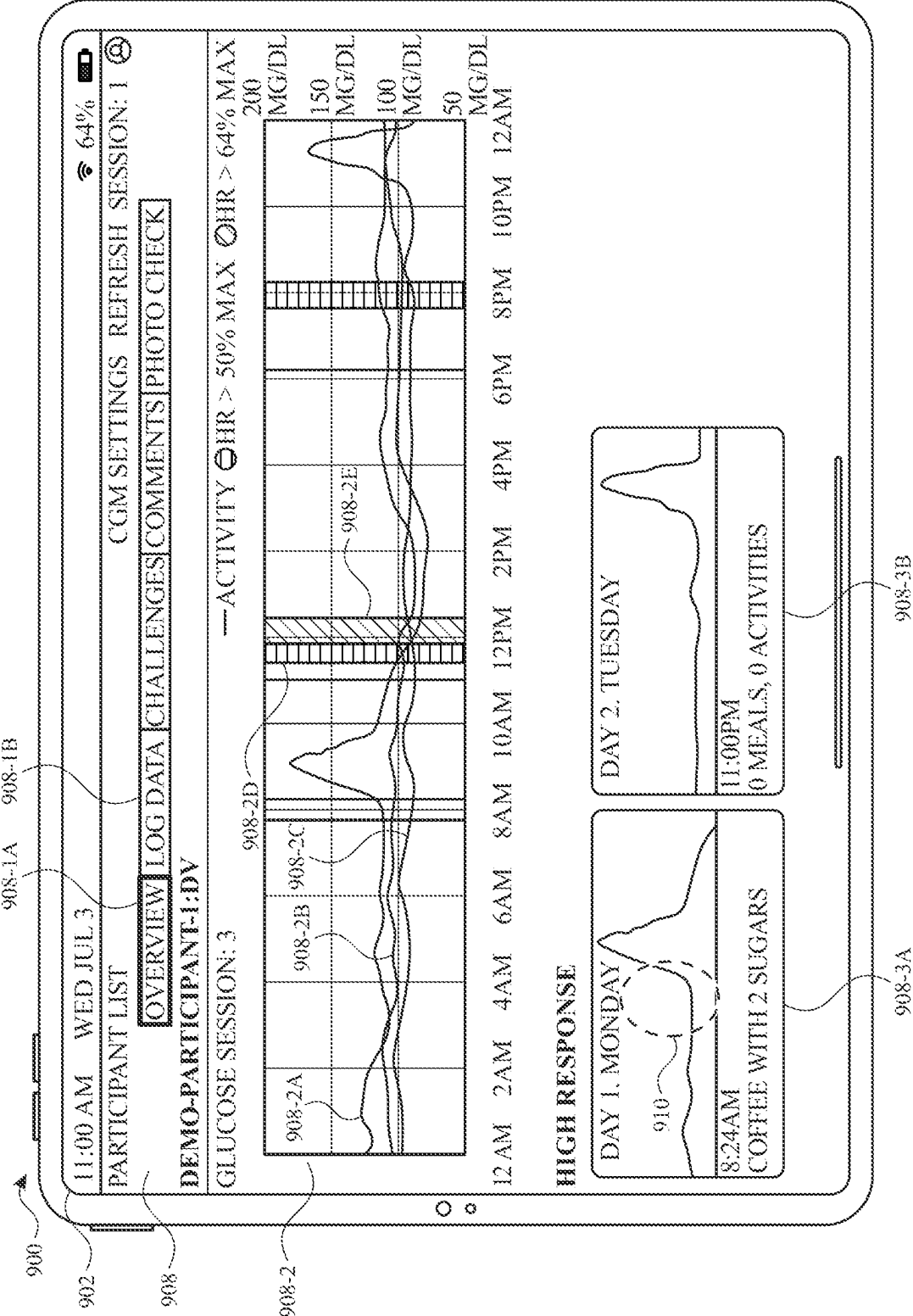


FIG. 9B

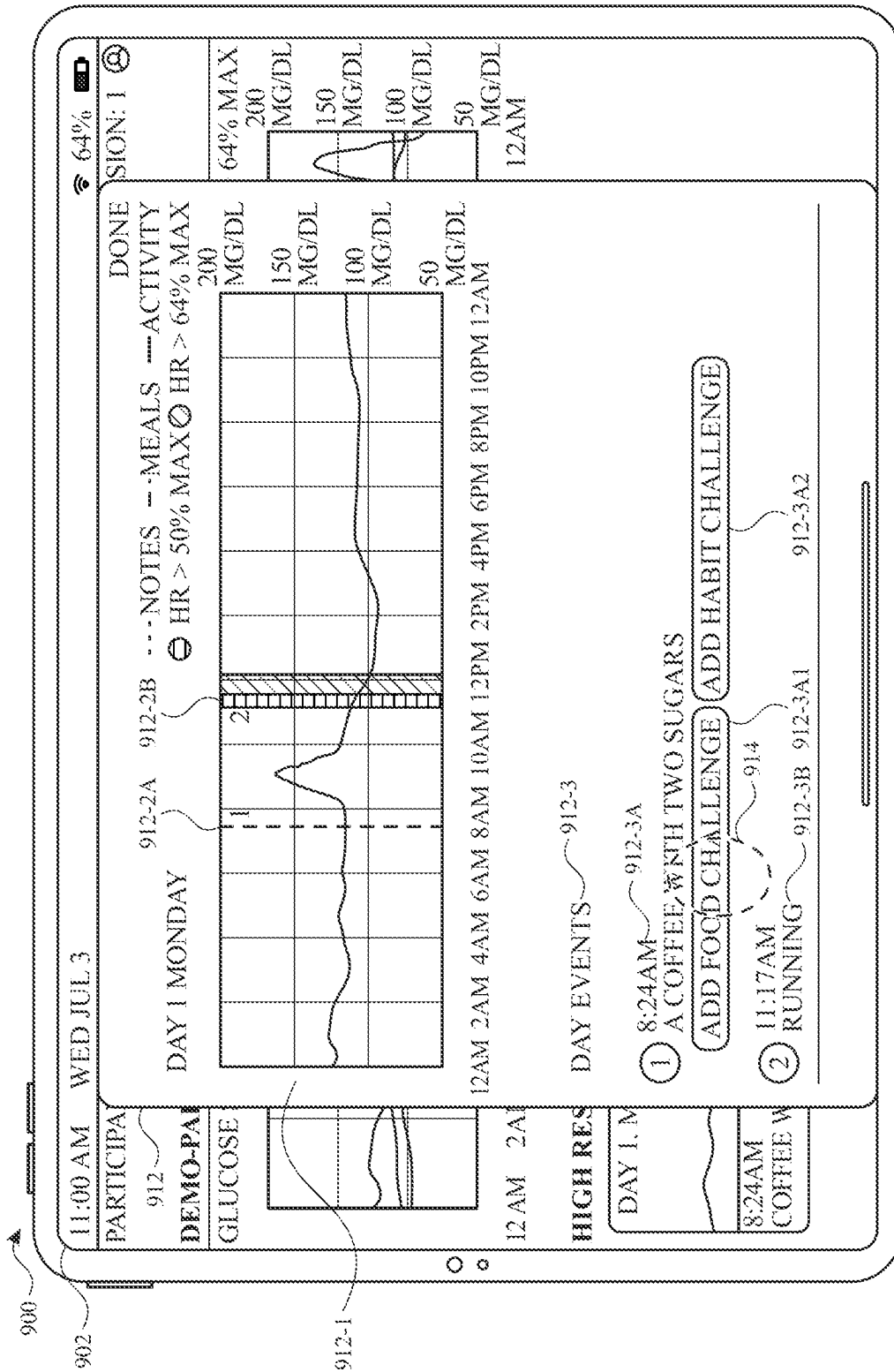


FIG. 9C

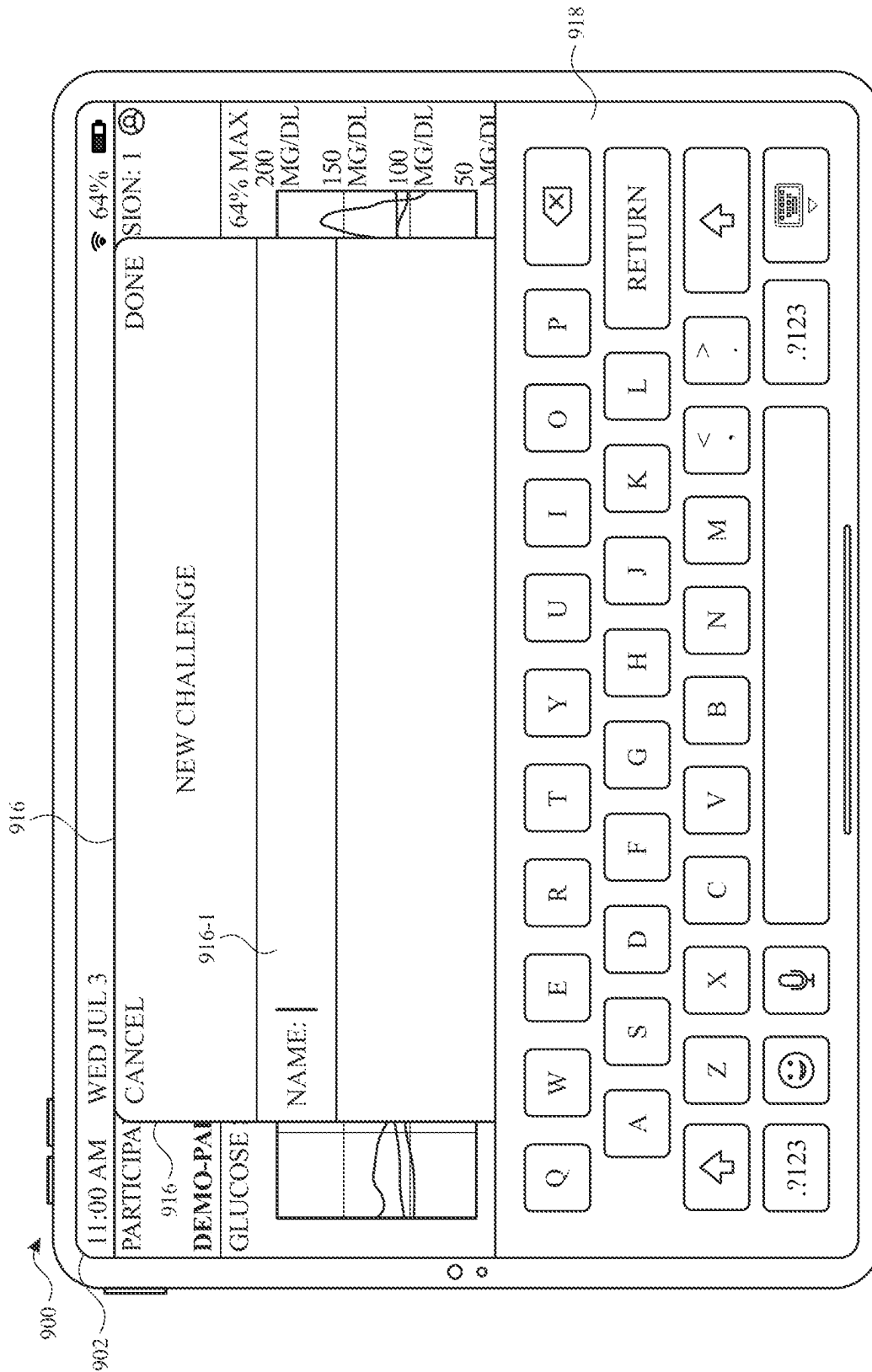


FIG. 9D

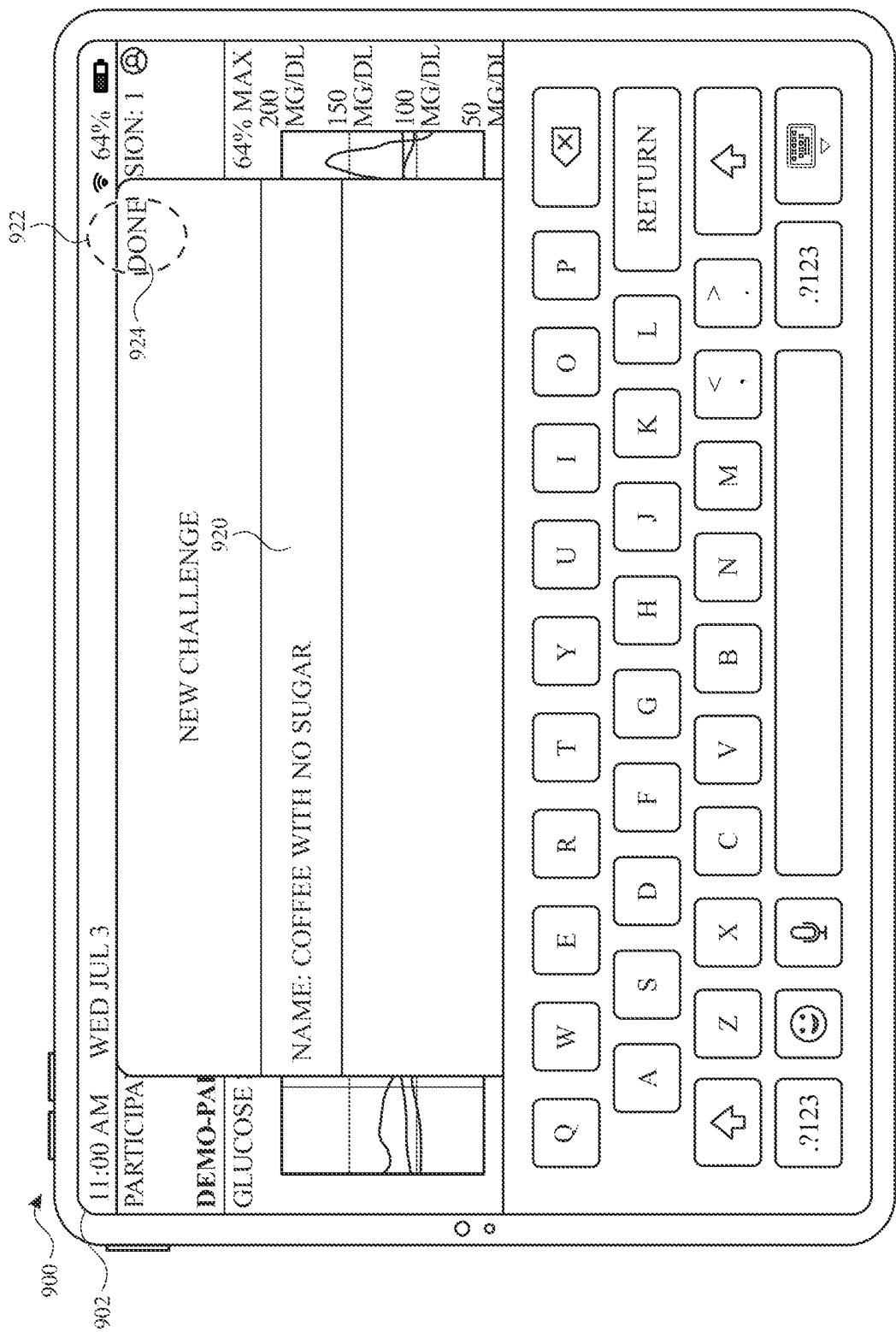


FIG. 9E

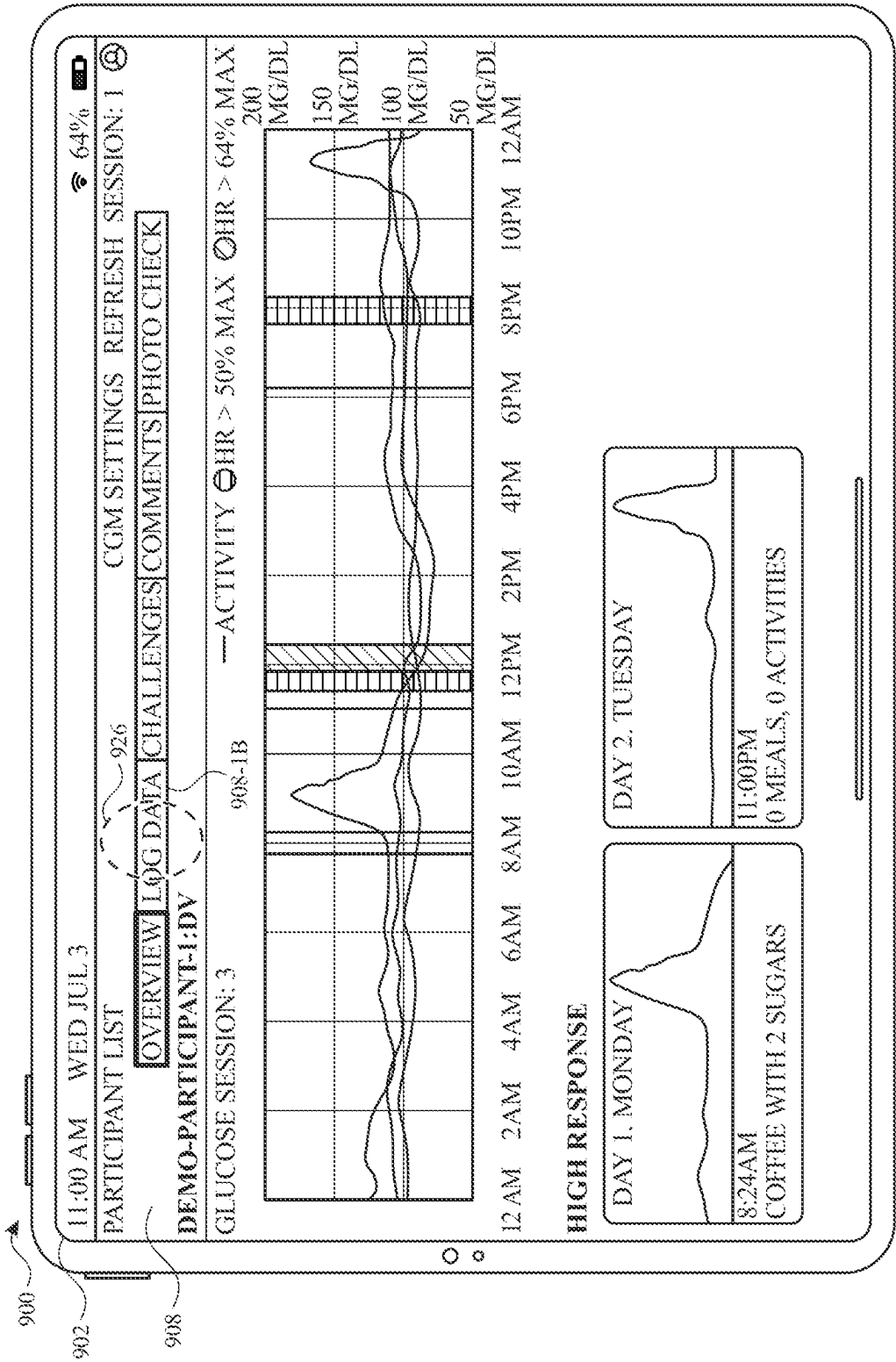


FIG. 9F

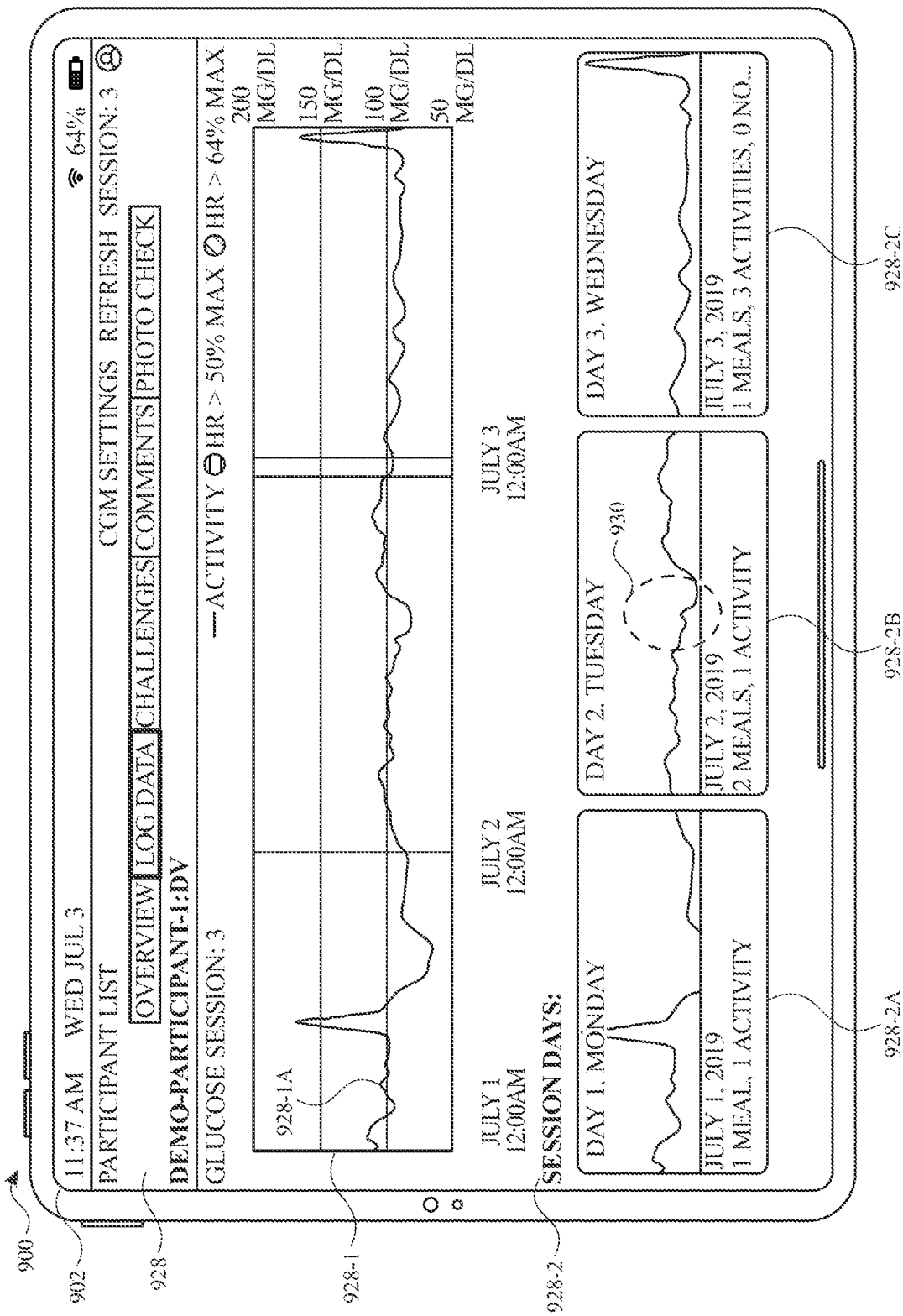


FIG. 9G

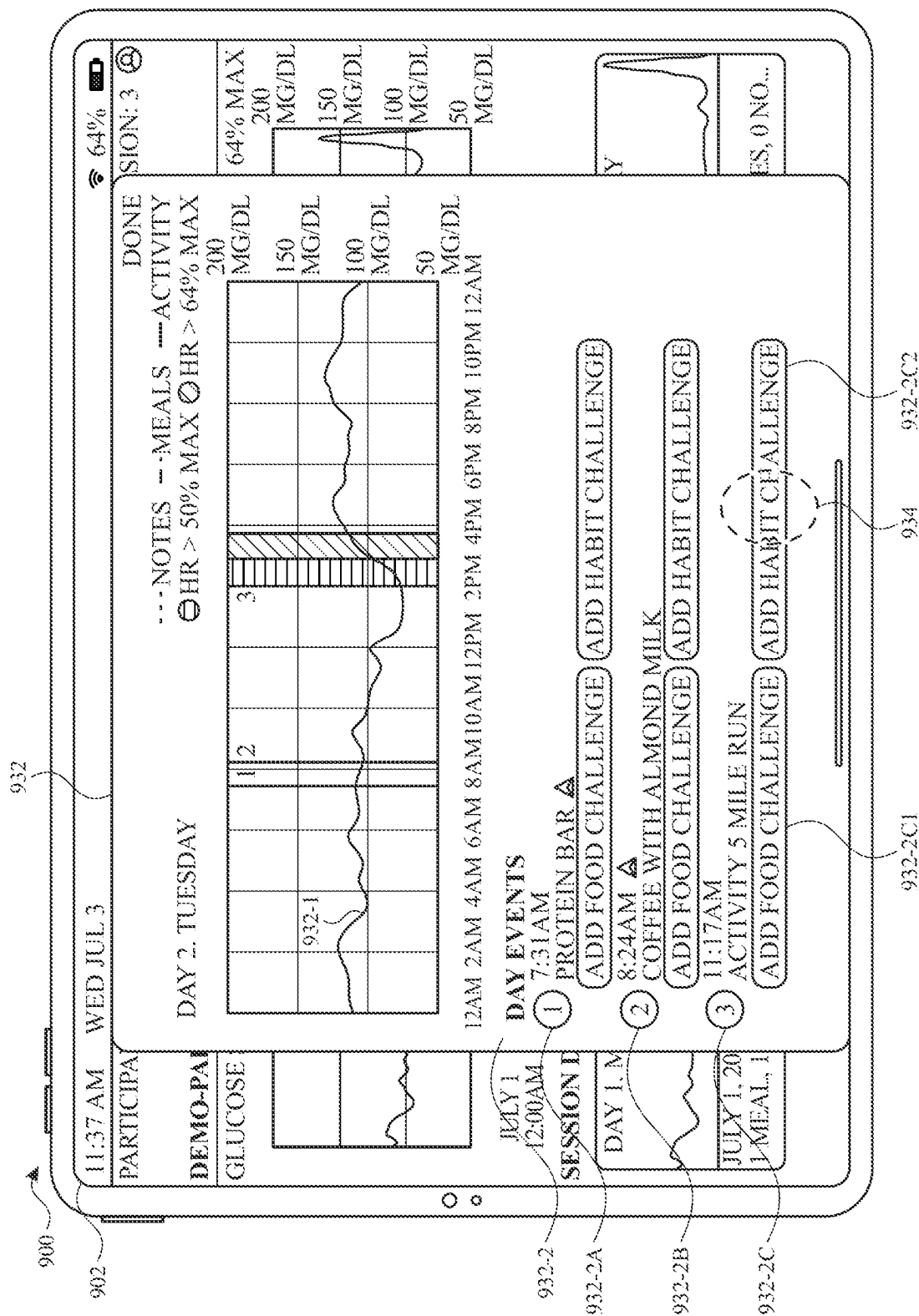


FIG. 9H

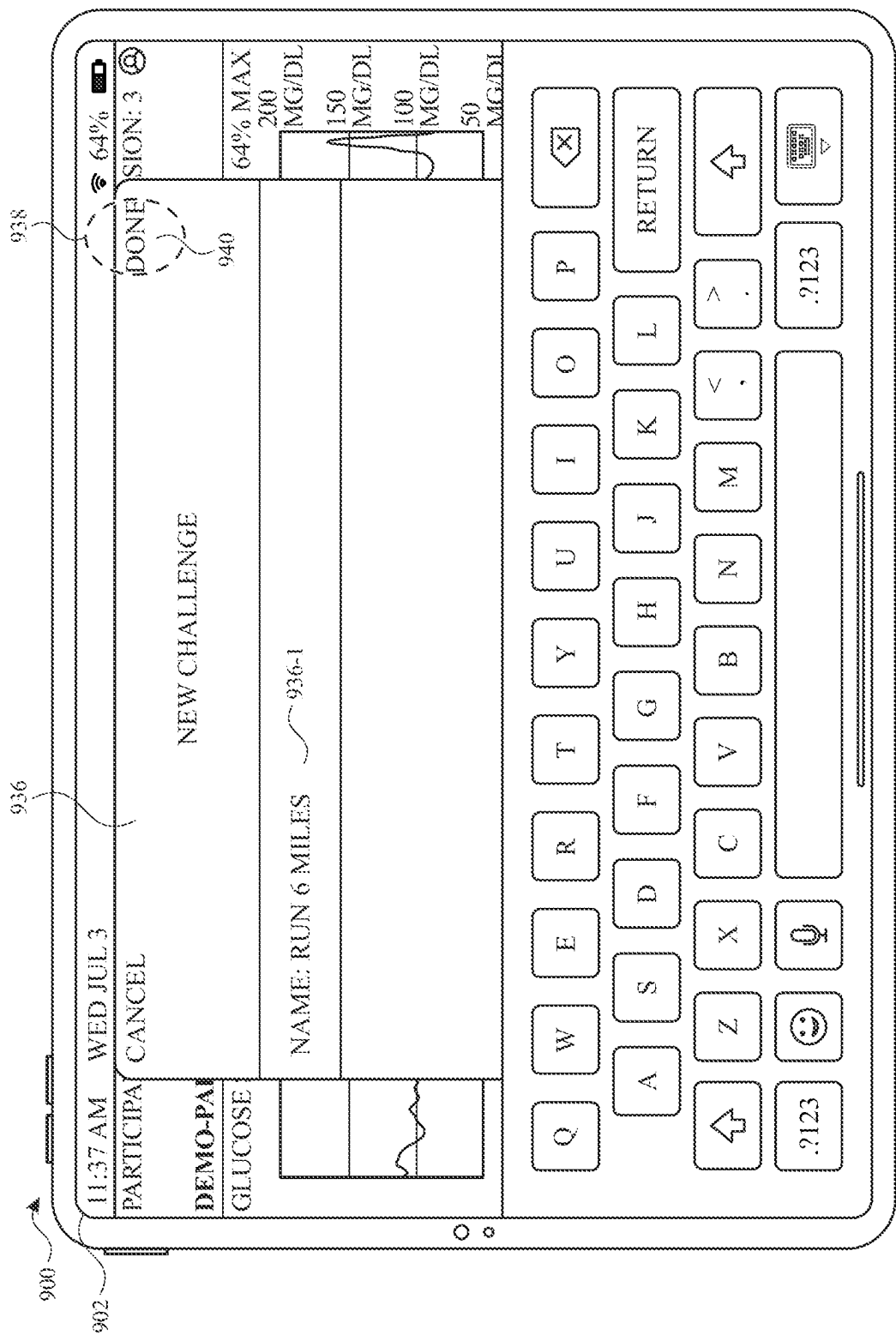


FIG. 91

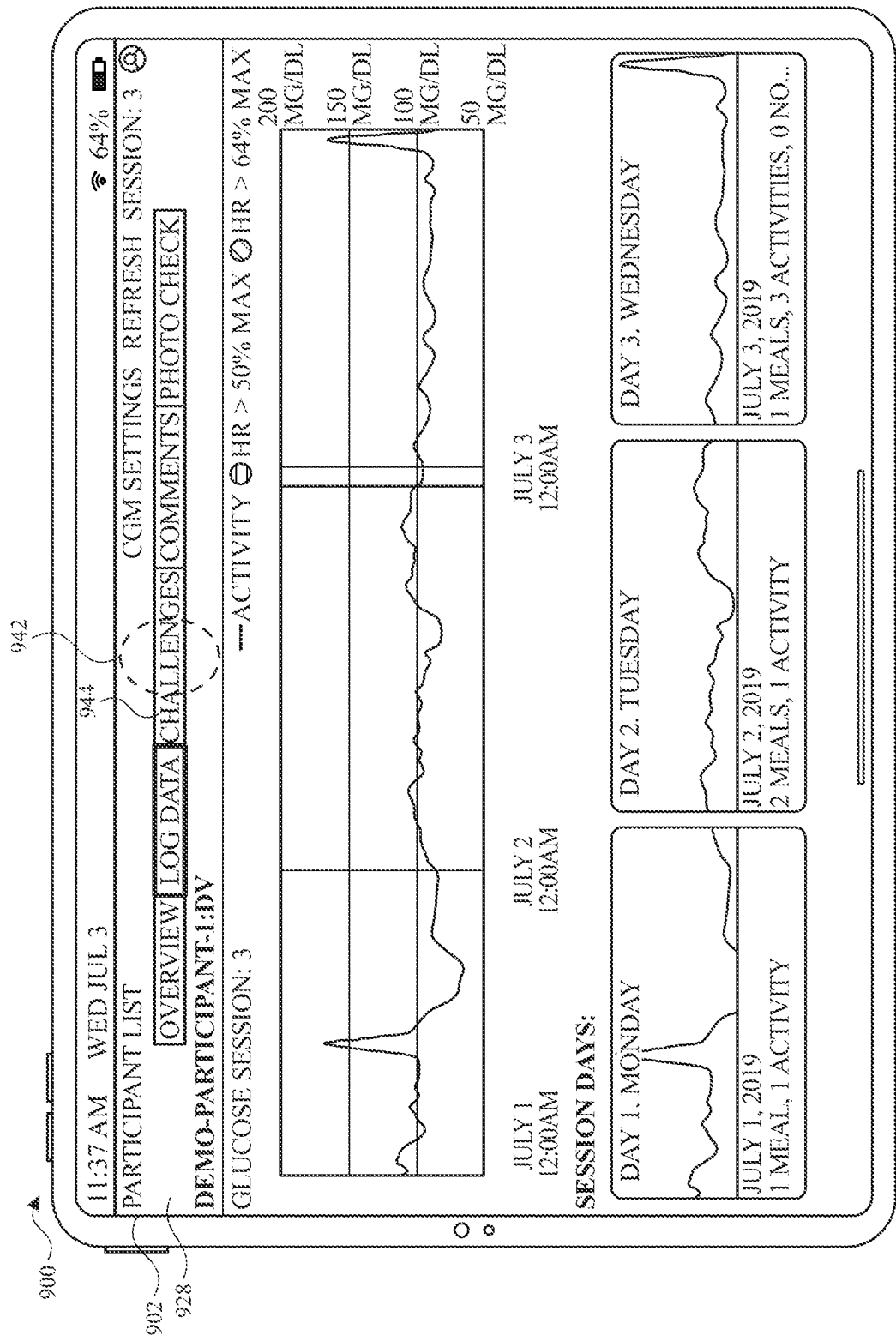


FIG. 9J

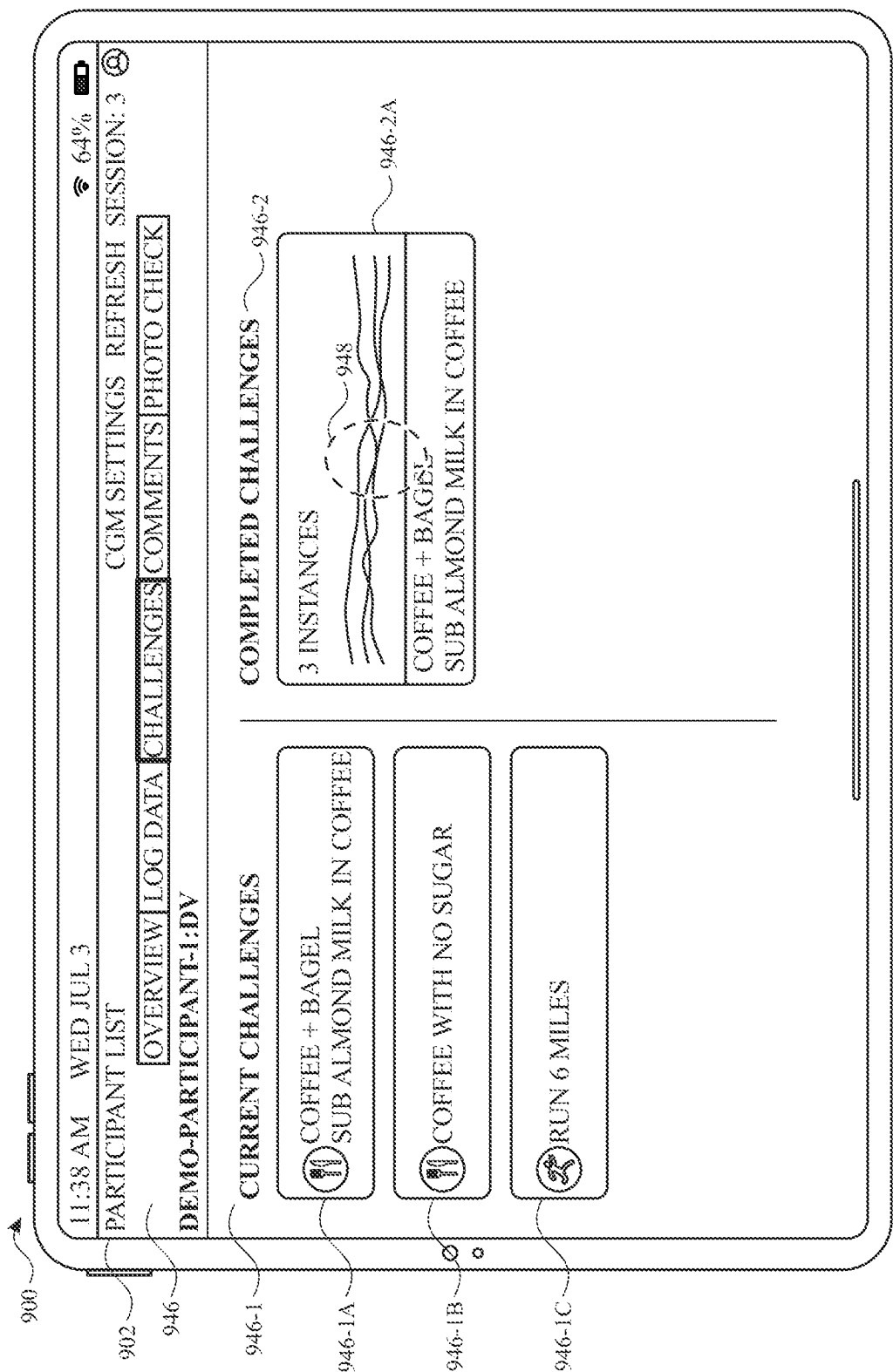


FIG. 9K

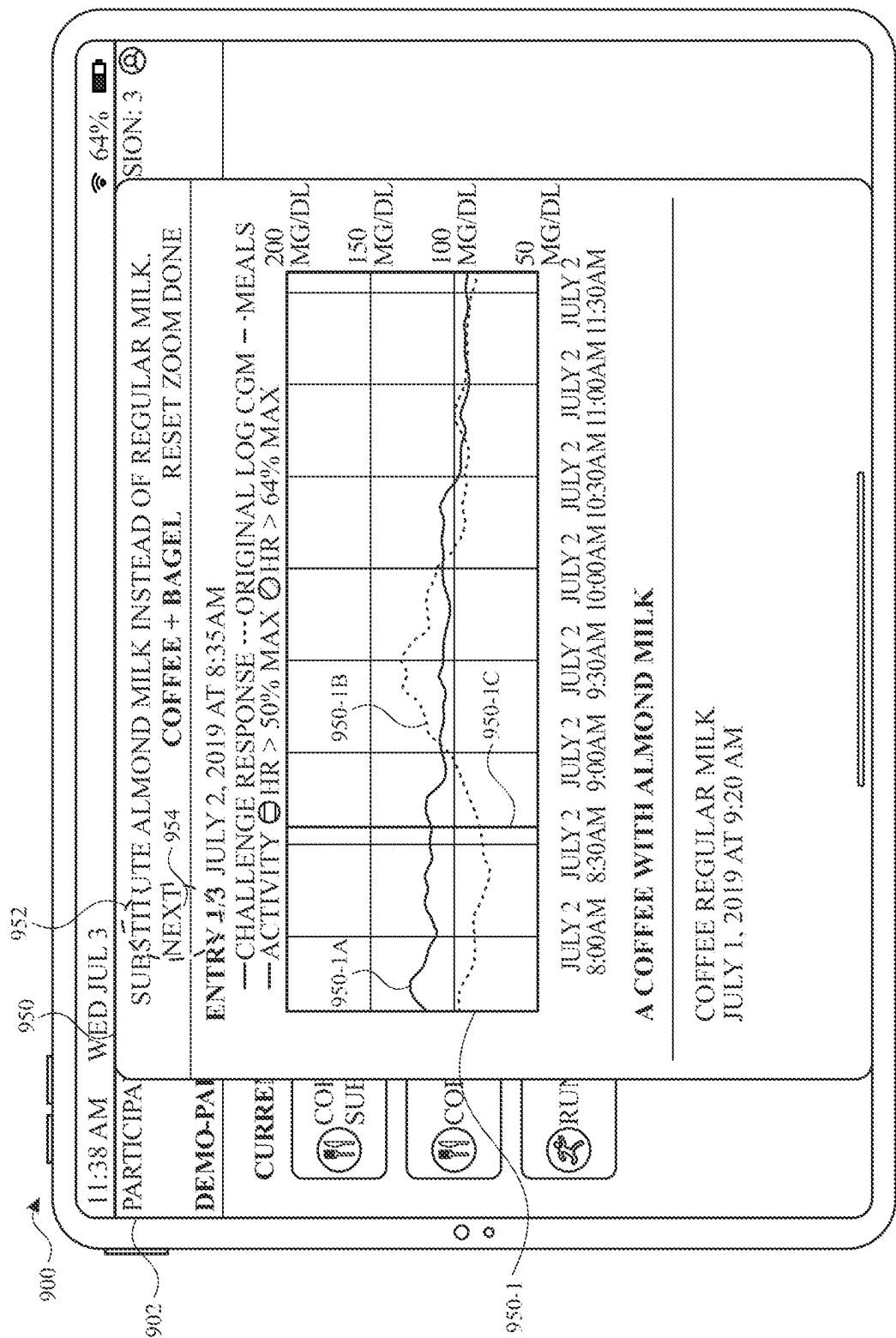


FIG. 9L

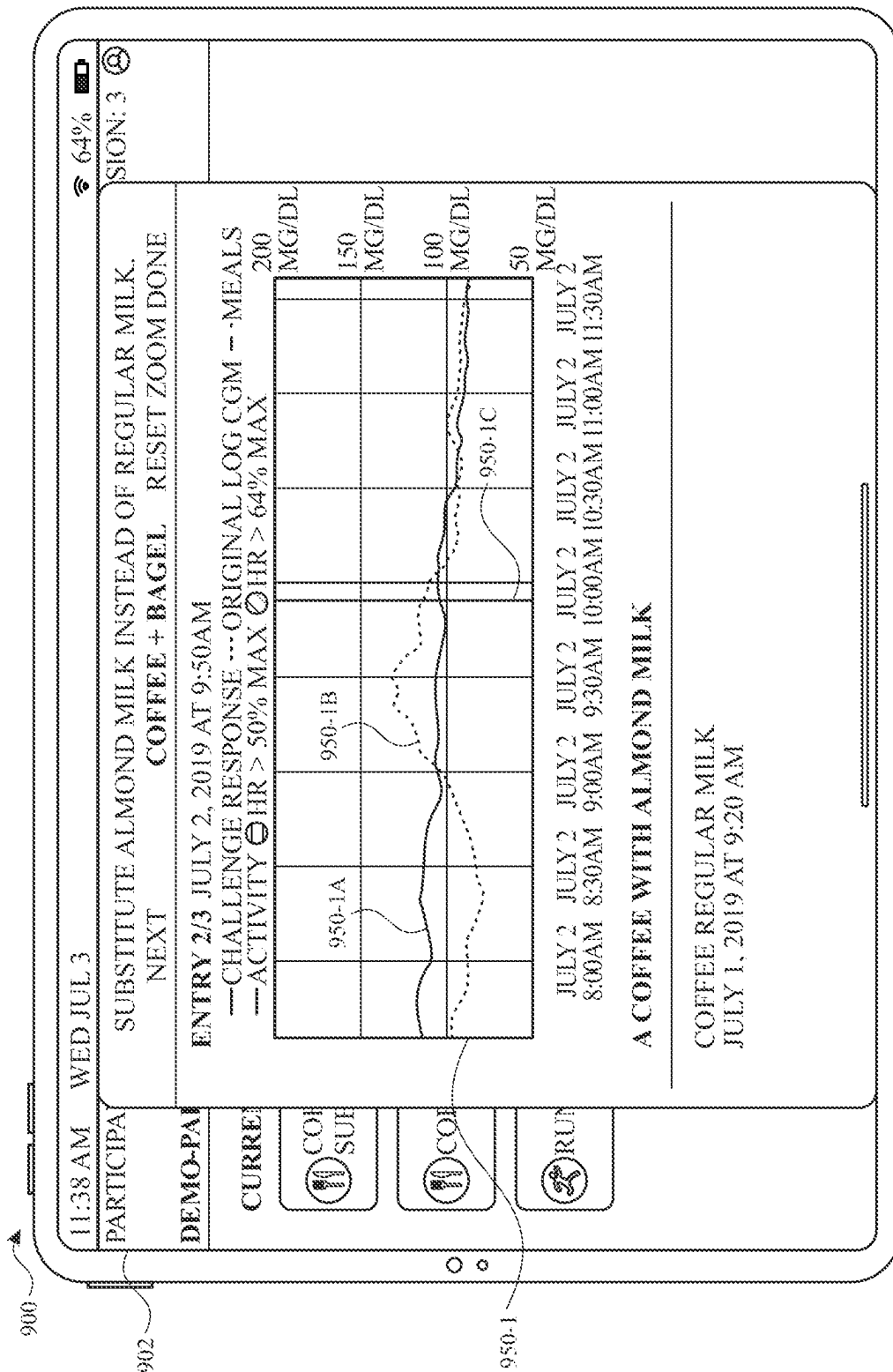
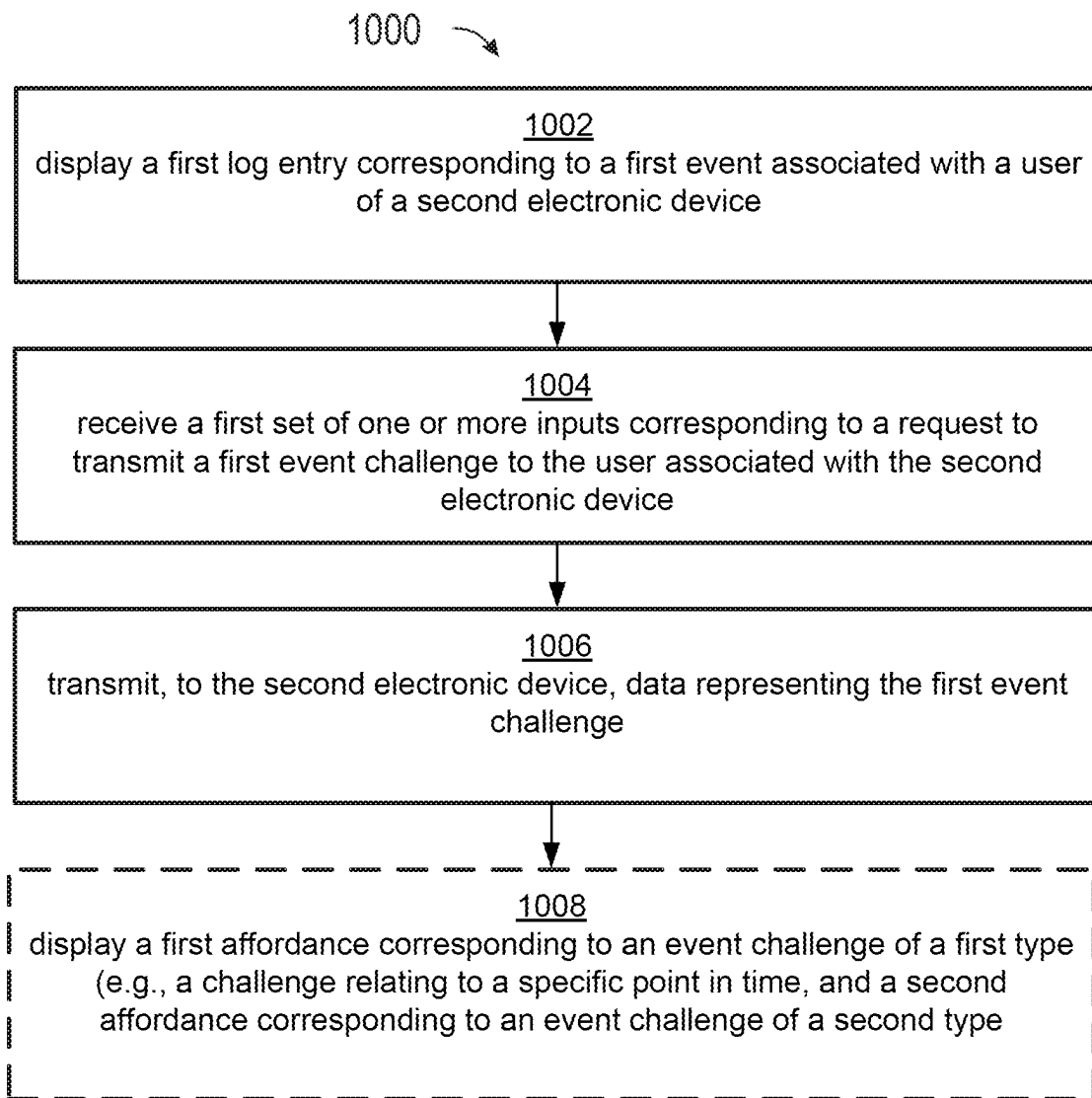


FIG. 9M

*FIG. 10*

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**HEALTH EVENT LOGGING AND
COACHING USER INTERFACES****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation of U.S. Nonprovisional application Ser. No. 16/851,451 titled “HEALTH EVENT LOGGING AND COACHING USER INTERFACES”, filed Apr. 17, 2020, which claims priority to U.S. Provisional Application No. 62/875,448 titled “HEALTH EVENT LOGGING AND COACHING USER INTERFACES”, filed Jul. 17, 2019, the content of which is hereby incorporated by reference in its entirety.

FIELD

The present disclosure relates generally to computer user interfaces, and more specifically to techniques and user interfaces for logging health-related events and for health-related coaching.

BACKGROUND

Users perform certain actions, or forgo from performing certain actions, that can have an impact on their health (e.g., on one or more physiological parameters relevant to their health). Electronic devices can collect relevant physiological data and also log data that can be relevant to understanding the health of users.

BRIEF SUMMARY

Some techniques for logging health-related events and for health-related coaching using electronic devices, however, are generally cumbersome and inefficient. For example, some existing techniques use a complex and time-consuming user interface, which may include multiple key presses or keystrokes. Existing techniques require more time than necessary, wasting user time and device energy. This latter consideration is particularly important in battery-operated devices.

Accordingly, the present technique provides electronic devices with faster, more efficient methods and interfaces for logging health-related events and for health-related coaching. Such methods and interfaces optionally complement or replace other methods for logging health-related events and for health-related coaching. Such methods and interfaces reduce the cognitive burden on a user and produce a more efficient human-machine interface. For battery-operated computing devices, such methods and interfaces conserve power and increase the time between battery charges. Such methods and interfaces can also assist users and professionals working with users to improve their health with viewing health-related events and assist users with modifying their behavior in ways that improve their health.

In accordance with some embodiments, a method is described. The method is performed at a first electronic device having a display device and one or more input devices. The method comprises: receiving, via a first input device of the one or more input devices, a first set of one or more inputs including: a first input that identifies a first action that was performed; and a second input that identifies a first user sentiment associated with the first action; in response to the first set of one or more inputs, generating a

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first log entry that includes the first action and the first user sentiment, wherein the first log entry is associated with a first time point.

In accordance with some embodiments, a non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a first electronic device having a display device and one or more input devices is described. The one or more programs include instructions for: receiving, via a first input device of the one or more input devices, a first set of one or more inputs including: a first input that identifies a first action that was performed; and a second input that identifies a first user sentiment associated with the first action; in response to the first set of one or more inputs, generating a first log entry that includes the first action and the first user sentiment, wherein the first log entry is associated with a first time point.

In accordance with some embodiments, a transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a first electronic device having a display device and one or more input devices is described. The one or more programs include instructions for: receiving, via a first input device of the one or more input devices, a first set of one or more inputs including: a first input that identifies a first action that was performed; and a second input that identifies a first user sentiment associated with the first action; in response to the first set of one or more inputs, generating a first log entry that includes the first action and the first user sentiment, wherein the first log entry is associated with a first time point.

In some accordance with some embodiments, an electronic device is described. The electronic device comprises a display device; one or more input devices; one or more processors; and memory storing one or more programs configured to be executed by the one or more processors. The one or more programs include instructions for: receiving, via a first input device of the one or more input devices, a first set of one or more inputs including: a first input that identifies a first action that was performed; and a second input that identifies a first user sentiment associated with the first action; in response to the first set of one or more inputs, generating a first log entry that includes the first action and the first user sentiment, wherein the first log entry is associated with a first time point.

In accordance with some an electronic device is described. The electronic device comprises: a display device; one or more input devices; means for receiving, via a first input device of the one or more input devices, a first set of one or more inputs including: a first input that identifies a first action that was performed; and a second input that identifies a first user sentiment associated with the first action; and means for, in response to the first set of one or more inputs, generating a first log entry that includes the first action and the first user sentiment, wherein the first log entry is associated with a first time point.

In accordance with some embodiments, a method is described. The method is performed at a first electronic device having a display device and one or more input devices. The method comprises: displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device; while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set

of one or more inputs includes a first input corresponding to a set of one or more event conditions; and in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge; wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a challenge response event based on the set of one or more event conditions.

In accordance with some embodiments, a non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a first electronic device having a display device and one or more input devices is described. The one or more programs include instructions for: displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device; while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge; wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a challenge response event based on the set of one or more event conditions.

In accordance with some embodiments, a transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a first electronic device having a display device and one or more input devices is described. The one or more programs include instructions for: displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device; while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge; wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a challenge response event based on the set of one or more event conditions.

In accordance with some embodiments, a first electronic device is described. The first electronic device comprises: a display device; one or more input devices; one or more processors; and memory storing one or more programs configured to be executed by the one or more processors. The one or more programs include instructions for: displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device; while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and in response to the first set of one or more inputs, transmitting, to the second electronic device, data represent-

ing the first event challenge; wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a challenge response event based on the set of one or more event conditions.

In accordance with some embodiments, a first electronic device is described. The first electronic device comprises: a display device; one or more input devices; means for displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device; means for, while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and means for, in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge; wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a challenge response event based on the set of one or more event conditions.

Executable instructions for performing these functions are, optionally, included in a non-transitory computer-readable storage medium or other computer program product configured for execution by one or more processors. Executable instructions for performing these functions are, optionally, included in a transitory computer-readable storage medium or other computer program product configured for execution by one or more processors.

Thus, devices are provided with faster, more efficient methods and interfaces for logging health-related events and for health-related coaching, thereby increasing the effectiveness, efficiency, and user satisfaction with such devices. Such methods and interfaces may complement or replace other methods for logging health-related events and for health-related coaching.

DESCRIPTION OF THE FIGURES

For a better understanding of the various described embodiments, reference should be made to the Description of Embodiments below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

FIG. 1A is a block diagram illustrating a portable multifunction device with a touch-sensitive display in accordance with some embodiments.

FIG. 1B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments.

FIG. 2 illustrates a portable multifunction device having a touch screen in accordance with some embodiments.

FIG. 3 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments.

FIG. 4A illustrates an exemplary user interface for a menu of applications on a portable multifunction device in accordance with some embodiments.

FIG. 4B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display in accordance with some embodiments.

FIG. 5A illustrates a personal electronic device in accordance with some embodiments.

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FIG. 5B is a block diagram illustrating a personal electronic device in accordance with some embodiments.

FIG. 6 is a schematic diagram of a set of interconnected users and electronic devices in accordance with some embodiments.

FIGS. 7A-7M depict devices and user interfaces for logging health-related events in accordance with some embodiments.

FIG. 8 is a flow diagram illustrating a method for logging health-related events in accordance with some embodiments.

FIGS. 9A-9M depict devices and user interfaces for health-related coaching using electronic devices in accordance with some embodiments.

FIG. 10 is a flow diagram illustrating a method for health-related coaching using electronic devices in accordance with some embodiments.

DESCRIPTION OF EMBODIMENTS

The following description sets forth exemplary methods, parameters, and the like. It should be recognized, however, that such description is not intended as a limitation on the scope of the present disclosure but is instead provided as a description of exemplary embodiments.

There is a need for electronic devices that provide efficient methods and interfaces for logging health-related events and for health-related coaching. Such techniques can reduce the cognitive burden on a user who log health-related events and/or who provide health-related coaching, thereby enhancing productivity. Further, such techniques can reduce processor and battery power otherwise wasted on redundant user inputs.

Below, FIGS. 1A-1B, 2, 3, 4A-4B, and 5A-5B provide a description of exemplary devices for performing the techniques for logging health-related events and for health-related coaching. FIGS. 7A-7M illustrate exemplary user interfaces for logging health-related events. FIG. 8 is a flow diagram illustrating methods of managing event notifications in accordance with some embodiments. The user interfaces in FIGS. 7A-7M are used to illustrate the processes described below, including the processes in FIG. 8. FIGS. 9A-9M illustrate exemplary user interfaces for providing health-related coaching (e.g., via issuing health-related challenges). FIG. 10 is a flow diagram illustrating methods for health-related coaching using electronic devices in accordance with some embodiments. The user interfaces in FIGS. 9A-9M are used to illustrate the processes described below, including the processes in FIG. 10.

Although the following description uses terms “first,” “second,” etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first touch could be termed a second touch, and, similarly, a second touch could be termed a first touch, without departing from the scope of the various described embodiments. The first touch and the second touch are both touches, but they are not the same touch.

The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations

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of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

Embodiments of electronic devices, user interfaces for such devices, and associated processes for using such devices are described. In some embodiments, the device is a portable communications device, such as a mobile telephone, that also contains other functions, such as PDA and/or music player functions. Exemplary embodiments of portable multifunction devices include, without limitation, the iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, California. Other portable electronic devices, such as laptops or tablet computers with touch-sensitive surfaces (e.g., touch screen displays and/or touchpads), are, optionally, used. It should also be understood that, in some embodiments, the device is not a portable communications device, but is a desktop computer with a touch-sensitive surface (e.g., a touch screen display and/or a touchpad).

In the discussion that follows, an electronic device that includes a display and a touch-sensitive surface is described. It should be understood, however, that the electronic device optionally includes one or more other physical user-interface devices, such as a physical keyboard, a mouse, and/or a joystick.

The device typically supports a variety of applications, such as one or more of the following: a drawing application, a presentation application, a word processing application, a website creation application, a disk authoring application, a spreadsheet application, a gaming application, a telephone application, a video conferencing application, an e-mail application, an instant messaging application, a workout support application, a photo management application, a digital camera application, a digital video camera application, a web browsing application, a digital music player application, and/or a digital video player application.

The various applications that are executed on the device optionally use at least one common physical user-interface device, such as the touch-sensitive surface. One or more functions of the touch-sensitive surface as well as corresponding information displayed on the device are, optionally, adjusted and/or varied from one application to the next and/or within a respective application. In this way, a common physical architecture (such as the touch-sensitive surface) of the device optionally supports the variety of applications with user interfaces that are intuitive and transparent to the user.

Attention is now directed toward embodiments of portable devices with touch-sensitive displays. FIG. 1A is a block diagram illustrating portable multifunction device 100 with touch-sensitive display system 112 in accordance with some embodiments. Touch-sensitive display 112 is sometimes called a “touch screen” for convenience and is sometimes known as or called a “touch-sensitive display system.”

Device 100 includes memory 102 (which optionally includes one or more computer-readable storage media), memory controller 122, one or more processing units (CPUs) 120, peripherals interface 118, RF circuitry 108, audio circuitry 110, speaker 111, microphone 113, input/output (I/O) subsystem 106, other input control devices 116, and external port 124. Device 100 optionally includes one or more optical sensors 164. Device 100 optionally includes one or more contact intensity sensors 165 for detecting intensity of contacts on device 100 (e.g., a touch-sensitive surface such as touch-sensitive display system 112 of device 100). Device 100 optionally includes one or more tactile output generators 167 for generating tactile outputs on device 100 (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system 112 of device 100 or touchpad 355 of device 300). These components optionally communicate over one or more communication buses or signal lines 103.

As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of

mass of the device that will be detected by a user with the user’s sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user’s hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user’s movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that device 100 is only one example of a portable multifunction device, and that device 100 optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. 1A are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

Memory 102 optionally includes high-speed random access memory and optionally also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller 122 optionally controls access to memory 102 by other components of device 100.

Peripherals interface 118 can be used to couple input and output peripherals of the device to CPU 120 and memory 102. The one or more processors 120 run or execute various software programs and/or sets of instructions stored in memory 102 to perform various functions for device 100 and to process data. In some embodiments, peripherals interface 118, CPU 120, and memory controller 122 are, optionally, implemented on a single chip, such as chip 104. In some other embodiments, they are, optionally, implemented on separate chips.

RF (radio frequency) circuitry 108 receives and sends RF signals, also called electromagnetic signals. RF circuitry 108 converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry 108 optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry 108 optionally communicates with networks, such as the Internet, also referred to

as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry **108** optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPDA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VOIP), Wi-MAX, a protocol for e-mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry **110**, speaker **111**, and microphone **113** provide an audio interface between a user and device **100**. Audio circuitry **110** receives audio data from peripherals interface **118**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **111**. Speaker **111** converts the electrical signal to human-audible sound waves. Audio circuitry **110** also receives electrical signals converted by microphone **113** from sound waves. Audio circuitry **110** converts the electrical signal to audio data and transmits the audio data to peripherals interface **118** for processing. Audio data is, optionally, retrieved from and/or transmitted to memory **102** and/or RF circuitry **108** by peripherals interface **118**. In some embodiments, audio circuitry **110** also includes a headset jack (e.g., **212**, FIG. 2). The headset jack provides an interface between audio circuitry **110** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem **106** couples input/output peripherals on device **100**, such as touch screen **112** and other input control devices **116**, to peripherals interface **118**. I/O subsystem **106** optionally includes display controller **156**, optical sensor controller **158**, depth camera controller **169**, intensity sensor controller **159**, haptic feedback controller **161**, and one or more input controllers **160** for other input or control devices. The one or more input controllers **160** receive/send electrical signals from/to other input control devices **116**. The other input control devices **116** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some alternate embodiments, input controller(s) **160** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **208**, FIG. 2) optionally include an up/down button for volume control of

speaker **111** and/or microphone **113**. The one or more buttons optionally include a push button (e.g., **206**, FIG. 2).

A quick press of the push button optionally disengages a lock of touch screen **112** or optionally begins a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **206**) optionally turns power to device **100** on or off. The functionality of one or more of the buttons are, optionally, user-customizable. Touch screen **112** is used to implement virtual or soft buttons and one or more soft keyboards.

Touch-sensitive display **112** provides an input interface and an output interface between the device and a user. Display controller **156** receives and/or sends electrical signals from/to touch screen **112**. Touch screen **112** displays visual output to the user. The visual output optionally includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output optionally corresponds to user-interface objects.

Touch screen **112** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **112** and display controller **156** (along with any associated modules and/or sets of instructions in memory **102**) detect contact (and any movement or breaking of the contact) on touch screen **112** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **112**. In an exemplary embodiment, a point of contact between touch screen **112** and the user corresponds to a finger of the user.

Touch screen **112** optionally uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology, although other display technologies are used in other embodiments. Touch screen **112** and display controller **156** optionally detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **112**. In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

A touch-sensitive display in some embodiments of touch screen **112** is, optionally, analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846 (Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **112** displays visual output from device **100**, whereas touch-sensitive touchpads do not provide visual output.

A touch-sensitive display in some embodiments of touch screen **112** is described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, "Multipoint Touch Surface Controller," filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, "Multipoint Touchscreen," filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, "Gestures For Touch Sensitive Input Devices," filed Jul. 30, 2004; (4) U.S. patent application Ser.

No. 11/048,264, "Gestures For Touch Sensitive Input Devices," filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, "Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices," filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, "Virtual Input Device Placement On A Touch Screen User Interface," filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, "Operation Of A Computer With A Touch Screen Interface," filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, "Activating Virtual Keys Of A Touch-Screen Virtual Keyboard," filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, "Multi-Functional Hand-Held Device," filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

Touch screen **112** optionally has a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user optionally makes contact with touch screen **112** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to the touch screen, device **100** optionally includes a touchpad for activating or deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is, optionally, a touch-sensitive surface that is separate from touch screen **112** or an extension of the touch-sensitive surface formed by the touch screen.

Device **100** also includes power system **162** for powering the various components. Power system **162** optionally includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Device **100** optionally also includes one or more optical sensors **164**. FIG. 1A shows an optical sensor coupled to optical sensor controller **158** in I/O subsystem **106**. Optical sensor **164** optionally includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor **164** receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction with imaging module **143** (also called a camera module), optical sensor **164** optionally captures still images or video. In some embodiments, an optical sensor is located on the back of device **100**, opposite touch screen display **112** on the front of the device so that the touch screen display is enabled for use as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is, optionally, obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor **164** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single

optical sensor **164** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device **100** optionally also includes one or more depth camera sensors **175**. FIG. 1A shows a depth camera sensor coupled to depth camera controller **169** in I/O subsystem **106**. Depth camera sensor **175** receives data from the environment to create a three dimensional model of an object (e.g., a face) within a scene from a viewpoint (e.g., a depth camera sensor). In some embodiments, in conjunction with imaging module **143** (also called a camera module), depth camera sensor **175** is optionally used to determine a depth map of different portions of an image captured by the imaging module **143**. In some embodiments, a depth camera sensor is located on the front of device **100** so that the user's image with depth information is, optionally, obtained for video conferencing while the user views the other video conference participants on the touch screen display and to capture selfies with depth map data. In some embodiments, the depth camera sensor **175** is located on the back of device, or on the back and the front of the device **100**. In some embodiments, the position of depth camera sensor **175** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a depth camera sensor **175** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device **100** optionally also includes one or more contact intensity sensors **165**. FIG. 1A shows a contact intensity sensor coupled to intensity sensor controller **159** in I/O subsystem **106**. Contact intensity sensor **165** optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor **165** receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **112**). In some embodiments, at least one contact intensity sensor is located on the back of device **100**, opposite touch screen display **112**, which is located on the front of device **100**.

Device **100** optionally also includes one or more proximity sensors **166**. FIG. 1A shows proximity sensor **166** coupled to peripherals interface **118**. Alternately, proximity sensor **166** is, optionally, coupled to input controller **160** in I/O subsystem **106**. Proximity sensor **166** optionally performs as described in U.S. patent application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No. 11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen **112** when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

Device **100** optionally also includes one or more tactile output generators **167**. FIG. 1A shows a tactile output generator coupled to haptic feedback controller **161** in I/O subsystem **106**. Tactile output generator **167** optionally includes one or more electroacoustic devices such as speak-

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ers or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor **165** receives tactile feedback generation instructions from haptic feedback module **133** and generates tactile outputs on device **100** that are capable of being sensed by a user of device **100**. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **112**) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of device **100**) or laterally (e.g., back and forth in the same plane as a surface of device **100**). In some embodiments, at least one tactile output generator sensor is located on the back of device **100**, opposite touch screen display **112**, which is located on the front of device **100**.

Device **100** optionally also includes one or more accelerometers **168**. FIG. 1A shows accelerometer **168** coupled to peripherals interface **118**. Alternately, accelerometer **168** is, optionally, coupled to an input controller **160** in I/O subsystem **106**. Accelerometer **168** optionally performs as described in U.S. Patent Publication No. 20050190059, "Acceleration-based Theft Detection System for Portable Electronic Devices," and U.S. Patent Publication No. 20060017692, "Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer," both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device **100** optionally includes, in addition to accelerometer(s) **168**, a magnetometer and a GPS (or GLONASS or other global navigation system) receiver for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device **100**.

In some embodiments, the software components stored in memory **102** include operating system **126**, communication module (or set of instructions) **128**, contact/motion module (or set of instructions) **130**, graphics module (or set of instructions) **132**, text input module (or set of instructions) **134**, Global Positioning System (GPS) module (or set of instructions) **135**, and applications (or sets of instructions) **136**. Furthermore, in some embodiments, memory **102** (FIG. 1A) or **370** (FIG. 3) stores device/global internal state **157**, as shown in FIGS. 1A and 3. Device/global internal state **157** includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display **112**; sensor state, including information obtained from the device's various sensors and input control devices **116**; and location information concerning the device's location and/or attitude.

Operating system **126** (e.g., Darwin, RTXC, LINUX, UNIX, OS X, IOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module **128** facilitates communication with other devices over one or more external ports **124** and also includes various software components for handling data received by RF circuitry **108** and/or external port **124**. External port **124** (e.g., Universal Serial Bus (USB), FIRE-

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WIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or similar to and/or compatible with, the 30-pin connector used on iPod® (trademark of Apple Inc.) devices.

Contact/motion module **130** optionally detects contact with touch screen **112** (in conjunction with display controller **156**) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module **130** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **130** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., "multitouch"/multiple finger contacts). In some embodiments, contact/motion module **130** and display controller **156** detect contact on a touchpad.

In some embodiments, contact/motion module **130** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has "clicked" on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device **100**). For example, a mouse "click" threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click "intensity" parameter).

Contact/motion module **130** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.

Graphics module **132** includes various known software components for rendering and displaying graphics on touch screen **112** or other display, including components for changing the visual impact (e.g., brightness, transparency,

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saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including, without limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, anima-

tions, and the like. In some embodiments, graphics module 132 stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module 132 receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller 156.

Haptic feedback module 133 includes various software components for generating instructions used by tactile output generator(s) 167 to produce tactile outputs at one or more locations on device 100 in response to user interactions with device 100.

Text input module 134, which is, optionally, a component of graphics module 132, provides soft keyboards for entering text in various applications (e.g., contacts module 137, e-mail client module 140, IM module 141, browser module 147, and any other application that needs text input).

GPS module 135 determines the location of the device and provides this information for use in various applications (e.g., to telephone module 138 for use in location-based dialing; to camera module 143 as picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

Applications 136 optionally include the following modules (or sets of instructions), or a subset or superset thereof:

Contacts module 137 (sometimes called an address book or contact list);

Telephone module 138;

Video conference module 139;

E-mail client module 140;

Instant messaging (IM) module 141;

Workout support module 142;

Camera module 143 for still and/or video images;

Image management module 144;

Video player module;

Music player module;

Browser module 147;

Calendar module 148;

Widget modules 149, which optionally include one or more of: weather widget 149-1, stocks widget 149-2, calculator widget 149-3, alarm clock widget 149-4, dictionary widget 149-5, and other widgets obtained by the user, as well as user-created widgets 149-6;

Widget creator module 150 for making user-created widgets 149-6;

Search module 151;

Video and music player module 152, which merges video player module and music player module;

Notes module 153;

Map module 154; and/or

Online video module 155.

Examples of other applications 136 that are, optionally, stored in memory 102 include other word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and

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text input module 134, contacts module 137 are, optionally, used to manage an address book or contact list (e.g., stored in application internal state 192 of contacts module 137 in memory 102 or memory 370), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physical address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone module 138, video conference module 139, e-mail client module 140, or IM module 141; and so forth.

In conjunction with RF circuitry 108, audio circuitry 110, speaker 111, microphone 113, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, telephone module 138 are optionally, used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module 137, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies.

In conjunction with RF circuitry 108, audio circuitry 110, speaker 111, microphone 113, touch screen 112, display controller 156, optical sensor 164, optical sensor controller 158, contact/motion module 130, graphics module 132, text input module 134, contacts module 137, and telephone module 138, video conference module 139 includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, e-mail client module 140 includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module 144, e-mail client module 140 makes it very easy to create and send e-mails with still or video images taken with camera module 143.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, the instant messaging module 141 includes executable instructions to enter a sequence of characters corresponding to an instant message, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages optionally include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, “instant messaging” refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, text input module 134, GPS module 135, map module 154, and music player module, workout support module 142 includes executable instructions to create workouts (e.g., with time, distance, and/or calorie burning goals);

communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

In conjunction with touch screen 112, display controller 156, optical sensor(s) 164, optical sensor controller 158, contact/motion module 130, graphics module 132, and image management module 144, camera module 143 includes executable instructions to capture still images or video (including a video stream) and store them into memory 102, modify characteristics of a still image or video, or delete a still image or video from memory 102.

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, text input module 134, and camera module 143, image management module 144 includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, browser module 147 includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, text input module 134, e-mail client module 140, and browser module 147, calendar module 148 includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, text input module 134, and browser module 147, widget modules 149 are mini-applications that are, optionally, downloaded and used by a user (e.g., weather widget 149-1, stocks widget 149-2, calculator widget 149-3, alarm clock widget 149-4, and dictionary widget 149-5) or created by the user (e.g., user-created widget 149-6). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, text input module 134, and browser module 147, the widget creator module 150 are, optionally, used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, search module 151 includes executable instructions to search for text, music, sound, image, video, and/or other files in memory 102 that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, audio circuitry 110, speaker 111, RF circuitry 108, and browser module 147, video and music player module 152 includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files,

and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen 112 or on an external, connected display via external port 124). In some embodiments, device 100 optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, notes module 153 includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, text input module 134, GPS module 135, and browser module 147, map module 154 are, optionally, used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, audio circuitry 110, speaker 111, RF circuitry 108, text input module 134, e-mail client module 140, and browser module 147, online video module 155 includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port 124), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module 141, rather than e-mail client module 140, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are, optionally, combined or otherwise rearranged in various embodiments. For example, video player module is, optionally, combined with music player module into a single module (e.g., video and music player module 152, FIG. 1A). In some embodiments, memory 102 optionally stores a subset of the modules and data structures identified above. Furthermore, memory 102 optionally stores additional modules and data structures not described above.

In some embodiments, device 100 is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device 100, the number of physical input control devices (such as push buttons, dials, and the like) on device 100 is, optionally, reduced.

The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodi-

ments, the touchpad, when touched by the user, navigates device **100** to a main, home, or root menu from any user interface that is displayed on device **100**. In such embodiments, a “menu button” is implemented using a touchpad. In some other embodiments, the menu button is a physical push button or other physical input control device instead of a touchpad.

FIG. **1B** is a block diagram illustrating exemplary components for event handling in accordance with some embodiments. In some embodiments, memory **102** (FIG. **1A**) or **370** (FIG. **3**) includes event sorter **170** (e.g., in operating system **126**) and a respective application **136-1** (e.g., any of the aforementioned applications **137-151**, **155**, **380-390**).

Event sorter **170** receives event information and determines the application **136-1** and application view **191** of application **136-1** to which to deliver the event information. Event sorter **170** includes event monitor **171** and event dispatcher module **174**. In some embodiments, application **136-1** includes application internal state **192**, which indicates the current application view(s) displayed on touch-sensitive display **112** when the application is active or executing. In some embodiments, device/global internal state **157** is used by event sorter **170** to determine which application(s) is (are) currently active, and application internal state **192** is used by event sorter **170** to determine application views **191** to which to deliver event information.

In some embodiments, application internal state **192** includes additional information, such as one or more of: resume information to be used when application **136-1** resumes execution, user interface state information that indicates information being displayed or that is ready for display by application **136-1**, a state queue for enabling the user to go back to a prior state or view of application **136-1**, and a redo/undo queue of previous actions taken by the user.

Event monitor **171** receives event information from peripherals interface **118**. Event information includes information about a sub-event (e.g., a user touch on touch-sensitive display **112**, as part of a multi-touch gesture). Peripherals interface **118** transmits information it receives from I/O subsystem **106** or a sensor, such as proximity sensor **166**, accelerometer(s) **168**, and/or microphone **113** (through audio circuitry **110**). Information that peripherals interface **118** receives from I/O subsystem **106** includes information from touch-sensitive display **112** or a touch-sensitive surface.

In some embodiments, event monitor **171** sends requests to the peripherals interface **118** at predetermined intervals. In response, peripherals interface **118** transmits event information. In other embodiments, peripherals interface **118** transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter **170** also includes a hit view determination module **172** and/or an active event recognizer determination module **173**.

Hit view determination module **172** provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display **112** displays more than one view. Views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which

a touch is detected optionally correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is, optionally, called the hit view, and the set of events that are recognized as proper inputs are, optionally, determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module **172** receives information related to sub-events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module **172** identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module **172**, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module **173** determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module **173** determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module **173** determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module **174** dispatches the event information to an event recognizer (e.g., event recognizer **180**). In embodiments including active event recognizer determination module **173**, event dispatcher module **174** delivers the event information to an event recognizer determined by active event recognizer determination module **173**. In some embodiments, event dispatcher module **174** stores in an event queue the event information, which is retrieved by a respective event receiver **182**.

In some embodiments, operating system **126** includes event sorter **170**. Alternatively, application **136-1** includes event sorter **170**. In yet other embodiments, event sorter **170** is a stand-alone module, or a part of another module stored in memory **102**, such as contact/motion module **130**.

In some embodiments, application **136-1** includes a plurality of event handlers **190** and one or more application views **191**, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view **191** of the application **136-1** includes one or more event recognizers **180**. Typically, a respective application view **191** includes a plurality of event recognizers **180**. In other embodiments, one or more of event recognizers **180** are part of a separate module, such as a user interface kit or a higher level object from which application **136-1** inherits methods and other properties. In some embodiments, a respective event handler **190** includes one or more of: data updater **176**, object updater **177**, GUI updater **178**, and/or event data **179** received from event sorter **170**. Event handler **190** optionally utilizes or calls data updater **176**, object updater **177**, or GUI updater **178** to update the application internal state **192**. Alternatively, one or more of the application views **191** include one or more respective event handlers **190**. Also, in

some embodiments, one or more of data updater **176**, object updater **177**, and GUI updater **178** are included in a respective application view **191**.

A respective event recognizer **180** receives event information (e.g., event data **179**) from event sorter **170** and identifies an event from the event information. Event recognizer **180** includes event receiver **182** and event comparator **184**. In some embodiments, event recognizer **180** also includes at least a subset of: metadata **183**, and event delivery instructions **188** (which optionally include sub-event delivery instructions).

Event receiver **182** receives event information from event sorter **170**. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information optionally also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator **184** compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub-event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator **184** includes event definitions **186**. Event definitions **186** contain definitions of events (e.g., predefined sequences of sub-events), for example, event **1** (**187-1**), event **2** (**187-2**), and others. In some embodiments, sub-events in an event (**187**) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event **1** (**187-1**) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event **2** (**187-2**) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display **112**, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers **190**.

In some embodiments, event definition **187** includes a definition of an event for a respective user-interface object. In some embodiments, event comparator **184** performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display **112**, when a touch is detected on touch-sensitive display **112**, event comparator **184** performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler **190**, the event comparator uses the result of the hit test to determine which event handler **190** should be activated. For example, event comparator **184** selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event (**187**) also includes delayed actions that delay delivery of the event information until after it has been determined

whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer **180** determines that the series of sub-events do not match any of the events in event definitions **186**, the respective event recognizer **180** enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer **180** includes metadata **183** with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata **183** includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata **183** includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer **180** activates event handler **190** associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer **180** delivers event information associated with the event to event handler **190**. Activating an event handler **190** is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer **180** throws a flag associated with the recognized event, and event handler **190** associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions **188** include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater **176** creates and updates data used in application **136-1**. For example, data updater **176** updates the telephone number used in contacts module **137**, or stores a video file used in video player module. In some embodiments, object updater **177** creates and updates objects used in application **136-1**. For example, object updater **177** creates a new user-interface object or updates the position of a user-interface object. GUI updater **178** updates the GUI. For example, GUI updater **178** prepares display information and sends it to graphics module **132** for display on a touch-sensitive display.

In some embodiments, event handler(s) **190** includes or has access to data updater **176**, object updater **177**, and GUI updater **178**. In some embodiments, data updater **176**, object updater **177**, and GUI updater **178** are included in a single module of a respective application **136-1** or application view **191**. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate multifunction devices **100** with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact

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movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. 2 illustrates a portable multifunction device 100 having a touch screen 112 in accordance with some embodiments. The touch screen optionally displays one or more graphics within user interface (UI) 200. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers 202 (not drawn to scale in the figure) or one or more styluses 203 (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device 100. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.

Device 100 optionally also include one or more physical buttons, such as “home” or menu button 204. As described previously, menu button 204 is, optionally, used to navigate to any application 136 in a set of applications that are, optionally, executed on device 100. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen 112.

In some embodiments, device 100 includes touch screen 112, menu button 204, push button 206 for powering the device on/off and locking the device, volume adjustment button(s) 208, subscriber identity module (SIM) card slot 210, headset jack 212, and docking/charging external port 124. Push button 206 is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device 100 also accepts verbal input for activation or deactivation of some functions through microphone 113. Device 100 also, optionally, includes one or more contact intensity sensors 165 for detecting intensity of contacts on touch screen 112 and/or one or more tactile output generators 167 for generating tactile outputs for a user of device 100.

FIG. 3 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device 300 need not be portable. In some embodiments, device 300 is a laptop computer, a desktop computer, a tablet computer, a multimedia player device, a navigation device, an educational device (such as a child’s learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device 300 typically includes one or more processing units (CPUs) 310, one or more network or other communications interfaces 360, memory 370, and one or more communication buses 320 for interconnecting these components. Communication buses 320 optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device 300 includes input/output (I/O) interface 330 comprising display

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340, which is typically a touch screen display. I/O interface 330 also optionally includes a keyboard and/or mouse (or other pointing device) 350 and touchpad 355, tactile output generator 357 for generating tactile outputs on device 300 (e.g., similar to tactile output generator(s) 167 described above with reference to FIG. 1A), sensors 359 (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) 165 described above with reference to FIG. 1A). Memory 370 includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory 370 optionally includes one or more storage devices remotely located from CPU(s) 310. In some embodiments, memory 370 stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory 102 of portable multifunction device 100 (FIG. 1A), or a subset thereof. Furthermore, memory 370 optionally stores additional programs, modules, and data structures not present in memory 102 of portable multifunction device 100. For example, memory 370 of device 300 optionally stores drawing module 380, presentation module 382, word processing module 384, website creation module 386, disk authoring module 388, and/or spreadsheet module 390, while memory 102 of portable multifunction device 100 (FIG. 1A) optionally does not store these modules.

Each of the above-identified elements in FIG. 3 is, optionally, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are, optionally, combined or otherwise rearranged in various embodiments. In some embodiments, memory 370 optionally stores a subset of the modules and data structures identified above. Furthermore, memory 370 optionally stores additional modules and data structures not described above.

Attention is now directed towards embodiments of user interfaces that are, optionally, implemented on, for example, portable multifunction device 100.

FIG. 4A illustrates an exemplary user interface for a menu of applications on portable multifunction device 100 in accordance with some embodiments. Similar user interfaces are, optionally, implemented on device 300. In some embodiments, user interface 400 includes the following elements, or a subset or superset thereof:

- Signal strength indicator(s) 402 for wireless communication(s), such as cellular and Wi-Fi signals;
- Time 404;
- Bluetooth indicator 405;
- Battery status indicator 406;
- Tray 408 with icons for frequently used applications, such as:
 - Icon 416 for telephone module 138, labeled “Phone,” which optionally includes an indicator 414 of the number of missed calls or voicemail messages;
 - Icon 418 for e-mail client module 140, labeled “Mail,” which optionally includes an indicator 410 of the number of unread e-mails;
 - Icon 420 for browser module 147, labeled “Browser;” and

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Icon **422** for video and music player module **152**, also referred to as iPod (trademark of Apple Inc.) module **152**, labeled “iPod;” and

Icons for other applications, such as:

Icon **424** for IM module **141**, labeled “Messages;”

Icon **426** for calendar module **148**, labeled “Calendar;”

Icon **428** for image management module **144**, labeled “Photos;”

Icon **430** for camera module **143**, labeled “Camera;”

Icon **432** for online video module **155**, labeled “Online Video;”

Icon **434** for stocks widget **149-2**, labeled “Stocks;”

Icon **436** for map module **154**, labeled “Maps;”

Icon **438** for weather widget **149-1**, labeled “Weather;”

Icon **440** for alarm clock widget **149-4**, labeled “Clock;”

Icon **442** for workout support module **142**, labeled “Workout Support;”

Icon **444** for notes module **153**, labeled “Notes;” and

Icon **446** for a settings application or module, labeled “Settings;” which provides access to settings for device **100** and its various applications **136**.

It should be noted that the icon labels illustrated in FIG. 4A are merely exemplary. For example, icon **422** for video and music player module **152** is labeled “Music” or “Music Player.” Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

FIG. 4B illustrates an exemplary user interface on a device (e.g., device **300**, FIG. 3) with a touch-sensitive surface **451** (e.g., a tablet or touchpad **355**, FIG. 3) that is separate from the display **450** (e.g., touch screen display **112**). Device **300** also, optionally, includes one or more contact intensity sensors (e.g., one or more of sensors **359**) for detecting intensity of contacts on touch-sensitive surface **451** and/or one or more tactile output generators **357** for generating tactile outputs for a user of device **300**.

Although some of the examples that follow will be given with reference to inputs on touch screen display **112** (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. 4B. In some embodiments, the touch-sensitive surface (e.g., **451** in FIG. 4B) has a primary axis (e.g., **452** in FIG. 4B) that corresponds to a primary axis (e.g., **453** in FIG. 4B) on the display (e.g., **450**). In accordance with these embodiments, the device detects contacts (e.g., **460** and **462** in FIG. 4B) with the touch-sensitive surface **451** at locations that correspond to respective locations on the display (e.g., in FIG. 4B, contact **460** corresponds to **468** and contact **462** corresponds to **470**). In this way, user inputs (e.g., contacts **460** and **462**, and movements thereof) detected by the device on the touch-sensitive surface (e.g., **451** in FIG. 4B) are used by the device to manipulate the user interface on the display (e.g., **450** in FIG. 4B) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input

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device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

FIG. 5A illustrates exemplary personal electronic device **500**. Device **500** includes body **502**. In some embodiments, device **500** can include some or all of the features described with respect to devices **100** and **300** (e.g., FIGS. 1A-4B). In some embodiments, device **500** has touch-sensitive display screen **504**, hereafter touch screen **504**. Alternatively, or in addition to touch screen **504**, device **500** has a display and a touch-sensitive surface. As with devices **100** and **300**, in some embodiments, touch screen **504** (or the touch-sensitive surface) optionally includes one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen **504** (or the touch-sensitive surface) can provide output data that represents the intensity of touches. The user interface of device **500** can respond to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **500**.

Exemplary techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled “Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application,” filed May 8, 2013, published as WIPO Publication No. WO/2013/169849, and International Patent Application Serial No. PCT/US2013/069483, titled “Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships,” filed Nov. 11, 2013, published as WIPO Publication No. WO/2014/105276, each of which is hereby incorporated by reference in their entirety.

In some embodiments, device **500** has one or more input mechanisms **506** and **508**. Input mechanisms **506** and **508**, if included, can be physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **500** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **500** with, for example, hats, eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device **500** to be worn by a user.

FIG. 5B depicts exemplary personal electronic device **500**. In some embodiments, device **500** can include some or all of the components described with respect to FIGS. 1A, 1B, and 3. Device **500** has bus **512** that operatively couples I/O section **514** with one or more computer processors **516** and memory **518**. I/O section **514** can be connected to display **504**, which can have touch-sensitive component **522** and, optionally, intensity sensor **524** (e.g., contact intensity sensor). In addition, I/O section **514** can be connected with communication unit **530** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device **500** can include input

mechanisms **506** and/or **508**. Input mechanism **506** is, optionally, a rotatable input device or a depressible and rotatable input device, for example. Input mechanism **508** is, optionally, a button, in some examples.

Input mechanism **508** is, optionally, a microphone, in some examples. Personal electronic device **500** optionally includes various sensors, such as GPS sensor **532**, accelerometer **534**, directional sensor **540** (e.g., compass), gyroscope **536**, motion sensor **538**, and/or a combination thereof, all of which can be operatively connected to I/O section **514**.

Memory **518** of personal electronic device **500** can include one or more non-transitory computer-readable storage mediums, for storing computer-executable instructions, which, when executed by one or more computer processors **516**, for example, can cause the computer processors to perform the techniques described below, including processes **800** and **1000** (FIGS. **8** and **10**). A computer-readable storage medium can be any medium that can tangibly contain or store computer-executable instructions for use by or in connection with the instruction execution system, apparatus, or device. In some examples, the storage medium is a transitory computer-readable storage medium. In some examples, the storage medium is a non-transitory computer-readable storage medium. The non-transitory computer-readable storage medium can include, but is not limited to, magnetic, optical, and/or semiconductor storages. Examples of such storage include magnetic disks, optical discs based on CD, DVD, or Blu-ray technologies, as well as persistent solid-state memory such as flash, solid-state drives, and the like. Personal electronic device **500** is not limited to the components and configuration of FIG. **5B**, but can include other or additional components in multiple configurations.

As used here, the term “affordance” refers to a user-interactive graphical user interface object that is, optionally, displayed on the display screen of devices **100**, **300**, and/or **500** (FIGS. **1A**, **3**, and **5A-5B**). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each optionally constitute an affordance.

As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad **355** in FIG. **3** or touch-sensitive surface **451** in FIG. **4B**) while the cursor is over a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system **112** in FIG. **1A** or touch screen **112** in FIG. **4A**) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus regard to the specific form taken by the focus selector, the

focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user’s intended interaction with the user interface (e.g., by indicating, to the device, the element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally, based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds optionally includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to perform a respective operation or forgo performing the respective operation), rather than being used to determine whether to perform a first operation or a second operation.

In some embodiments, a portion of a gesture is identified for purposes of determining a characteristic intensity. For example, a touch-sensitive surface optionally receives a continuous swipe contact transitioning from a start location and reaching an end location, at which point the intensity of the contact increases. In this example, the characteristic intensity of the contact at the end location is, optionally, based on only a portion of the continuous swipe contact, and not the entire swipe contact (e.g., only the portion of the swipe contact at the end location). In some embodiments, a smoothing algorithm is, optionally, applied to the intensities of the swipe contact prior to determining the characteristic

intensity of the contact. For example, the smoothing algorithm optionally includes one or more of: an unweighted sliding-average smoothing algorithm, a triangular smoothing algorithm, a median filter smoothing algorithm, and/or an exponential smoothing algorithm. In some circumstances, these smoothing algorithms eliminate narrow spikes or dips in the intensities of the swipe contact for purposes of determining a characteristic intensity.

The intensity of a contact on the touch-sensitive surface is, optionally, characterized relative to one or more intensity thresholds, such as a contact-detection intensity threshold, a light press intensity threshold, a deep press intensity threshold, and/or one or more other intensity thresholds. In some embodiments, the light press intensity threshold corresponds to an intensity at which the device will perform operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, the deep press intensity threshold corresponds to an intensity at which the device will perform operations that are different from operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, when a contact is detected with a characteristic intensity below the light press intensity threshold (e.g., and above a nominal contact-detection intensity threshold below which the contact is no longer detected), the device will move a focus selector in accordance with movement of the contact on the touch-sensitive surface without performing an operation associated with the light press intensity threshold or the deep press intensity threshold. Generally, unless otherwise stated, these intensity thresholds are consistent between different sets of user interface figures.

An increase of characteristic intensity of the contact from an intensity below the light press intensity threshold to an intensity between the light press intensity threshold and the deep press intensity threshold is sometimes referred to as a “light press” input. An increase of characteristic intensity of the contact from an intensity below the deep press intensity threshold to an intensity above the deep press intensity threshold is sometimes referred to as a “deep press” input. An increase of characteristic intensity of the contact from an intensity below the contact-detection intensity threshold to an intensity between the contact-detection intensity threshold and the light press intensity threshold is sometimes referred to as detecting the contact on the touch-surface. A decrease of characteristic intensity of the contact from an intensity above the contact-detection intensity threshold to an intensity below the contact-detection intensity threshold is sometimes referred to as detecting liftoff of the contact from the touch-surface. In some embodiments, the contact-detection intensity threshold is zero. In some embodiments, the contact-detection intensity threshold is greater than zero.

In some embodiments described herein, one or more operations are performed in response to detecting a gesture that includes a respective press input or in response to detecting the respective press input performed with a respective contact (or a plurality of contacts), where the respective press input is detected based at least in part on detecting an increase in intensity of the contact (or plurality of contacts) above a press-input intensity threshold. In some embodiments, the respective operation is performed in response to detecting the increase in intensity of the respective contact above the press-input intensity threshold (e.g., a “down stroke” of the respective press input). In some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the press-input intensity threshold, and the respective opera-

tion is performed in response to detecting the subsequent decrease in intensity of the respective contact below the press-input threshold (e.g., an “up stroke” of the respective press input).

In some embodiments, the device employs intensity hysteresis to avoid accidental inputs sometimes termed “jitter,” where the device defines or selects a hysteresis intensity threshold with a predefined relationship to the press-input intensity threshold (e.g., the hysteresis intensity threshold is X intensity units lower than the press-input intensity threshold or the hysteresis intensity threshold is 75%, 90%, or some reasonable proportion of the press-input intensity threshold). Thus, in some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the hysteresis intensity threshold that corresponds to the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the hysteresis intensity threshold (e.g., an “up stroke” of the respective press input). Similarly, in some embodiments, the press input is detected only when the device detects an increase in intensity of the contact from an intensity at or below the hysteresis intensity threshold to an intensity at or above the press-input intensity threshold and, optionally, a subsequent decrease in intensity of the contact to an intensity at or below the hysteresis intensity, and the respective operation is performed in response to detecting the press input (e.g., the increase in intensity of the contact or the decrease in intensity of the contact, depending on the circumstances).

For ease of explanation, the descriptions of operations performed in response to a press input associated with a press-input intensity threshold or in response to a gesture including the press input are, optionally, triggered in response to detecting either: an increase in intensity of a contact above the press-input intensity threshold, an increase in intensity of a contact from an intensity below the hysteresis intensity threshold to an intensity above the press-input intensity threshold, a decrease in intensity of the contact below the press-input intensity threshold, and/or a decrease in intensity of the contact below the hysteresis intensity threshold corresponding to the press-input intensity threshold. Additionally, in examples where an operation is described as being performed in response to detecting a decrease in intensity of a contact below the press-input intensity threshold, the operation is, optionally, performed in response to detecting a decrease in intensity of the contact below a hysteresis intensity threshold corresponding to, and lower than, the press-input intensity threshold.

As used herein, an “installed application” refers to a software application that has been downloaded onto an electronic device (e.g., devices 100, 300, and/or 500) and is ready to be launched (e.g., become opened) on the device. In some embodiments, a downloaded application becomes an installed application by way of an installation program that extracts program portions from a downloaded package and integrates the extracted portions with the operating system of the computer system.

As used herein, the terms “open application” or “executing application” refer to a software application with retained state information (e.g., as part of device/global internal state 157 and/or application internal state 192). An open or executing application is, optionally, any one of the following types of applications:

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an active application, which is currently displayed on a display screen of the device that the application is being used on;

a background application (or background processes), which is not currently displayed, but one or more processes for the application are being processed by one or more processors; and

a suspended or hibernated application, which is not running, but has state information that is stored in memory (volatile and non-volatile, respectively) and that can be used to resume execution of the application.

As used herein, the term “closed application” refers to software applications without retained state information (e.g., state information for closed applications is not stored in a memory of the device). Accordingly, closing an application includes stopping and/or removing application processes for the application and removing state information for the application from the memory of the device. Generally, opening a second application while in a first application does not close the first application. When the second application is displayed and the first application ceases to be displayed, the first application becomes a background application. Attention is now directed towards embodiments of user interfaces (“UI”) and associated processes that are implemented on an electronic device, such as portable multifunction device 100, device 300, or device 500.

FIG. 6 illustrates an exemplary set of interconnected users and devices. FIGS. 7A-7M illustrate exemplary systems and user interfaces for logging health-related events, in accordance with some embodiments. The user interfaces in these figures are used to illustrate the processes described below, including the processes in FIG. 8.

FIG. 6 is a schematic diagram of a set 600 of interconnected users (e.g., participants and a coach) and electronic devices. Set 600 includes Participant 1, Participant 2, and Participant N, wherein N represents any number of additional participants. Participants 1-N are connected, via internet 602 to Coach 604. In some embodiments, each participant is connected to a different coach than coach 604. In some embodiments, one or more participants are connected to additional coaches, in addition to coach 604.

As shown in FIG. 6, Participant 1 is a user of device 606-1 (e.g., a smart phone), device 608-1 (e.g., a smart watch), and device 610-1 (e.g., a continuous glucose monitor). Participant 2 is a user of device 606-2 (e.g., a smart phone), device 608-2 (e.g., a smart watch), and device 610-2 (e.g., a continuous glucose monitor). Participant N is a user of device 606-3 (e.g., a smart phone), device 608-3 (e.g., a smart watch), and device 610-3 (e.g., a continuous glucose monitor).

Device 606-1 of Participant 1 is in communication with devices 608-1 and 610-1. In some embodiments, devices 608-1 and 610-1 are also in direct communication. Similarly, the devices of Participant 2 and Participant N, respectively, are in communication.

Coach 604 is a user of device 612 (e.g., a tablet computer) which is configured to receive from, and transmit data to, one or more devices of Participants 1-N, via internet 602. In some embodiments, a coach can work with one or more participants over a series of sessions (e.g., a set of days (e.g., 10 days)) to monitor health-related events and data, such as diet and blood glucose data, to assist the one or more participants with better understanding their physiological reaction to health-related events and, potentially, to assist the one or more participants with modifying their behavior to improve their health. For example, a coach can work with a pre-diabetic participant to, in a first session, first understand

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the impact of various events, that the participant would log, on their blood glucose. The coach can then work with the pre-diabetic participant, in subsequent sessions, to modify their behavior, through the use of issued event challenges, so as to positively affect their blood glucose levels and/or general health.

FIGS. 7A-7M depict devices and user interfaces for logging health-related events.

FIG. 7A depicts device 700, which is a smart phone. In some embodiments, device 700 is, or includes one or more features of, devices 100, 300, 500, and/or 606-1. Device 700 includes a touch-sensitive display 702, which serves as both a display and an input device. Device 700 also includes one or more wireless communication antennas for communicating with external electronic devices, such as devices 608-1, 610-1, and 612 of FIG. 6.

In FIG. 7A, device 700 is displaying daily blood glucose interface 704 that includes a graph 704-1 depicting the user's blood glucose data throughout the current day (e.g., May 10). In some embodiments, the blood glucose data is transmitted to device 700 from device 610-1, a continuous blood glucose monitor sensor operatively connected to the user of device 700. As shown, graph 704-1 has an x-axis of time (e.g., time of day) and a y-axis of blood glucose level (e.g., in units of mg/dL or mmol/L). Interface 704 also includes affordances 704-2, 704-3, and 704-4 for navigating to other days. Interface 704 further includes region 704-5, which displays the current (e.g., most recent) blood glucose reading for the user (e.g., the most recent data received from device 610-1) and an indication that the displayed day, May 10, is the current day.

Interface 704 also includes logged event region 704-6 that displays affordance(s) corresponding to any logged events for the currently displayed day. As shown in FIG. 7A, region 704-6 includes an affordance 704-6A corresponding to a logged food event, as indicated by the fork-and-knife graphical icon. Affordance 704-6A also includes an indication of an action (consumption of “a coffee with almond milk”) associated with the logged food event and a time (6:35 AM) that the event was logged, indicating when the action was taken. In some embodiments, physiological data other than blood glucose is tracked in addition to, or in place of blood glucose. Other physiological data can, for example, include cholesterol, body weight, body temperature, viral load, and blood pressure.

Interface 704 further includes a navigation region 704-7 that includes affordances 704-7A, 704-7B, and 704-7C for navigating to various views, as discussed in more detail below. Affordance 704-7A, corresponding to the daily view of user interface 704 is currently visually highlighted, to indicate that the daily view is currently active.

In FIG. 7A, device 700 detects an input 706 (e.g., a touch) on add event affordance 704-8. In response to detecting the input, device 700 displays user interface 708 of FIG. 7B.

In FIG. 7B, device 700 displays event type interface 708 as an overlay over interface 704, the remaining displayed portions of which are visually deemphasized (e.g., greyed out), while interface 708 is displayed. Interface 708 includes a plurality of affordances 708-1 to 708-4 for selecting a type of event to log. For example, affordance 708-1 can be selected to log a food/drink related event. 708-4 can be selected to log a new event based on a previously favored event.

In FIG. 7B, device 700 detects an input 710 on affordance 708-2 corresponding to a past activity event. In response to detecting the input, device 700 displays user interface 712 of FIG. 7C.

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In FIG. 7C, device 700 displays new past activity user interface 712 for logging a new past event after detecting input 710 and after receiving further inputs from the user. Interface 712 includes an activity/action input region 712-1 (already populated with the text “5 MILE RUN” via additional input) and date/time region 712-2 (already populated with “May 10, 2019 at 8:00” via additional input). Interface 712 also includes sentiment selection region 712-3 for selecting a sentiment associated with the past event. Region 712-3 includes affordances 712-A1 to 712-A3 for indicating a sentiment associated with the event and action being logged. In some embodiments, the sentiments are mutually exclusive and only one can be selected. In some embodiments, multiple sentiments can be selected. As shown in FIG. 7C, the sentiment “FATIGUED” has been selected via previous input. In some embodiments, sentiments are used by a participant and/or a coach on subsequent review to understand whether subsequent changes in behavior affect the given sentiment for later occurrences of the related event.

In FIG. 7C, device 700 detects an input 714 (e.g., a touch) on save affordance 716. In response to detecting the input, device 700 re-displays user interface 704 as seen in FIG. 7D.

In FIG. 7D, device 700 displays interface 704 with the newly logged event displayed as affordance 704-6B in region 704-6. Affordance 704-6B includes an indication of the action associated with the event (“5 MILE RUN”) and the time associated with the event (“8:00 AM”). In some embodiments, the affordance can also include an indication of the sentiment associated with the event (e.g., “FATIGUED”).

In FIG. 7D, device 700 detects an input 718 (e.g., a touch) on affordance 704-6A corresponding to the previously logged food/drink event. In response to detecting the input, device 700 displays user interface 720 as seen in FIG. 7E.

In FIG. 7E, device 700 displays event detail interface 720. Interface 720 includes a graph 720-1 depicting the user’s blood glucose data throughout the day associated with the logged event. Because the event is associated with the current day, the blood glucose data is the same as that shown in graph 704-1 of interface 704 (FIG. 7A). Graph 720-1 also includes an indication 720-2 indicating the time associated with the event. As seen in FIG. 7E, the user’s blood glucose level rose soon after the event (e.g., soon after the consumption of the coffee with almond milk). Interface 720 also includes an events detail region 720-3 that includes further details of the event. For example, region 720-3 includes an icon 720-3A that indicates that the user provided input indicating the sentiment (e.g., a prediction) that the event would cause the user’s blood glucose to go up a little, as discussed in more detail below. Region 720-3 also includes an icon 720-3B indicating that the event has been favorited. Region 720-3 also includes an indication 720-3C of whether any photos were associated with the event (e.g., photos of the food or drink that was consumed).

In FIG. 7E, device 700 detects an input 722 (e.g., a touch) on edit affordance 724 corresponding to a request to edit the event shown in FIG. 7E. In response to detecting the input, device 700 displays user interface 726 as seen in FIG. 7F.

In FIG. 7F, device 700 displays event edit interface 726 for editing the event after detecting input 722 and after receiving further inputs from the user. As seen in FIG. 7F, device 700 has detected further inputs to change the action, shown at 726-1, to “A COFFEE WITH ALMOND MILK AND SUGAR,” as the user now recalls that the coffee also included added sugar. Interface 726 also includes a region 726-2 for indicating that the event is a response to an event

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challenge that has been issued, as discussed in more detail below. Interface 726 further includes region 726-3 for indicating a sentiment (e.g., an expected physiological response) associated with the event. Region 726-3 includes affordance 726-3A for indicating that the expected response will be “GOES UP A LOT” (e.g., blood glucose will go up a large amount) and affordance 726-3B for indicating that the expected response will be “GOES UP A LITTLE” (e.g., blood glucose will only increase by a small amount) in response to the event (e.g., in response to the consumption of the food/drink).

In FIG. 7F, device 700 detects an input 728 (e.g., a touch) on affordance 726-3A corresponding to “GOES UP A LOT”. In response to detecting the input, device 700 modifies user interface 726 as shown in FIG. 7G.

In FIG. 7G, device 700 displays interface 726 with a check mark in affordance 726-3A to indicate that the event is now associated with the user sentiment that blood glucose will go up a lot in response to the food/drink event.

In FIG. 7G, device 700 detects an input 730 (e.g., a touch) on save affordance 732. In response to detecting the input, device 700 updates the record for the event and re-displays interface 704 as shown in FIG. 7H.

In FIG. 7H, device 700 displays user interface 704 with affordance 704-6A now indicating the event as “A COFFEE WITH ALMOND MILK AND SUGAR” based on the edits made in FIG. 7G.

In FIG. 7H, device 700 detects an input 734 (e.g., a touch) on affordance 704-7B for navigating to a challenges view. In response to detecting the input, device 700 displays challenges user interface 736 of FIG. 7I.

In 7I, device 700 displays challenges user interface 736. Interface 736 displays active challenges, if any, that have been received at device 700. The creation of challenges is discussed in more detail, below, with reference to FIGS. 9A-9M and FIG. 10. Interface 736 includes a region 736-1 for displaying active food challenges, if any, and a region 736-2 for displaying active habit challenges, if any. As seen in FIG. 7I, two food challenges, corresponding to affordances 736-1A and 736-1B, are currently active. Affordance 746-1A, corresponding to the first food challenge, includes an indication of the action to be taken in response to the challenge (“REPLACE PROTEIN BAR WITH FRUIT AND NUTS”) and the original food/drink event that the challenge was created in response to (“PROTEIN BAR AFTER RUN”). In response to a food challenge, a user can log a response event (e.g., a food/drink consumption event) in a manner similar to that discussed with respect to FIGS. 7D-7G, including indicating a specific time within the day for the response event. For example, the user of device 700 can choose to replace a protein bar that the user might normally consume as part of a routine with fruit and nuts and then log a food event that is responsive to the challenge.

As seen in FIG. 7I, one habit challenge corresponding to affordance 736-2A, is currently active. Device 700 detects an input 738 (e.g., a touch) on affordance 736-2A corresponding to the active habit challenge. In response to detecting the input, device 700 displays habit challenge response interface 740 as seen in FIG. 7J.

In FIG. 7J, device 700 displays interface 740 for marking one or more days for which the user determines that the habit challenge was successfully met. In some embodiments, the habit challenge can specify one or more actions (e.g., “EAT PROTEIN BAR IMMEDIATELY AFTER EXERCISE”) that are to be performed throughout the day to meet the challenge. In some embodiments, the habit challenge can specify one or more actions that should be avoided to meet

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the challenge (e.g., “DO NOT DRINK SUGARED SODA”). Interface **740** includes a plurality of affordances **740-1**, including affordance **740-1A** corresponding to May 11, 2019.

In FIG. 7J, device **700** detects an input **742** (e.g., a touch) on affordance **740-1A**. In response to detecting the input, device **700** modifies interface **740** as shown in FIG. 7K.

In FIG. 7K, device **700** displays interface **740** with affordance **740-1A** checked to indicate that the habit challenge was met for that day. In some embodiments, the user can select a plurality of affordances **740-1** to indicate a plurality of days on which the habit challenge was met. In some embodiments, responses to habit challenges only indicate a day, without indicating any specific time of day, as habit challenges can only be met on a daily basis.

In FIG. 7K, device **700** detects an input **744** (e.g., a touch) on save affordance **746**. In response to detecting the input, device **700** modifies saves the user’s selection of date(s) on which the habit challenge was met.

FIG. 7L depicts a device **750**, which is a smart watch. In some embodiments, device **60** is, or includes one or more features of, devices **100**, **300**, **500**, and/or **608-1**. Device **750** includes a touch-sensitive display **752**, which serves as both a display and an input device. Device **750** also includes one or more wireless communication antennas for communicating with external electronic devices, such as devices **606-1**, **610-1**, and **612** of FIG. 6.

In FIG. 7L, device **750** displays user interface **754** for logging a food/drink event at a specific time. Data for such a logged event can then be transmitted to an external device (e.g., device **700**) to enter additional details (e.g., via the editing methods depicted in FIGS. 7D-7G). In some embodiments, device **750** is more easily or readily accessible to the user than device **700** and, since it is important to accurately record the time at which a food/drink event occurred, device **750** can be used to facilitate more accurate logging of food/drink events. In FIG. 7M, device **750** displays an indication that logging of an event was successful.

FIG. 8 is a flow diagram illustrating a method for logging health-related events using an electronic device in accordance with some embodiments. Method **800** is performed at a first electronic device (e.g., **100**, **300**, **500**; **606-1**, **700**) (e.g., a personal computer, a laptop computer, a smartphone, a tablet computer) with a display device (e.g., **702**) (e.g., an integrated touch-sensitive display) and one or more input devices (e.g., **702**) (e.g., a mouse, a touchpad, a touch sensitive display). Some operations in method **800** are, optionally, combined, the orders of some operations are, optionally, changed, and some operations are, optionally, omitted.

As described below, method **800** provides an intuitive way for logging health-related events. The method reduces the cognitive burden on a user for logging health-related events, thereby creating a more efficient human-machine interface. For battery-operated computing devices, enabling a user to log health-related events faster and more efficiently conserves power and increases the time between battery charges.

The first electronic device (e.g., **700**) receives (**802**), via a first input device (e.g., **702**) of the one or more input devices, a first set of one or more inputs. The first set of one or more inputs includes a first input (e.g., input entering text at **712-1** or **726-1**) that identifies a first action (e.g., consumption of food or drink, performance of exercise) that was performed (e.g., performed by a user of the first electronic device), and a second input (e.g., input selecting **712-3B**; input **728**)) that identifies a first user sentiment associated

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with the first action (e.g., current sense of physical well-being of the user (e.g., when the first action is the performance of exercise) or prediction of future result (e.g., a prediction of future impact of the first action on a physiological parameter (e.g., blood glucose or heart rate) of the first action (e.g., when the first action is consumption of food or drink)).

The first electronic device, in response to the first set of one or more inputs, generates (**804**) a first log entry (e.g., event **704-6A** or **704-6B**) that includes the first action and the first user sentiment, wherein the first log entry is associated with a first time point (e.g., a time at which the first set of one or more inputs was received). Generating a log entry that includes an inputted action and sentiment provides the user with an improved input method for creating log entries with desired information which enhances the operability of the first electronic device. Enhancing the operability of the device makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first log entry includes an indication (e.g., icon of **704-6A**) that the first log entry is associated with the consumption of food or drink and the first user sentiment (e.g., **726-3A** or **726-3B**) includes a prediction of the impact (e.g., effect of; prediction that it will increase the physiological parameter by a first amount or a second amount) of the consumption of the food or drink on a first physiological parameter (e.g., blood glucose, heart rate, cholesterol) of the user of the first electronic device. Generating a log entry that is associated with the consumption of food or drink and a user prediction of the impact of the consumption provides the user with memorialized feedback as to a sentiment of predicted impact of a consumption event for later reference and review. Providing improved feedback enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first set of one or more inputs includes a third input (e.g., **704-8**) and the first electronic device displays (**806**) (e.g., prior to receiving the first set of one or more inputs), via the display device a first affordance (e.g., **708-1**) corresponding to a log entry of a first type (e.g., a log entry corresponding to the consumption of food or drink), and a second affordance (e.g., **708-2**) corresponding to a log entry of a second type (e.g., performance of a past activity (e.g., performance of exercise)) different from the first type. In accordance with a determination that the third input corresponds to selection of the first affordance, the first log entry includes an indication the first log entry is of the first type. In accordance with a determination that the third input corresponds to the first affordance, the first log entry includes an indication that the first log entry is of the second type. Including, in the log entry, an indication of whether the log entry is of the first type or the second type based on user input reduces the number of inputs required to designate a characteristic of the log entry. Reducing the number of inputs needed to perform this operation enhances the operability of the first electronic device. Enhancing the operability of the device makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs

and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, after generating the first log entry, the first electronic device displays (808), via the display device, a first user interface (e.g., 704) including a representation (e.g., 704-1) (e.g., a graph with time as a first axis and the value of a physiological parameter as a second axis) of a first period of time (e.g., 24-hour day, January 1st) that includes the first time point, a representation of the first log entry (e.g., 705-6A), and a representation of a second log entry (e.g., 704-6B) that is associated with a second time point, different from the first time point, within the first period of time.

In some embodiments, the first user interface includes a representation (e.g., a line graph) of the value of a second physiological parameter (e.g., blood glucose, heart rate, cholesterol) of the user of the first electronic device over the first period of time.

In some embodiments, the value of the second physiological parameter over the first period of time is received from a second electronic device (e.g., 610-1 or 608-1) (e.g., an external electronic device; a heart rate monitor, a continuous blood glucose monitor).

In some embodiments, the first user interface includes a representation (e.g., 704-5) of a current value (e.g., present value, a real-time value) of a third physiological parameter (e.g., blood glucose, heart rate, cholesterol) of the user of the first electronic device.

In some embodiments, the first user interface includes a representation of a third log entry (e.g., a log entry entered using interface 754) that was generated at a third electronic device (e.g., 750) (e.g., a personal computer, a laptop computer, a smartphone, a tablet computer) and transmitted to the first electronic device. In some embodiments, the third log entry can be edited (e.g., via set of one or more inputs received at the first electronic device) to add additional details corresponding to the log entry.

In some embodiments, the first set of one or more inputs includes fourth input (e.g., 738) that corresponds to a first event challenge (e.g., 736-2A) (e.g., a set of one or more actions to be performed or to not be performed (e.g., to forgo)) that was generated at a fourth electronic device (e.g., 612 or 900) (e.g., a personal computer, a laptop computer, a smartphone, a tablet computer) and transmitted to the first electronic device, and the first log entry includes an indication of the first event challenge. Generating a log entry that includes an indication of a relevant event challenge received from a fourth electronic device memorializes a relationship between an action performed by the user and a received event challenge, thereby providing enhanced feedback. Providing improved feedback enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first electronic device displays (710), via the display device, a second user interface (e.g., 736) that includes a representation (e.g., 736-1A) (e.g., an affordance) of a second event challenge (e.g., a set of one or more actions to be performed or to not be performed (e.g., to forgo)) that was generated at the fourth electronic device and transmitted to the first electronic device, and a representation (e.g., an affordance) of a third event challenge,

different from the second event challenge, that was generated at the fourth electronic device and transmitted to the first electronic device.

In some embodiments, the second event challenge (e.g., 736-1A) is an event challenge of a first type (e.g., a food/drink consumption challenge) and the third event challenge (e.g., 736-2A) is an event challenge of a second type (e.g., a habit challenge) different from the first type. The first electronic device, while displaying the second user interface (e.g., 736), receives a second set of one or more inputs (e.g., set of inputs including input 738). The first electronic device, in accordance with a determination that the second set of one or more inputs includes selection of the representation of the second event challenge, displays a third user interface (e.g., 726) configured to generate an event challenge response associated with a third time point (a specific point in time within a day/date (e.g., 12:00 PM on January 1st)). The first electronic device, in accordance with a determination that the second set of one or more inputs includes selection of the representation of the third event challenge, displaying a fourth user interface (e.g., 740) configured to generate an event challenge response associated with a first date (e.g., January 1st (e.g., but not associated with any specific time point of the date)). Displaying challenge response interfaces that are different based on the type of event challenge reduces the number of inputs required for a user to memorialize data relevant to the specific challenge event type. Reducing the number of inputs needed to perform this operation enhances the operability of the first electronic device. Enhancing the operability of the device makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first electronic device, after generating the first log entry, transmits the first log entry to a fifth electronic device (e.g., 612 or 900) (e.g., a personal computer, a laptop computer, a smartphone, a tablet computer) (e.g., an electronic device that receives log entries generated by the first electronic device and that transmits one or more event challenges to the first electronic device).

Note that details of the processes described above with respect to method 800 (e.g., FIG. 8) are also applicable in an analogous manner to the methods described below. For example, method 1000 optionally includes one or more of the characteristics of the various methods described above with reference to method 800. For example, the log entry displayed at 1002 of method 1000 can be based on an event generated at 804 of method 800. For brevity, these details are not repeated below.

FIGS. 9A-9M illustrate exemplary systems and user interfaces for health-related coaching using electronic devices.

FIG. 9A depicts device 900, which is a tablet computer. In some embodiments, device 900 is, or includes one or more features of, devices 100, 300, 500, and/or 612. Device 900 includes a touch-sensitive display 902, which serves as both a display and an input device. Device 900 also includes one or more wireless communication antennas for communicating with external electronic devices, such as devices 606-1, 608-1, and 610-1 of FIG. 6.

In FIG. 9A, device 900 is displaying a participant list interface 904. Interface 904 includes affordances 904-1A, 904-1B, and 904-1C corresponding to different participant devices currently sharing health-related data (e.g., blood glucose data and health-related event data) with device 900.

Interface **904** also includes region **904-2** that includes a plurality of affordances for sorting, searching, and modifying the participants listed in interface **904**.

In FIG. 9A, device **900** detects an input **906** (e.g., a touch) on affordance **904-1A**, corresponding to Participant 1 (e.g., Participant **1** of FIG. 6). In response to the input, device **900** displays overview interface **908** for Participant 1 as seen in FIG. 9B.

In FIG. 9B, device **900** displays overview interface **908**. Interface **908** includes a navigation region **908-1**, including affordances, such as affordance **908-1A** (“OVERVIEW”) and **908-1B** (“LOG DATA”) for navigating between certain interfaces. Affordance **908-1A** is currently visually distinguished to indicate that the current user interface is overview interface **908**. Interface **908** also includes an affordance (“COMMENTS”) for navigating to a view for entering comments, by a coach using device **900**, regarding Participant 1’s data. Interface **908** also includes an affordance (“PHOTO CHECK”) for navigating to a view for reviewing event-related photos transmitted by Participant 1.

Interface **908** also includes a graph **908-2** depicting blood glucose data for Participant 1. In some embodiments, the blood glucose data is transmitted to device **900** by one or more of devices **606-1**, **608-1**, and **610-1** of FIG. 6 that are associated with Participant 1. As shown, graph **908-2** has an x-axis showing a recurring 24-hour period for a plurality of days (e.g., days of a session 3, as shown in the graph). Graph **908-2** has a y-axis As shown, graph **908-2** includes three lines of blood glucose data, **908-2A**, **908-2B**, **908-2C**, one for each of three days currently logged in session 3 for Participant 1. Thus, graph **908-2** provides the ability to compare Participant 1’s blood glucose levels at the same time of day (e.g., 9 AM) across multiple days, as blood glucose levels often follow a daily cycle. In some embodiments, device **900** can scale or modify the graph based on input detected (e.g., pinch or swipe gestures) on display **902**. Graph **908-2** also includes indication **908-2D** that depicts periods of time during which Participant 1’s heart rate exceeded 50% of the maximum heart rate (e.g., the maximum heart rate for the depicted period). Graph **908-2** also includes indication **908-2E** that depicts periods of time during which Participant 1’s heart rate exceeded a higher threshold—64% of maximum. In some embodiments, providing heart rate data in addition to blood glucose data can provide the participant or a coach with information showing the impact of cardiovascular exercise on blood glucose levels.

Interface **908** also includes a high response region **908-3** that includes affordances for subsets of glucose data that meet reporting criteria (also referred to as excursions). In some embodiments, device **900** identifies and surfaces glucose data that indicates a notable rate of increase in blood glucose or a notable absolute value of blood glucose that exceeds a threshold. In FIG. 9B, device **900** has identified two notable incidents and displays affordances **908-3A** and **908-3B**. As shown, each affordance can also include pertinent information. For example, affordance **908-3A** includes the indication “COFFEE WITH 2 SUGARS” to indicate that a food/drink event was logged in or near the time associated with the notable data. Affordance **908-3B** includes the indication “0 MEALS, 0 ACTIVITIES” to indicate that no events were logged in or near the time associated with the notable data.

In FIG. 9B, device **900** detects an input **910** (e.g., a touch) on affordance **908-3A**, corresponding to the first high response data. In response to the input, device **900** displays day interface **912** for Participant 1 as seen in FIG. 9C.

In FIG. 9C, device **900** displays day interface **912** overlaid on interface **908**. Interface **912** includes graph **912-1** that corresponds to the data shown by line **908-2A** of interface **908**. As seen in FIG. 9C, that line corresponds to Participant 1’s blood glucose data for Day 1 (a Monday) of session 3. Graph **912-1** also includes indicators **912-2A** and **912-2B** corresponding to events logged on that by Participant 1. The logged events are also depicted in region **912-3** that includes indications of the two respective events as **912-3A** (a food/drink event (A COFFEE WITH TWO SUGARS)) and **912-3B** (an activity event (“RUNNING”)). In some embodiments, region **912-3** is scrollable to show additional events logged for the currently displayed day. Region **912-3** also includes affordance **912-3A1** and **912-3A2**, both associated with event **912-3A**. Affordance **912-3A1** can be used to initiate a process for generating a food challenge. Affordance **912-3A2** can be used to initiate a process for generating a habit challenge. In some embodiments, each event displayed in region **912-3** includes associated affordances for creating food and event challenges; scrolling region **912-3** up would reveal the affordances for event **912-3B**. As shown in graph **912-1**, Participant 1’s blood glucose level rose dramatically after event **912-3A** and decreased slightly after event **912-3B**. In some embodiments, a coach using device **900** can choose to issue a food challenge to guide Participant 1 to modify his or her consumption activities to mitigate dramatic increases in blood sugar.

In FIG. 9C, device **900** detects an input **914** (e.g., a touch) on affordance **912-3A1**. In response to the input, device **900** displays food challenge interface **916**, as seen in FIG. 9D.

In FIG. 9D, device **900** displays interface **916** for creating a new food challenge for Participant 1. Interface **916** includes region **916-1** for inputting a name for the challenge that also serves as the food/drink-related action that Participant 1 should perform to meet the challenge. In some embodiments, interface **916** includes separate fields for the name of the challenge and the action(s) required to meet the challenge. Interface **916** is displayed together with keyboard **918** for entry of text.

In FIG. 9E, device **900** has detected keyboard input and, in response, displays text **920** (“COFFEE WITH NO SUGAR”) as the action that Participant 1 should take to meet the challenge that was issued in response to the event “COFFEE WITH TWO SUGARS”. In FIG. 9E, device **900** detects input **922** on done affordance **924**. In response to the input, device **900** creates the food challenge and transmits data corresponding to the challenge to device(s) associated with Participant 1 (e.g., devices **606-1**, **608-1**, and **610-1**). In some embodiments, the transmitted data causes Participant 1’s devices to prompt Participant 1 to log a response event, as described above with respect to FIGS. 7I to 7K.

FIG. 9F depicts device **900** redisplaying interface **908** after generating and transmitting the food challenge associated with event **912-3A**. In FIG. 9F, device **900** detects input **926** on affordance **908-1B**. In response to the input, device **900** displays log data interface **928**, as seen in FIG. 9G.

In FIG. 9G, device **900** displays log data interface **928**. Interface **928** includes graph **928-1** that depicts the blood glucose data for Participant 1 over the period of the third logged days of session 3. Thus, the x-axis is the entire range of logged data for session 3, with the y-axis being blood glucose. The single line **928-1A** is a composite of the data shown by lines **908-2A**, **908-2B**, and **908-2C** of interface **908**.

Interface **928** includes region **928-2** with affordance(s) for the logged days of the currently selected session. In FIG. 9G,

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includes affordances **928-2A**, **928-2B**, and **928-2C**, one for each of the three logged days of session 3. Each affordance also includes a summary of events logged for that day (e.g., “1 MEAL, 1 ACTIVITY”).

In FIG. 9G, device **900** detects an input **930** (e.g., a touch) on affordance **928-2B** corresponding to Day 2 of session 3. In response to the input, device **900** displays day interface **932**, as seen in FIG. 9H.

In FIG. 9H, device **900** displays day interface **932**. Interface **932** includes elements similar to day interface **912**, but provides data for Day 2, rather than Day 1 of session 3. For example, interface **932** also includes a graph, graph **932-1**. Region **932-2** of interface **932** includes events logged by Participant 1 on Day 2 including events **932-2A**, **932-2B**, and **932-2C**. As shown in FIG. 9H, each event is associated with its own affordance for creating a corresponding food challenge or a corresponding habit challenge. For example, event **932-2C** is associated with food challenge affordance **932-2C1** and habit challenge affordance **932-2C2**.

In FIG. 9H, device **900** detects an input **934** (e.g., a touch) on affordance **932-2C2** corresponding to Day 2 of session 3. In response to the input, device **900** displays habit challenge interface **936**, as seen in FIG. 9I.

FIG. 9I depicts device **900** displaying habit challenge interface **936** after input **934** and after further inputs. Specifically, device **900** has detected keyboard input corresponding to entry of the text string “RUN 6 MILES” in field **936-1** which serves as both the name of the challenge and the action that should be taken to meet the habit challenge. In some embodiments, interface **936** includes separate fields for the name of the challenge and the action(s) that should be taken (or forgone) to meet the habit challenge.

In FIG. 9I, device **900** detects an input **938** (e.g., a touch) on done affordance **940**. In response to the input, device **900** creates the habit challenge and transmits data corresponding to the challenge to device(s) associated with Participant 1 (e.g., devices **606-1**, **608-1**, and **610-1**). In some embodiments, the transmitted data causes Participant 1’s devices to prompt Participant 1 to log a response the habit challenge, as described above with respect to FIGS. 7I to 7K.

In FIG. 9J, device **900** redisplay interface **928** after creating and transmitting the habit challenge associated with event **932-2C**. In FIG. 9J, device **900** detects input **942** on challenges affordance **944**. In response to the input, device **900** displays challenges interface **946**.

In FIG. 9K, device displays challenges interface **946**. Interface **946** includes current challenges region **946-1** that includes affordances **946-1A**, **946-1B**, and **946-1C**, corresponding to currently active challenges (e.g., both food and habit challenges) transmitted to device(s) of Participant 1. As shown, region **946-1** includes affordances for the newly created food challenge (**946-1B**) and the newly created habit challenge (**946-1C**).

Interface **946** also includes region **946-2** that includes affordance(s) for any challenges that have received at least one response. In FIG. 9K, the previously created challenge associated with affordance **946-1A** (“SUB ALMOND MILK IN COFFEE”) has responded to/met three times. Accordingly, region **946-2** includes an affordance **946-2A** for viewing data corresponding to the challenge responses.

In FIG. 9K, device **900** detects an input **948** (e.g., a touch) on affordance **946-2A** corresponding to Day 2 of session 3. In response to the input, device **900** displays challenge response interface **950**, as seen in FIG. 9L.

In FIG. 9L, device **900** displays challenge response interface **950**. Interface **950** includes a graph **950-1** of blood glucose data for Participant 1 for a time of day relevant to

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the responsive event and to the original event for which the challenge was issued. Graph **950-1** includes a blood glucose data line **950-1A** corresponding to Participant 1’s blood glucose data around the time when the 1st response event (of three) was logged. Graph **950-1** also includes blood glucose data line **950-1B** corresponding to Participant 1’s blood glucose data around the time when the original event for which the challenge was issued. Graph **950-1** also includes indication **950-1C** that marks the time when the 1st response event was logged. Thus, graph **950-1** provides a view for readily comparing the impact of the challenge action on Participant 1’s blood glucose levels.

In FIG. 9L, device **900** detects an input **952** (e.g., a touch) on next affordance **954**. In response to the input, device **900** updates challenge response interface **950** to show data relevant to the 2nd response event (of three) for the selected challenge, as shown in FIG. 9M.

In FIG. 9M, blood glucose data line **950-1A** now reflects Participant 1’s blood glucose data around the time when the 2nd response event was logged. Similarly, indicator **950-1C** now marks the time when the 2nd response event was logged. Note that blood glucose data line **950-1B**, corresponding to Participant 1’s blood glucose data around the time when the original event for which the challenge was issued, remains the same.

FIG. 10 is a flow diagram illustrating a method for health-related coaching using an electronic device in accordance with some embodiments. Method **1000** is performed at a first electronic device (e.g., **100**, **300**, **500**, **612**, **900**) ((e.g., a personal computer, a laptop computer, a smartphone, a tablet computer) with a display device (e.g., **902**) (e.g., an integrated touch-sensitive display) and one or more input devices (e.g., **902**) (e.g., a mouse, a touchpad, a touch sensitive display). Some operations in method **1000** are, optionally, combined, the orders of some operations are, optionally, changed, and some operations are, optionally, omitted.

As described below, method **1000** provides an intuitive way for logging health-related events. The method reduces the cognitive burden on a user for health-related coaching, thereby creating a more efficient human-machine interface. For battery-operated computing devices, enabling a user to log health-related events faster and more efficiently conserves power and increases the time between battery charges.

The first electronic device displays (**1002**), via the display device (e.g., **902**), a first log entry (e.g., **912-3A**) corresponding to a first event (e.g., a food-related event, an activity or exercise related event) associated with a user of a second electronic device (e.g., **606-1** or **700**) (e.g., a personal computer, a laptop computer, a smartphone, a tablet computer that has transmitted data to the first electronic device).

In some embodiments, displaying the first log entry (e.g., **912-3A**) includes displaying a first time point (e.g., 12 PM on January 1st) associated with (e.g., displayed in conjunction with, adjacent to) the first log entry, and first physiological data (e.g., **912-1**) (e.g., a blood glucose value, a heart rate, physiological data specifically associated with the first time point) associated with the user of the second electronic device, the first physiological data associated with (e.g., displayed in conjunction with, adjacent to) the first log entry.

In some embodiments, the first physiological data includes blood glucose data (e.g., as seen in **912**).

In some embodiments, displaying the first physiological data includes displaying a plurality of values of the first

physiological that is associated with a continuous range of time points over a first period of time (e.g., a day) (e.g., as seen in **912**). The first electronic device displays, via the display device and concurrently with displaying the first physiological data, second physiological data (e.g., heart rate data **912-2B**) associated with the user of the second electronic device, different from the first physiological data (e.g., the first physiological data is blood glucose and the second physiological data is heart rate), wherein the second physiological data is associated with a subset of time points (e.g., time points associated with **912-2B**) within the continuous range of time points (e.g., a subset of a day during which heart rate was greater than threshold amount (e.g., greater than 50% of max heart rate for the day)). Displaying the first and second physiological data, concurrently, provides the user with feedback regarding the relationship between the first and second physiological data. Providing improved feedback enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first event (e.g., **912-3A**) is associated with the consumption of food or drink by the user of the second electronic device.

The first electronic device, while displaying the first log entry corresponding to the first event (e.g., a food-related event, an activity or exercise related event), receives (**1004**), via a first input device (e.g., **902**) of the one or more input devices, a first set of one or more inputs (e.g., including input **914**) (e.g., a series or sequence of inputs) corresponding to a request to transmit a first event challenge (e.g., the challenge created in FIG. 9E) to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input (e.g., input entering text **920** in **916-1**) corresponding to (e.g., the input generates, creates, establishes) a set of one or more event conditions (e.g., set of one or more actions to be performed or to not be performed (e.g., to forgo)).

The first electronic device, in response to the first set of one or more inputs, transmits (**1006**), to the second electronic device, data representing the first event challenge. The data representing the first event challenge includes a prompt (e.g., prompt seen in **736**) to the user of the second electronic device (e.g., the transmitted challenge data include instructions that, when executed on the second electronic device, cause the device to issue a prompt) to log a challenge response event (e.g., an event that is logged in response to the challenge) based on the set of one or more event conditions (e.g., the issued prompt includes an indication of the set of one or more event conditions). In some embodiments, the data representing the event challenge includes data associating (e.g., linking and/or identifying) the event challenge with the first event. Transmitting data representing the first event challenge that includes a prompt to the second device to log a challenge response event reduces the number of inputs required to affect coordination between the first and second electronic devices thereby enhancing the operability of both devices. Enhancing the operability of the devices makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first electronic device, subsequent to displaying the first log entry, displays (**1008**), via the display device, a first affordance (e.g., **912-3A1**) corresponding to an event challenge of a first type (e.g., a challenge relating to a specific point in time (e.g., a challenge relating to the consumption of food or drink at a specific point in time or time within a day)), and a second affordance (e.g., **912-3B**) corresponding to an event challenge of a second type (e.g., a challenge relating to a day, but not relating to a specific time within that day), different than the first type, wherein the first set of one or more inputs includes a second input (e.g., **914**) received while displaying the first affordance and the second affordance. The first electronic device, in response to the second input of the first set of one or more inputs and in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, includes, in the data transmitted to the second electronic device, an indication that the first event challenge is of the first type. The first electronic device, in response to the second input of the first set of one or more inputs and in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, includes, in the data transmitted to the second electronic device, an indication that the first event challenge is of the second type. In some embodiments, the transmitted data causes the second electronic device to issue differing prompts, depending on whether the challenge is of the first type or the second type. Including, in the transmitted challenge data an indication of whether the event challenge is of the first type or the second type based on user input reduces the number of inputs required to designate a characteristic of the event challenge. Reducing the number of inputs needed to perform this operation enhances the operability of the first electronic device. Enhancing the operability of the device makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first event challenge is a challenge of the first type, the prompt includes a prompt to the user to log the second event (e.g., at a specific time (e.g., a specific time of day, a specific time within a day, a time that is more precise than the day)) based on (e.g., based on satisfaction of) the set of one or event conditions, and the transmitted data representing the set of one or more event conditions of the first event challenge includes an indication (e.g., as seen in **736**) of a first set of one or more actions (e.g., actions related to the consumption of food or drink at a specific point in time or time within a day) to the to be performed.

In some embodiments, the first event challenge is a challenge of the second type and the prompt includes a prompt to the user to log the challenge response event (e.g., at a specific date (e.g., a date that does not include a specific time within the day of that date)) based on the set of one or event conditions. The transmitted data representing the set of one or more event conditions of the first event challenge includes an indication (e.g., as seen in **736**) of a second set of one or more actions (e.g., actions related to the consumption of food or drink or actions related to the performance of physical activity (e.g., exercise)), and an indication of whether the second set of one or more actions are to be performed or are not to be performed.

In some embodiments, the first electronic device displays, via the display device and concurrently with displaying the first log entry, a second log entry (e.g., **932-2B**) corresponding to a second event associated with the user of the second electronic device (e.g., a second event logged on the same day as the first event), a third affordance (e.g., food challenge affordance associated with **932-2B**) associated with the second log entry that, when selected, initiates a process for transmitting a second event challenge (e.g., a challenge of the first or second type) to the user associated with the second electronic device, a third log entry corresponding to a third event associated with the user of the second electronic device, and a fourth affordance (e.g., habit challenge affordance associated with **932-2B**) associated with the third log entry that, when selected, initiates a process for transmitting a third event challenge to the user associated with the second electronic device. In some embodiments, the first electronic device displays a plurality of log entries corresponding to the same time period (e.g., to the same day). Each of the log entries includes one or more affordances for issuing event challenges associated with the respective log entry. Displaying the first log entry in conjunction with the second log entry, with its associated affordances, provides improved feedback as to what log entries have been received by the first electronic device and improved options for performing related functions with less inputs. Providing improved feedback enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first electronic device, after transmitting the data representing the first event challenge, displays, via the display device, a first user interface (e.g., **946**) (e.g., a transmitted challenges user interface). The first user interface includes a first region (e.g., **946-1**) (e.g., a pending challenges region) that includes a representation (e.g., **946-1A**) (e.g., an affordance (e.g., a selectable representation); a representation that includes one or more details (e.g., details regarding the set of one or more event conditions; details regarding the first event) of the first event challenge) of the first event challenge, and a second region (e.g., **946-2**) (e.g., a completed/responded challenge region). The second region, in accordance with a determination that data representing a first challenge response event has been received (e.g., received in response to the first event challenge or in response to a second event challenge different from the first event challenge) from the second electronic device, includes a representation (e.g., **946-2A**) of the first challenge response event. The second region, in accordance with a determination that data representing the first challenge response event has not been received the second electronic device, does not include the representation of the first challenge response event (In some embodiments, the second region only includes representations associated with issued/transmitted event challenges for which at least one challenge response event has been received).

In some embodiments, the first challenge response event is responsive to the first event challenge. The first electronic device, while displaying the representation of the first challenge response event, receives a second set of one or more inputs (e.g., **948**) corresponding to a request to display a response event user interface (e.g., **950**) corresponding to the first challenge response event. The first electronic device, in response to the second set of one or more inputs, displays,

via the display device, the response event user interface. The response event user interface includes third physiological data (e.g., **950-1B**) (e.g., as a line graph) associated with the user of the second electronic device that corresponds to the first event (e.g., physiological data associated with the first event (e.g., physiological data at the time the first event was logged at the second electronic device)), and fourth physiological data (e.g., **950-1A**) (e.g., as a line graph), different from the second physiological data, associated with the user of the second electronic device that corresponds to the first challenge response event (e.g., physiological data associated with the first challenge event response (e.g., physiological data at the time the first challenge event response was logged at the second electronic device)). Displaying the third and fourth physiological data, concurrently, provides the user with feedback regarding the relationship between the second and third physiological data. Providing improved feedback enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first electronic device displays, via the display device, a second user interface (e.g., **908**) (e.g., an overview user interface that includes physiological data for multiple days for the same range of time of a day). The second user interface includes a plurality of representations (e.g., **908-2A**, **908-2B**, **908-2C**) (e.g., a plurality of lines on a graph) of physiological data associated with the user of the second electronic device, including a first representation (e.g., a first line of a graph) of physiological data for a first occurrence (e.g., a first day (e.g., January 1st)) of a first recurring time period (e.g., morning, 12 AM to 11:9 PM, a 24-hour day) and a second representation of physiological data for a second occurrence (e.g., a first day (e.g., January 1st)) of the first recurring time period, and a fifth affordance (e.g., **908-3A**) (e.g., a tile affordance that includes a representation of the subset of physiological data) corresponding to a subset of the physiological data for the first occurrence that meets a set of reporting criteria (e.g., criteria based on one or more factors selected from a rate of change (e.g., specifically a rate of increase) of the physiological parameter of the data and an absolute value of the physiological data that exceeds a threshold). Displaying a fifth affordance that corresponds to a subset of data that meets a set of reporting criteria provides the user with feedback as to portions of data that are potentially significant and reduces the number of inputs required to identify and display potentially significant data. Providing improved feedback enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

In some embodiments, the first electronic device receives a third set of one or more inputs (e.g., **910**) corresponding to selection of the fifth affordance. The first electronic device, in response to receiving the third set of one or more inputs, displays, via the display device, a third user interface (e.g., **912**) (e.g., an interface associated with first occurrence (e.g., the day)). The third user interface includes a third representation (e.g., **912-1**) (a graph line) of physiological data for the first occurrence of the first recurring time period. The

third user interface, in accordance with a determination that the first occurrence of the recurring time period is associated with a third log entry corresponding to a fourth event, includes a representation (e.g., **912-3A**) of the third log entry (e.g., representation on the graph of a time at which the fourth event was logged). The third user interface, in accordance with a determination that the first occurrence of the first recurring time period is not associated with the third log entry corresponding to the fourth event, does not include the representation of the third log entry.

In some embodiments, the first electronic device displays, via the display device, a fourth user interface (e.g., **928**) (e.g., an overview user interface that includes physiological data for multiple days for the same range of time of a day). The fourth user interface includes a fourth representation (e.g., **928-1A**) of physiological data (e.g., a line of graph that spans multiple days) for a first occurrence (e.g., a first day, January 1st) of a second recurring time period (e.g., 24-hour day) and a second occurrence (e.g., a second day, January 2nd) of the second recurring time period associated with the user of the second electronic device, and a sixth affordance corresponding to the first occurrence of the second recurring time period (e.g., and not associated with the second occurrence). In some embodiments, selection of the sixth affordance causes display of physiological data associated with the first occurrence of the recurring time period without display of physiological data corresponding to the second occurrence of the second recurring time period.

In some embodiments, the first electronic device, displays, via the display device, a fourth log entry corresponding to a fifth event (e.g., a food-related event, an activity or exercise related event) associated with a user of a third electronic device (e.g., **606-2**) (e.g., a personal computer, a laptop computer, a smartphone, a tablet computer that has transmitted data to the first electronic device) different from the first electronic device and the second electronic device.

Note that details of the processes described above with respect to method **1000** (e.g., FIG. **10**) are also applicable in an analogous manner to the methods described above. For example, method **800** optionally includes one or more of the characteristics of the various methods described above with reference to method **1000**. For example, the first log entry generated at **804** of method **800** can be responsive to the first challenge event transmitted at **1006** of method **1000**. For brevity, these details are not repeated below.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.

As described above, one aspect of the present technology is the gathering and use of data available from various sources to improve the delivery to users of invitational content or any other content that may be of interest to them.

The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to record and deliver personalized health information to a user. Accordingly, use of such personal information data enables users to improve logging and viewing of health information. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of health monitoring services, the present technology can be configured to allow users to select to "opt in" or "opt out" of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide health-

associated data for targeted content delivery services. In yet another example, users can select to limit the length of time health-associated data is maintained or entirely prohibit the development of a baseline health profile. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user’s privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, content can be selected and delivered to users by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the content delivery services, or publicly available information.

What is claimed is:

1. A first electronic device, comprising:
 - a display device;
 - one or more input devices;
 - one or more processors; and
 - memory storing one or more programs configured to be executed by the one or more processors, the one or more programs including instructions for:
 - displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device;
 - while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and
 - in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge;
 - wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a first challenge response event based on the set of one or more event conditions;
 - after transmitting the data representing the first event challenge, displaying, via the display device, a first user interface that includes:

- a first region that includes a representation of the first event challenge; and

- a second region that:

- in accordance with a determination that data representing the first challenge response event has been received from the second electronic device, includes a representation of the first challenge response event, wherein the representation of the first challenge response event includes an affordance for viewing data corresponding to the first challenge response event that, when selected, causes the display of first physiological data associated with the first challenge response event wherein the first challenge response event is responsive to the first event challenge; and

- in accordance with a determination that data representing the first challenge response event has not been received from the second electronic device, does not include the representation of the first challenge response event;

- while displaying the representation of the first challenge response event, receiving a second set of one or more inputs that includes a first input corresponding to the affordance for viewing data corresponding to the first challenge response event; and

- in response to the second set of one or more inputs, displaying, via the display device, a response event user interface, wherein the response event user interface includes:

- second physiological data associated with the user of the second electronic device that corresponds to the first event; and

- third physiological data, different from the second physiological data,

- associated with the user of the second electronic device that corresponds to the first challenge response event.

2. The first electronic device of claim 1, the one or more programs further including instructions for:

- subsequent to displaying the first log entry, displaying, via the display device:

- a first affordance corresponding to an event challenge of a first type; and

- a second affordance corresponding to an event challenge of a second type, different than the first type, wherein the first set of one or more inputs includes a second input received while displaying the first affordance and the second affordance;

- in response to the second input of the first set of one or more inputs:

- in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, including, in the data transmitted to the second electronic device, an indication that the first event challenge is of the first type; and

- in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, including, in the data transmitted to the second electronic device, an indication that the first event challenge is of the second type.

3. The first electronic device of claim 2, wherein:
 - the first event challenge is a challenge of the first type;
 - the prompt includes a prompt to the user to log a second event based on the set of one or event conditions; and

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the transmitted data representing the set of one or more event conditions of the first event challenge includes an indication of a first set of one or more actions to be performed.

4. The first electronic device of claim 2, wherein:

the first event challenge is a challenge of the second type; the prompt includes a prompt to the user to log the first challenge response event based on the set of one or more event conditions; and

the transmitted data representing the set of one or more event conditions of the first event challenge includes: an indication of a second set of one or more actions; and

an indication of whether the second set of one or more actions are to be performed or are not to be performed.

5. The first electronic device of claim 1, the one or more programs further including instructions for:

displaying, via the display device and concurrently with displaying the first log entry:

a second log entry corresponding to a second event associated with the user of the second electronic device;

a third affordance associated with the second log entry that, when selected, initiates a process for transmitting a second event challenge to the user associated with the second electronic device;

a third log entry corresponding to a third event associated with the user of the second electronic device; and

a fourth affordance associated with the third log entry that, when selected, initiates a process for transmitting a third event challenge to the user associated with the second electronic device.

6. The first electronic device of claim 1, wherein:

displaying the first log entry includes displaying:

a first time point associated with the first log entry; and

fourth physiological data associated with the user of the second electronic device, the fourth physiological data associated with the first log entry.

7. The first electronic device of claim 6, wherein the fourth physiological data includes blood glucose data.

8. The first electronic device of claim 6, wherein displaying the fourth physiological data includes displaying a plurality of values of the fourth physiological data that is associated with a continuous range of time points over a first period of time, the one or more programs further including instructions for:

displaying, via the display device and concurrently with displaying the fourth physiological data, fifth physiological data associated with the user of the second electronic device, different from the fourth physiological data, wherein the fifth physiological data is associated with a subset of time points within the continuous range of time points.

9. The first electronic device of claim 1, the one or more programs further including instructions for:

displaying, via the display device, a second user interface that includes:

a plurality of representations of physiological data associated with the user of the second electronic device, including a first representation of physiological data for a first occurrence of a first recurring time period and a second representation of physiological data for a second occurrence of the first recurring time period; and

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a fifth affordance corresponding to a subset of the physiological data for the first occurrence that meets a set of reporting criteria.

10. The first electronic device of claim 9, the one or more programs further including instructions for:

receiving a third set of one or more inputs corresponding to selection of the fifth affordance;

in response to receiving the third set of one or more inputs, displaying, via the display device, a third user interface that:

includes a third representation of physiological data for the first occurrence of the first recurring time period; in accordance with a determination that the first occurrence of the first recurring time period is associated with a third log entry corresponding to a fourth event, includes a representation of the third log entry; and

in accordance with a determination that the first occurrence of the first recurring time period is not associated with the third log entry corresponding to the fourth event, does not include the representation of the third log entry.

11. The first electronic device of claim 1, the one or more programs further including instructions for:

displaying, via the display device, a fourth user interface that includes:

a fourth representation of physiological data for a first occurrence of a second recurring time period and a second occurrence of the second recurring time period associated with the user of the second electronic device; and

a sixth affordance corresponding to the first occurrence of the second recurring time period.

12. The first electronic device of claim 1, wherein the first event is associated with the consumption of food or drink by the user of the second electronic device.

13. The first electronic device of claim 1, the one or more programs further including instructions for:

displaying, via the display device, a fourth log entry corresponding to a fifth event associated with a user of a third electronic device different from the first electronic device and the second electronic device.

14. A non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a first electronic device having a display device and one or more input devices, the one or more programs including instructions for:

displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device;

while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and

in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge;

wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a first challenge response event based on the set of one or more event conditions;

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after transmitting the data representing the first event challenge, displaying, via the display device, a first user interface that includes:

a first region that includes a representation of the first event challenge; and

a second region that:

in accordance with a determination that data representing the first challenge response event has been received from the second electronic device, includes a representation of the first challenge response event, wherein the representation of the first challenge response event includes an affordance for viewing data corresponding to the first challenge response event that, when selected, causes the display of first physiological data associated with the first challenge response event wherein the first challenge response event is responsive to the first event challenge; and
in accordance with a determination that data representing the first challenge response event has not been received from the second electronic device, does not include the representation of the first challenge response event;

while displaying the representation of the first challenge response event, receiving a second set of one or more inputs that includes a first input corresponding to the affordance for viewing data corresponding to the first challenge response event; and

in response to the second set of one or more inputs, displaying, via the display device, a response event user interface, wherein the response event user interface includes:

second physiological data associated with the user of the second electronic device that corresponds to the first event; and

third physiological data, different from the second physiological data, associated with the user of the second electronic device that corresponds to the first challenge response event.

15. The non-transitory computer-readable storage medium of claim **14**, the one or more programs further including instructions for:

subsequent to displaying the first log entry, displaying, via the display device:

a first affordance corresponding to an event challenge of a first type; and

a second affordance corresponding to an event challenge of a second type, different than the first type, wherein the first set of one or more inputs includes a second input received while displaying the first affordance and the second affordance;

in response to the second input of the first set of one or more inputs:

in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, including, in the data transmitted to the second electronic device, an indication that the first event challenge is of the first type; and

in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, including, in the data transmitted to the second electronic device, an indication that the first event challenge is of the second type.

16. The non-transitory computer-readable storage medium of claim **15**, wherein:

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the first event challenge is a challenge of the first type; the prompt includes a prompt to the user to log a second event based on the set of one or event conditions; and the transmitted data representing the set of one or more event conditions of the first event challenge includes an indication of a first set of one or more actions to be performed.

17. The non-transitory computer-readable storage medium of claim **15**, wherein:

the first event challenge is a challenge of the second type; the prompt includes a prompt to the user to log the first challenge response event based on the set of one or event conditions; and

the transmitted data representing the set of one or more event conditions of the first event challenge includes: an indication of a second set of one or more actions; and

an indication of whether the second set of one or more actions are to be performed or are not to be performed.

18. The non-transitory computer-readable storage medium of claim **14**, the one or more programs further including instructions for:

displaying, via the display device and concurrently with displaying the first log entry:

a second log entry corresponding to a second event associated with the user of the second electronic device;

a third affordance associated with the second log entry that, when selected, initiates a process for transmitting a second event challenge to the user associated with the second electronic device;

a third log entry corresponding to a third event associated with the user of the second electronic device; and

a fourth affordance associated with the third log entry that, when selected, initiates a process for transmitting a third event challenge to the user associated with the second electronic device.

19. The non-transitory computer-readable storage medium of claim **14**, wherein:

displaying the first log entry includes displaying:

a first time point associated with the first log entry; and fourth physiological data associated with the user of the second electronic device, the fourth physiological data associated with the first log entry.

20. The non-transitory computer-readable storage medium of claim **19**, wherein the fourth physiological data includes blood glucose data.

21. The non-transitory computer-readable storage medium of claim **19**, wherein displaying the fourth physiological data includes displaying a plurality of values of the fourth physiological that is associated with a continuous range of time points over a first period of time, the one or more programs further including instructions for:

displaying, via the display device and concurrently with displaying the fourth physiological data, fifth physiological data associated with the user of the second electronic device, different from the fourth physiological data, wherein the fifth physiological data is associated with a subset of time points within the continuous range of time points.

22. The non-transitory computer-readable storage medium of claim **14**, the one or more programs further including instructions for:

displaying, via the display device, a second user interface that includes:

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a plurality of representations of physiological data associated with the user of the second electronic device, including a first representation of physiological data for a first occurrence of a first recurring time period and a second representation of physiological data for a second occurrence of the first recurring time period; and

a fifth affordance corresponding to a subset of the physiological data for the first occurrence that meets a set of reporting criteria.

23. The non-transitory computer-readable storage medium of claim 22, the one or more programs further including instructions for:

receiving a third set of one or more inputs corresponding to selection of the fifth affordance;

in response to receiving the third set of one or more inputs, displaying, via the display device, a third user interface that:

includes a third representation of physiological data for the first occurrence of the first recurring time period;

in accordance with a determination that the first occurrence of the first recurring time period is associated with a third log entry corresponding to a fourth event, includes a representation of the third log entry; and

in accordance with a determination that the first occurrence of the first recurring time period is not associated with the third log entry corresponding to the fourth event, does not include the representation of the third log entry.

24. The non-transitory computer-readable storage medium of claim 14, the one or more programs further including instructions for:

displaying, via the display device, a fourth user interface that includes:

a fourth representation of physiological data for a first occurrence of a second recurring time period and a second occurrence of the second recurring time period associated with the user of the second electronic device; and

a sixth affordance corresponding to the first occurrence of the second recurring time period.

25. The non-transitory computer-readable storage medium of claim 14, wherein the first event is associated with the consumption of food or drink by the user of the second electronic device.

26. The non-transitory computer-readable storage medium of claim 14, the one or more programs further including instructions for:

displaying, via the display device, a fourth log entry corresponding to a fifth event associated with a user of a third electronic device different from the first electronic device and the second electronic device.

27. A method comprising:

at a first electronic device having a display device and one or more input devices:

displaying, via the display device, a first log entry corresponding to a first event associated with a user of a second electronic device;

while displaying the first log entry corresponding to the first event, receiving, via a first input device of the one or more input devices, a first set of one or more inputs corresponding to a request to transmit a first event challenge to the user associated with the second electronic device, wherein the first set of one or more inputs includes a first input corresponding to a set of one or more event conditions; and

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in response to the first set of one or more inputs, transmitting, to the second electronic device, data representing the first event challenge;

wherein the data representing the first event challenge includes a prompt to the user of the second electronic device to log a first challenge response event based on the set of one or more event conditions;

after transmitting the data representing the first event challenge, displaying, via the display device, a first user interface that includes:

a first region that includes a representation of the first event challenge; and

a second region that:

in accordance with a determination that data representing the first challenge response event has been received from the second electronic device, includes a representation of the first challenge response event, wherein the representation of the first challenge response event includes an affordance for viewing data corresponding to the first challenge response event that, when selected, causes the display of first physiological data associated with the first challenge response event wherein the first challenge response event is responsive to the first event challenge; and

in accordance with a determination that data representing the first challenge response event has not been received the second electronic device, does not include the representation of the first challenge response event;

while displaying the representation of the first challenge response event, receiving a second set of one or more inputs that includes a first input corresponding to the affordance for viewing data corresponding to the first challenge response event; and

in response to the second set of one or more inputs, displaying, via the display device, a response event user interface, wherein the response event user interface includes:

second physiological data associated with the user of the second electronic device that corresponds to the first event; and

third physiological data, different from the second physiological data, associated with the user of the second electronic device that corresponds to the first challenge response event.

28. The method of claim 27, further comprising:

subsequent to displaying the first log entry, displaying, via the display device:

a first affordance corresponding to an event challenge of a first type; and

a second affordance corresponding to an event challenge of a second type, different than the first type, wherein the first set of one or more inputs includes a second input received while displaying the first affordance and the second affordance;

in response to the second input of the first set of one or more inputs:

in accordance with a determination that the second input of the first set of one or more inputs corresponds to the first affordance, including, in the data transmitted to the second electronic device, an indication that the first event challenge is of the first type; and

in accordance with a determination that the second input of the first set of one or more inputs corre-

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sponds to the first affordance, including, in the data transmitted to the second electronic device, an indication that the first event challenge is of the second type.

29. The method of claim 28, wherein:
the first event challenge is a challenge of the first type;
the prompt includes a prompt to the user to log a second event based on the set of one or event conditions; and
the transmitted data representing the set of one or more event conditions of the first event challenge includes an indication of a first set of one or more actions to be performed.

30. The method of claim 28, wherein:
the first event challenge is a challenge of the second type;
the prompt includes a prompt to the user to log the first challenge response event based on the set of one or event conditions; and
the transmitted data representing the set of one or more event conditions of the first event challenge includes:
an indication of a second set of one or more actions;
and
an indication of whether the second set of one or more actions are to be performed or are not to be performed.

31. The method of claim 27, further comprising:
displaying, via the display device and concurrently with displaying the first log entry:
a second log entry corresponding to a second event associated with the user of the second electronic device;
a third affordance associated with the second log entry that, when selected, initiates a process for transmitting a second event challenge to the user associated with the second electronic device;
a third log entry corresponding to a third event associated with the user of the second electronic device;
and
a fourth affordance associated with the third log entry that, when selected, initiates a process for transmitting a third event challenge to the user associated with the second electronic device.

32. The method of claim 27, wherein:
displaying the first log entry includes displaying:
a first time point associated with the first log entry; and
fourth physiological data associated with the user of the second electronic device, the fourth physiological data associated with the first log entry.

33. The method of claim 32, wherein the fourth physiological data includes blood glucose data.

34. The method of claim 32, wherein displaying the fourth physiological data includes displaying a plurality of values of the fourth physiological that is associated with a continuous range of time points over a first period of time, the one or more programs further including instructions for:

displaying, via the display device and concurrently with displaying the fourth physiological data, fifth physiological data associated with the user of the second

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electronic device, different from the fourth physiological data, wherein the fifth physiological data is associated with a subset of time points within the continuous range of time points.

35. The method of claim 27, further comprising:
displaying, via the display device, a second user interface that includes:

a plurality of representations of physiological data associated with the user of the second electronic device, including a first representation of physiological data for a first occurrence of a first recurring time period and a second representation of physiological data for a second occurrence of the first recurring time period; and

a fifth affordance corresponding to a subset of the physiological data for the first occurrence that meets a set of reporting criteria.

36. The method of claim 35, further comprising:
receiving a third set of one or more inputs corresponding to selection of the fifth affordance;

in response to receiving the third set of one or more inputs, displaying, via the display device, a third user interface that:

includes a third representation of physiological data for the first occurrence of the first recurring time period;
in accordance with a determination that the first occurrence of the first recurring time period is associated with a third log entry corresponding to a fourth event, includes a representation of the third log entry; and

in accordance with a determination that the first occurrence of the first recurring time period is not associated with the third log entry corresponding to the fourth event, does not include the representation of the third log entry.

37. The method of claim 27, further comprising:
displaying, via the display device, a fourth user interface that includes:

a fourth representation of physiological data for a first occurrence of a second recurring time period and a second occurrence of the second recurring time period associated with the user of the second electronic device; and

a sixth affordance corresponding to the first occurrence of the second recurring time period.

38. The method of claim 27, wherein the first event is associated with the consumption of food or drink by the user of the second electronic device.

39. The method of claim 27, further comprising:
displaying, via the display device, a fourth log entry corresponding to a fifth event associated with a user of a third electronic device different from the first electronic device and the second electronic device.

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